Polynomial Upper Bounds on Characteristic-Data Length for Multiple Context-Free Languages under Fixed Finite-Semigroup Typing

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Abstract

We bound the total encoded length of characteristic samples for positive-data learning of multiple context-free languages that are tuple-substitutable under a fixed finite-semigroup typing, relative to their MCFG presentations; qualitative identification follows Yoshinaka's distributional framework. With the alphabet, fan-out, rule rank and typing fixed, a reduced good grammar G of rule thickness $\bar{\theta}_G$ admits characteristic data of length $O(\|G\|^2\bar{\theta}_G)$ for prefix–suffix-window typings combined with any finite, possibly noncommutative, group, and more generally whenever the positive typing image is a local group. The key step shortens interleaved tuple components simultaneously while preserving ordered group products. Conversely, the shortest word of each realized type is a presentation-independent lower bound. At fan-out one this yields a characterization: with rule rank two allowed, the pseudovarieties of finite semigroups admitting uniform polynomial characteristic-data length are exactly the subpseudovarieties of local groups; moreover, under every fixed typing some targets with asymptotically smallest presentations need characteristic data of length $\Omega(\|G\|\bar{\theta}_G)$. At fan-out two the local-group condition is no longer necessary for individual typings: for the binary typing $\mathtt{a} \mapsto 1$, $\mathtt{b} \mapsto \zeta$ onto $U_1 \notin \mathbf{LG}$, characteristic data of length $O(\|G\|^3\bar{\theta}_G^5)$ always exist, and a power-transfer lemma shows that the fan-out-one lower-bound family does not lift.

Keywords: grammatical inference; multiple context-free grammars; positive data; finite-semigroup typing; multidimensional substitutability; Gold identification; rule rank.

1 Introduction

Positive-data grammatical inference asks when a language can be identified in the limit from positive examples alone [1,2]. Distributional learning treats words, or tuples of words, that occur in common contexts as candidate constituents. It began with substitutable CFGs [3, 12] and was extended to multiple context-free grammars (MCFGs), whose tuple-valued nonterminals express discontinuous and cross-serial dependencies [13,14,27,28]. Yoshinaka's finite-signature framework [15] gives positive-data identification with characteristic samples of polynomially many words. Polynomially many words may still be exponentially long, so we measure a finite sample K by its total encoded length $\|K\|_+ = \sum_{w\in K}(|w| + 1)$, following the thick-data perspective of [10, Section 2.5.2]. Qualitative identification is inherited from Yoshinaka's framework; our contribution is to bound $\|\cdot\|_+$ of characteristic samples relative to an MCFG presentation of the target.

The setting. Before any data are seen, we fix a fan-out bound f and a surjective finite-semigroup morphism $s : \Sigma^+ \twoheadrightarrow \mathsf{S}$, a *typing*. The learner compares observed nonempty tuple components only when their s-values agree, and the target class $\mathcal{C}^{\mathrm{mcf}}_{f,s}$ consists of the f-MCFLs that are tuple-substitutable under this restriction (Definition 2.8). The typing is a class-level bias in the sense of [5], not chosen per target; other biases restrict derivations by control sets [6, 7] or restrict the evidence used to compare factors [9, 11]. Typical typings are prefix–suffix windows, values in a finite group, their products, and DFA transition morphisms. For instance, one fixed window–group typing places every language

$$L_R = \{\mathtt{a}^m \mathtt{p}_i \mathtt{b}^n \mathtt{c}^m \mathtt{q}_j \mathtt{d}^n : m, n \geq 1,\ (i, j) \in R\}, \qquad R \subseteq \{0, 1\}^2,$$

in $\mathcal{C}^{\mathrm{mcf}}_{2,s}$: the group component records the feature values i, j, while the licensing relation R and the crossing dependencies must be learned from data; untyped and boundary-only substitutability both fail for some R (Proposition 6.3). Group values compose in the order of the blocks, so noncommutative groups such as S_3 matter: some targets admit a window–S_3 typing but no window–abelian one (Proposition 6.6). Local groups, finite semigroups T with $e\mathsf{T}e$ a group for every idempotent e, add group contributions of sliding factors, such as the parity of occurrences of a bigram (Proposition B.5).

Approach. Refining the grammar by types and applying untyped estimates to the refinement can be expensive: for a fixed two-element typing there are presentations G_n of $\{\mathtt{a}, \mathtt{c}\}^+$ with $\|G_n\| = O(n)$ and rule thickness at most 2 whose full yield-type refinement has thickness at least 2^n (Proposition 4.9). We therefore bound characteristic data in terms of the original presentation. To obtain short tuples of prescribed type, we replace long subderivations by short ones that need not preserve the group values of the replaced subtrees. In an MCFG derivation a subtree contributes several blocks to different root components, interleaved with blocks of other subtrees, and for noncommutative groups the effect of simultaneous replacements depends on this order. The key step is an *owner-wise simultaneous selection* of replacements that preserves the ordered group product of every root component (Proposition 4.12). For local-group typings the group value is a product over sliding factors, which straddle block boundaries; selecting owners whose blocks lie far apart and enlarging each block by unchanged boundary letters reduces this case to the same selection (Lemma B.6).

Main results. For a reduced good MCFG G we use its rule thickness $\bar{\theta}_G$, the largest length of a shortest sentence using a given rule. Theorem 4.4 reduces all bounds to one parameter, the shortest typed tuple yield $t_{\mathrm{typ}}(G, s)$. With the alphabet, fan-out, rule-rank bound $r_{\max}$ and typing fixed, the bounds on the total encoded length of characteristic samples for the Const/Link/Comp reconstruction operator are:

Fixed typing / rank	Bound	Source
Rule rank ≤ 1; any typing	$O(\|G\|^3)$	Thm. 5.1
Prefix–suffix window with a finite group (possibly trivial or noncommutative)	$O(\|G\|^2 \bar{\theta}_G)$	Thm. 4.15
Positive image in **LG**	$O(\|G\|^2 \bar{\theta}_G)$	Thm. 4.19
Binary typing $\mathtt{a} \mapsto 1$, $\mathtt{b} \mapsto \zeta$ onto U_1; $f = 2$	$O(\|G\|^3 \bar{\theta}_G^5)$	Thm. 6.10

The bounds hold for every $L \in \mathcal{C}^{\mathrm{mcf}}_{f,s}$ with a presentation of the stated form, with constants depending only on f, $r_{\max}$, $|\Sigma|$ and the typing. They are polynomial in $\|G\|$ and $\bar{\theta}_G$ only: the constants contain the factor $|\mathsf{S}^\sharp|^{f r_{\max}}$, which counts typed rules, and factors such as $(|M_{k,\ell}||\mathbb{G}|)^f$, all exponential in

f and $r_{\max}$; for noncommutative groups they also contain $M_{\mathbb{G},f}$, which is doubly exponential in f, while the abelian case is sharper (Remark 4.16). A separate conservative learner gives Gold identification.

Conversely, every realized s-type of the target must occur in every characteristic sample, so its shortest word is a presentation-independent lower bound (Proposition 4.6, Corollary 4.7). At fan-out one this yields Theorem 4.22: for $r_{\max} \geq 2$, the pseudovarieties of finite semigroups with uniform polynomial characteristic-data length (Definition 4.21) are precisely the subpseudovarieties of **LG**; the lower bound uses a bracketed doubling family (Proposition 4.20). For individual typings, a three-letter variant gives the same exponential lower bound whenever two letters of one idempotent type and one letter of an absorbing idempotent type realize U_1 (Corollary 4.25); among typings onto U_1, only those with a single letter of type 1 remain open. Relative to the presentation, the product $\|G\|\bar{\theta}_G$ cannot be reduced at fan-out one: for every fixed typing, finite targets with asymptotically smallest presentations need characteristic data of length $\Omega(\|G\|\bar{\theta}_G)$ (Proposition 4.8). Local groups strictly extend $\mathbf{LI} \vee \mathbf{Grp}$ (Proposition B.5), and Section 6 separates the typing regimes by explicit non-context-free targets.

What is proved at fan-out two or more. At fan-out $f \geq 2$ the new content is threefold: (a) the window–group and local-group bounds hold for components that interleave across subtrees, which does not arise at fan-out one; (b) a two-hole power-transfer lemma shows that the fan-out-one lower-bound family is not (f, s)-tuple-substitutable for any fixed finite typing at any $f \geq 2$ (Theorem 6.8); (c) at fan-out two the binary typing $\mathtt{a} \mapsto 1$, $\mathtt{b} \mapsto \zeta$, whose positive image U_1 is not a local group, admits characteristic data of length $O(\|G\|^3\bar{\theta}_G^5)$ for every fixed rank (Theorem 6.10). Item (c) rests on two-hole contexts around a long run of the neutral letter, which either keep every insertion of $\mathtt{b}$ shallow or force the target to be Parikh-saturated. Whether **LG** is necessary at the level of pseudovarieties for $f \geq 2$ remains Open Problem 7.1.

Relation to prior work. Yoshinaka bounded characteristic data of (k, ℓ)-substitutable CFGs with a thickness parameter [12], learned multidimensionally substitutable MCFGs from positive data [13, 14], and gave the finite-signature learning principle with finite witness cardinality [15]; the qualitative results and the Const/Link/Comp construction used here are his. The finite-semigroup structure theory and local-group representations are classical [32–34]. Our contribution is the encoded-length analysis: the noncommutative owner-wise shortening and the window–group and local-group bounds (Theorems 4.15 and 4.19). Other mildly context-sensitive formalisms are learned from positive data in [19], and MCFGs with queries or dual context sets in [16, 18, 20]; the MCFL substitution lemma of [21] is a membership condition, not a learning criterion.

Relation to the companion CFG manuscript. The companion manuscript [23] treats fan-out one. Table 1 separates its results from ours (numbers in the second column refer to the companion). No result of the companion is a premise here, and its expressive results (counter and Dyck obstructions, comparison with congruential grammars) are not used.

Scope of the claims. All bounds concern the total encoded length of characteristic samples for the specified set-driven operator, relative to the stated presentation; they bound neither an intrinsic grammar size nor a stage of Gold convergence, and their exponents are discussed at the end of Section 7. Lower bounds on data come only from distinguishing targets (Propositions 4.6 and 4.8); thick refinements (Proposition 4.9) limit the witness construction, not the data (Corollary 4.10). The typing is fixed before the target is seen; refining it enlarges the class but changes the reconstruction

Table 1: Results of the companion CFG manuscript [23] (fan-out one) and of the present paper.

Topic	Companion [23]	Here
Typed reconstruction, Gold learner, typed-thickness witness bound	Lemma 2.2; Sections 4–5; Lemma 6.2, Corollary 6.3	Tuple versions (Section 3, Theorem 4.4); qualitative results credited to Yoshinaka; not new
Window and window–group bounds	Proposition 3.5, Lemma 7.1, Theorem 7.2, Lemma 7.4, Corollary 7.5	Not new at $f = 1$; new for $f \geq 2$: owner-wise replacement across interleaved components (Proposition 4.12)
Local-group typings	Not treated (only $\mathbf{LI} \vee \mathbf{Grp}$; Lemma 7.6)	New at every f (Theorems B.4 and 4.19); the semigroup algebra is classical
Bracketed doubling family under s_U	Proposition 6.4, Corollary 6.6, Appendix C	Not new; reproved for self-containment (Proposition 4.20)
Rare-type lower bound	Proposition 6.4(iii)–(iv), for a single word	Every realized type and every f (Proposition 4.6)
Fan-out-one pseudovariety boundary	Sufficiency for $\mathbf{LI} \vee \mathbf{Grp}$ only (Remark 7.7)	New (Theorem 4.22); typing-level obstruction new (Corollary 4.25)
Rank one	Theorem 8.3 (linear CFG targets)	A variant at $f = 1$; new for $f \geq 2$ (Theorem 5.1)
Necessity of the factor $\|G\|\bar{\theta}_G$	—	New at $f = 1$ (Proposition 4.8)
Polynomial data outside $\mathbf{LG}$ at $f = 2$	—	New (Theorem 6.10)
Typing separations, power transfer	—	New (Section 6)

operator, so we neither infer nor optimize it, and the separations of Section 6 separate fixed biases rather than giving lower bounds for individual targets.

Organization. Section 2 fixes the setting, Section 3 defines the reconstruction operator and its qualitative guarantees, Sections 4 and 5 prove the quantitative bounds, and Section 6 compares the typed classes with multidimensional substitutability and develops the examples. Appendices A–F prove the technical shortening, normalization and reconstruction statements.

2 Preliminaries

Throughout, Σ is a finite alphabet, ε the empty word and $\Sigma^+ := \Sigma^* \setminus \{\varepsilon\}$. If $\Sigma = \emptyset$, the only languages are $\emptyset$ and $\{\varepsilon\}$ and are handled directly. For a finite $K \subseteq \Sigma^*$ put $\|K\|_+ := \sum_{w \in K}(|w| + 1)$, and for a tuple $\vec{u} = (u_1, \ldots, u_d)$ put $\|\vec{u}\|_1 := \sum_i |u_i|$. Every polynomial bound below states whether it concerns cardinality or $\|\cdot\|_+$. The table fixes symbols used across sections; all others (sliding-factor labellings, auxiliary morphisms, padding lengths) are defined where they occur.

Symbol	Meaning
G, G_{in}	a (reduced good) MCFG; an arbitrary standard input MCFG
B, C, X, Y	nonterminals; X, Y also denote typed nonterminals where specified
$\|G\|$, θ_G, t_G	encoding size (Definition 2.5), rule thickness, nonterminal thickness
s, $\widehat{s}$, S	positive typing, its fresh-identity completion, its positive image
$\preceq$	kernel refinement (Definition 2.3), also between monoid morphisms on Σ^*
$\mathbb{G}$, $\mathbb{H}$	finite groups ($\mathbb{G}$ the group component of a typing, $\mathbb{H}$ a Rees-matrix group or an auxiliary group)
g	a monoid morphism $\Sigma^* \to \mathbb{G}$ into a finite group
S, T, N	finite semigroups (S the positive image of the typing) and the finite monoid receiving an auxiliary morphism λ
T	a derivation tree
n	the index of a grammar family

2.1 Positive-Data Learning

Definition 2.1 (Text, TxtEx identification, conservativeness)**.** Fix a pause symbol $\perp \notin \Sigma$. A *text* for $L \subseteq \Sigma^*$ is an infinite sequence over $L \cup \{\perp\}$ in which every element of L occurs; the *content* of a finite prefix is the set of elements of Σ^* occurring in it. A *learner* is a computable map from finite prefixes to grammars. It *TxtEx-identifies* L [1] if on every text for L its outputs are, from some stage on, one fixed grammar H with $L(H) = L$; it TxtEx-identifies a class if it does so for each member. It is *conservative* if it changes its hypothesis only on a datum $w \in \Sigma^*$ not generated by the current hypothesis. We fix a hypothesis H_0 with $L(H_0) = \emptyset$.

Definition 2.2 (Set-driven operator and characteristic sample)**.** A *set-driven reconstruction operator* $\mathcal{B}$ maps each finite $K \subseteq \Sigma^*$ to a grammar $\mathcal{B}(K)$ depending only on the set K; it is *sample-consistent* if $K \subseteq L(\mathcal{B}(K))$ for all K. A finite $\mathcal{F} \subseteq L$ is a *characteristic sample for $\mathcal{B}$ and L* if $L(\mathcal{B}(K)) = L$ for every finite K with $\mathcal{F} \subseteq K \subseteq L$ (cf. [4]). This constrains generated languages only: $\mathcal{B}(K)$ need not stabilize syntactically as K grows.

2.2 Fixed Finite-Semigroup Typing

Definition 2.3 (Typing, fresh completion, refinement)**.** A *typing* is a surjective semigroup morphism $s : \Sigma^+ \twoheadrightarrow \mathsf{S}$ onto a finite semigroup, given by the multiplication table of S and the images of the letters, so that $s(w)$ is computable in time $O(|w|)$; S is its *positive image*. (A morphism into a larger finite semigroup is replaced by its corestriction to the image.) Let $\mathsf{S}^\sharp := \mathsf{S} \uplus \{\mathbf{1}_\sharp\}$ be obtained by adjoining a *fresh* identity, even if S already has one, and let $\widehat{s} : \Sigma^* \twoheadrightarrow \mathsf{S}^\sharp$ be the monoid morphism with $\widehat{s}(\varepsilon) = \mathbf{1}_\sharp$ and $\widehat{s}|_{\Sigma^+} = s$. For $\vec{w} \in (\Sigma^*)^d$ put $\widehat{s}^{(d)}(\vec{w}) := (\widehat{s}(w_1), \ldots, \widehat{s}(w_d))$; on $(\Sigma^+)^d$ this is $s^{(d)}(\vec{w})$. For typings s, s' write $s \preceq s'$ if $\ker s' \subseteq \ker s$ on Σ^+, and let (s, s') be the typing $w \mapsto (s(w), s'(w))$ onto its image in $\mathsf{S} \times \mathsf{T}$, where $\mathsf{T} := s'(\Sigma^+)$; thus $s, s' \preceq (s, s')$. Write s_{triv} for the one-element typing.

The fresh completion evaluates grammar templates (including empty terminal gaps) and types tuple yields in the refinement of Section 3.2; its new identity is not a positive-word type. Algebraic conditions such as **LG** concern S, not $\mathsf{S}^\sharp$. A finite-monoid morphism $q : \Sigma^* \to \mathsf{N}$ determines the positive typing $q|_{\Sigma^+}$: by Lemma D.1(i), it gives the same target class and set-driven reconstruction rules as the fresh completion of that typing. Lemma D.1(ii) additionally identifies their trimmed yield-type refinements for *good* grammars. This last identification is not asserted for arbitrary standard grammars with empty-component yields.

Definition 2.4 (Prefix–suffix typing)**.** For $k, \ell \geq 0$ put $m_0 := \max\{1, k+\ell\}$, and for $|w| \geq m_0$ let $\mathrm{pref}_k(w)$, $\mathrm{suf}_\ell(w)$ be the prefix and suffix of the indicated lengths (length-zero windows are ε). Define

$$\widehat{s}_{k,\ell}(w) = \begin{cases} w, & |w| < m_0, \\ \langle \mathrm{pref}_k(w), \mathrm{suf}_\ell(w) \rangle, & |w| \geq m_0, \end{cases}$$

with disjoint tags for short words and pairs, and put $M_{k,\ell} := \widehat{s}_{k,\ell}(\Sigma^*)$. For short x, y and pairs $\langle u, v \rangle, \langle u', v' \rangle$ define

$$x \cdot y := \widehat{s}_{k,\ell}(xy), \qquad x \cdot \langle u, v \rangle := \langle \mathrm{pref}_k(xu), v \rangle,$$
$$\langle u, v \rangle \cdot y := \langle u, \mathrm{suf}_\ell(vy) \rangle, \quad \langle u, v \rangle \cdot \langle u', v' \rangle := \langle u, v' \rangle.$$

The *boundary typing* is $s_{k,\ell} := \widehat{s}_{k,\ell}|_{\Sigma^+}$; a typing s with $s \preceq s_{k,\ell}$ is (k,ℓ)*-boundary-determined*, or briefly a *boundary typing*.

A case check on the lengths of $x, y \in \Sigma^*$ gives $\widehat{s}_{k,\ell}(xy) = \widehat{s}_{k,\ell}(x) \cdot \widehat{s}_{k,\ell}(y)$. Since $\widehat{s}_{k,\ell}$ is onto $M_{k,\ell}$, the product is associative and $\widehat{s}_{k,\ell}(\varepsilon)$ is an identity, so $\widehat{s}_{k,\ell}$ is a finite-monoid morphism. The short-word tag separates ε from $\widehat{s}_{k,\ell}(\Sigma^+)$, so $\widehat{s}_{k,\ell}$ is isomorphic to the completion $\widehat{s_{k,\ell}}$.

2.3 Multiple Context-Free Grammars

Definition 2.5 (Standard MCFG presentation and size)**.** An MCFG [27, 28] is $G = (V, \Sigma, P, S, \mathrm{fan})$ with finite V, $S \in V$, $\mathrm{fan}(S) = 1$ and $\mathrm{fan}(A) \geq 1$. A rule of rank $r \geq 0$ is $A \to \rho(B_1, \ldots, B_r)$ with template $\rho = (\alpha_1, \ldots, \alpha_{d_0})$, $d_0 = \mathrm{fan}(A)$, $d_j = \mathrm{fan}(B_j)$, where each α_i is a word over $\Gamma_\rho := \Sigma \uplus \Xi_1 \uplus \cdots \uplus \Xi_r$, $\Xi_j := \{\xi_j^1, \ldots, \xi_j^{d_j}\}$, and every variable occurs at most once in $\alpha_1 \cdots \alpha_{d_0}$ (*linearity*). For $\vec{u}_j \in (\Sigma^*)^{d_j}$, $\rho(\vec{u}_1, \ldots, \vec{u}_r)$ substitutes $u_{j,\kappa}$ for ξ_j^κ; for $r = 0$ the rule is a *constant rule*. The tuple languages $L_A(G) \subseteq (\Sigma^*)^{\mathrm{fan}(A)}$ form the least family closed under the rules, and $L(G) := \{w : (w) \in L_S(G)\}$. We write $A \Rightarrow_G^* \vec{u}$ for a *terminal derivation* (a finite derivation tree rooted at A, evaluated bottom-up); one from S is *successful*. G is an f-MCFG if $\mathrm{fan}(A) \leq f$ for all A, and has rule-rank cap $r_{\max}$ if all ranks are at most $r_{\max}$. Such a presentation, with no further restriction, is called *standard*. Its *size* is

$$\|G\| := |V| + \sum_{(A \to \rho(B_1, \ldots, B_r)) \in P} \Big(1 + r + \textstyle\sum_i |\alpha_i|\Big),$$

where $|\alpha_i|$ counts terminal and variable occurrences. All bounds use this symbol-level size; a bit-level encoding with binary nonterminal names changes it by at most a factor $O(\log \|G\|)$.

$\mathsf{MCFL}(f, r)$ is the class of languages of MCFGs of fan-out $\leq f$ and rule rank $\leq r$; f-MCFL $= \bigcup_{r<\infty} \mathsf{MCFL}(f, r)$, and the minimum rule rank of a language is taken over all its fan-out-$\leq f$ presentations. For nonterminals of equal fan-out d, $A \to B$ abbreviates $A \to (\xi_1^1, \ldots, \xi_1^d)(B)$, and $A \to \vec{c}$ a constant rule. Rule variables are always ξ_j^κ; terminals in examples are in `typewriter` font. Fan-out and rank are distinct resources. At fixed fan-out at least two the rule-rank hierarchy does not collapse in general, although adjacent levels can coincide; in particular $\mathsf{MCFL}(2, 2) = \mathsf{MCFL}(2, 3)$ [22, 30]. By contrast, fan-out-one CFGs can be binarized. The reconstruction below therefore assumes neither a binary normal form nor a known rank cap.

Definition 2.6 (Grammar properties)**.** A rule is *nondeleting* if every variable occurs exactly once in its template; a nondeleting rule is *nonpermuting* if for each j the variables $\xi_j^1, \ldots, \xi_j^{d_j}$ occur in this order in $\alpha_1 \cdots \alpha_{d_0}$, and *nonmerging* if moreover no α_i contains a factor $\xi_j^\kappa \xi_j^{\kappa+1}$.

Following [14, Section 2.3], a rule is λ-*free* if no α_i is empty. A rule, template or grammar is *good* if it is λ-free, nondeleting, nonpermuting and nonmerging. A nonterminal A is *reachable* if there is a chain $S = A_0, \dots, A_m = A$ in which each A_{i+1} is a child in a rule for A_i, and *productive* if $L_A(G) \neq \emptyset$; G is *reduced* if every nonterminal is both.

In a λ-free grammar every component of every derived tuple is nonempty (induction on derivations). λ-freeness concerns template components and is distinct from $\varepsilon \notin L(G)$.

2.4 Tuple Contexts and Substitutability

A *sentence context of arity* d is a word $E = v_0 \Box_1 v_1 \cdots v_{d-1} \Box_d v_d$ with each hole occurring once, $v_0, v_d \in \Sigma^*$ and $v_1, \dots, v_{d-1} \in \Sigma^+$; $\mathcal{E}_d$ is the set of these *admissible* contexts [14, Section 3.1]. $E[\vec{w}]$ fills $\Box_i$ by w_i, $|E|_{\mathrm{fix}} := \sum_{i=0}^{d} |v_i|$, and E *accepts* $\vec{w}$ in L if $E[\vec{w}] \in L$. The *d-tuple distribution* of $\vec{x} \in (\Sigma^*)^d$ is $\mathcal{D}_L^{(d)}(\vec{x}) := \{E \in \mathcal{E}_d : E[\vec{x}] \in L\}$; at fan-out one it is the two-sided distribution $\{(u, v) : uxv \in L\}$.

Definition 2.7 (Tuple occurrence)**.** For $w = a_1 \cdots a_n$ number the cut positions $⓪ a_1 ① \cdots a_n ⓝ$ and write $w[i : i'] := a_{i+1} \cdots a_{i'}$. An *arity-$d$ tuple occurrence* in w is a sequence $0 \le i_1 < i'_1 < i_2 < i'_2 < \cdots < i_d < i'_d \le n$; it determines $\vec{x} = (w[i_1 : i'_1], \dots, w[i_d : i'_d]) \in (\Sigma^+)^d$ and the context $E \in \mathcal{E}_d$ obtained by replacing these intervals by $\Box_1, \dots, \Box_d$, so that $E[\vec{x}] = w$. For finite K, $\mathrm{Obs}_d(K) := \{\vec{x} \in (\Sigma^+)^d : E[\vec{x}] \in K \text{ for some } E \in \mathcal{E}_d\}$ is the set of tuples arising from such occurrences, and for a fixed fan-out bound $f \ge 1$, put $\mathrm{Obs}(K) := \bigcup_{1 \le d \le f} \mathrm{Obs}_d(K)$.

Definition 2.8 ((f, s)-tuple substitutability and the target class)**.** Let $f \ge 1$ and let s be a typing. A language $L \subseteq \Sigma^*$ is (f, s)-*tuple-substitutable* if for every $1 \le d \le f$ and all $\vec{x}, \vec{y} \in (\Sigma^+)^d$,

$$s^{(d)}(\vec{x}) = s^{(d)}(\vec{y}) \ \text{ and } \ \mathcal{D}_L^{(d)}(\vec{x}) \cap \mathcal{D}_L^{(d)}(\vec{y}) \neq \emptyset \implies \mathcal{D}_L^{(d)}(\vec{x}) = \mathcal{D}_L^{(d)}(\vec{y}).$$

$\mathcal{C}_{f,s}^{\mathrm{mcf}}$ is the class of (f, s)-tuple-substitutable f-MCFLs; a member of $\mathcal{C}_{f,s}^{\mathrm{mcf}}$ for a boundary typing s is *boundary-typed*. For $\vec{x}, \vec{y} \in (\Sigma^+)^d$ write $\vec{x} \equiv_{L,s}^{d} \vec{y}$ (or $\vec{x} \equiv_{L,s} \vec{y}$) if $s^{(d)}(\vec{x}) = s^{(d)}(\vec{y})$ and $\mathcal{D}_L^{(d)}(\vec{x}) = \mathcal{D}_L^{(d)}(\vec{y})$; this is an equivalence relation.

Empty components are never compared, following [14, Section 3.1 and footnote 3] (see [8] for CFGs), and $\varepsilon \in L$ is handled at the start symbol; otherwise already $\mathtt{a}^+$ would fail, since $\mathtt{a}$ and ε share a context but not their distributions. The bias consists of both s and f:

(a) If $s \preceq s'$, equal s'-types are equal s-types, so $\mathcal{C}_{f,s}^{\mathrm{mcf}} \subseteq \mathcal{C}_{f,s'}^{\mathrm{mcf}}$; in particular $\mathcal{C}_{f,s_{\mathrm{triv}}}^{\mathrm{mcf}} \subseteq \mathcal{C}_{f,s}^{\mathrm{mcf}}$. This compares classes only: the operators $\mathcal{B}_{f,s}$ and $\mathcal{B}_{f,s'}$ of Section 3 differ in general (they coincide when $\ker s = \ker s'$), and no monotonicity of characteristic-data length in the typing is claimed.

(b) $\mathcal{C}_{f+1,s}^{\mathrm{mcf}} \cap f\text{-MCFL} \subseteq \mathcal{C}_{f,s}^{\mathrm{mcf}}$, but the class is not monotone in f: for trivial typing $\{w\#w^R\}$ is 1- but not 2-dimensionally substitutable [14, Section 3.1, Example 5].

(c) For every $\mathsf{Acc} \subseteq \mathsf{S}^\sharp$, $\widehat{s}^{-1}(\mathsf{Acc}) \in \mathcal{C}_{f,s}^{\mathrm{mcf}}$: it is regular, and componentwise equal types give equal total types in every context.

Proposition 2.9 (Intersection and abelian filtering)**.** *Fix f and s.*

(i) If L_1, L_2 are (f, s)-tuple-substitutable, so is $L_1 \cap L_2$.

(ii) If L is (f, s)-tuple-substitutable, A is an abelian group, $\varphi : \Sigma^* \to \mathsf{A}$ a monoid morphism, $\mathsf{H} \le \mathsf{A}$ and $\gamma \in \mathsf{A}$, then $L \cap \varphi^{-1}(\gamma + \mathsf{H})$ is (f, s)-tuple-substitutable.

Whenever such an intersection is an f-MCFL, it belongs to $\mathcal{C}^{\mathrm{mcf}}_{f,s}$; in particular so does every f-MCFL $\widehat{s}^{-1}(\mathsf{Acc}) \cap \varphi^{-1}(\gamma + \mathsf{H})$. Such factorizations are not supplied to the learner.

Proof. (i) A context accepting equal-type $\vec{x}, \vec{y}$ in $L_1 \cap L_2$ accepts both in each L_i, so their distributions agree in each L_i, hence in the intersection. (ii) If some context E accepts $\vec{x}, \vec{y}$ in $\varphi^{-1}(\gamma + \mathsf{H})$, then $\delta := \sum_i (\varphi(y_i) - \varphi(x_i)) \in \mathsf{H}$, and by commutativity $\varphi(E'[\vec{y}]) - \varphi(E'[\vec{x}]) = \delta$ for every $E' \in \mathcal{E}_d$, so coset membership transfers in both directions. Thus $\varphi^{-1}(\gamma + \mathsf{H})$ is (f, s_{triv})-, hence (f, s)-tuple-substitutable; apply (i). The last two claims follow from (i), (ii) and item (c) above. □

3 Reconstruction and Qualitative Learning

This section fixes the reconstruction operator analysed later and records its qualitative guarantees, none of which is claimed as new. For fixed f, rank cap and typing, our operator follows the distributional reconstruction pattern for substitutable languages over finite signatures in [15]. We use the MCFG signature of [15, Example 3] with tuple sorts refined by s-types (cf. [15, Section 8]). The related qualitative results are

Here	Related prior result	Treatment here
Prop. 3.10 (soundness)	[15, Lemma 2]	—
Prop. 3.11 (completeness)	[15, Lemma 3]	arbitrary same-type representatives
Thm. 3.13 (exact reconstruction)	[15, Lemmas 2–3]	as for Prop. 3.11
Cor. 3.16 (identification)	[15, Theorem 1]	unbounded rank over the class

The table records qualitative analogues rather than literal identities of reconstruction operators. The concrete grammar statements and their adaptations are verified directly in Appendices C and D; no embedding into the signature framework is used as a premise. For a fixed sample the uncapped operator is the finite cap $\|K\|_+$, so soundness needs no addition; but identifying all of $\mathcal{C}^{\mathrm{mcf}}_{f,s}$, whose members have unbounded minimum rank, uses a sample-dependent cap, outside the bounded-arity assumption of [15]. Finally, [15] bounds the *number* of characteristic words by the number of target rules and nonterminals; their *length*, which this paper bounds, is not addressed there.

3.1 Set-Driven Reconstruction

All interaction between templates and contexts rests on one observation.

Lemma 3.1 (Induced child contexts)**.** *Let $\rho = (\alpha_1, \ldots, \alpha_e)$ be a good template of rank r with child arities $d_1, \ldots, d_r$, let $E \in \mathcal{E}_e$, let $1 \le j \le r$, and fix $\vec{u}_i \in (\Sigma^+)^{d_i}$ for $i \ne j$. Let E_j be the word obtained from $E[(\alpha_1, \ldots, \alpha_e)]$ by replacing ξ^κ_j by $\square_\kappa$ and every other variable ξ^ι_i by $u_{i,\iota}$. Then $E_j \in \mathcal{E}_{d_j}$ and $E_j[\vec{x}] = E[\rho(\vec{u}_1, \ldots, \vec{u}_{j-1}, \vec{x}, \vec{u}_{j+1}, \ldots, \vec{u}_r)]$ for every $\vec{x} \in (\Sigma^*)^{d_j}$.*

Proof. The identity holds because substitutions commute. By linearity and nondeletion each $\square_\kappa$ occurs exactly once in E_j, and by nonpermutation in the order $\kappa = 1, \ldots, d_j$, since E places $\alpha_1, \ldots, \alpha_e$ from left to right. Consider ξ^κ_j and $\xi^{\kappa+1}_j$. If both lie in one α_i, nonmerging puts a terminal or a variable ξ^ι_i with $i \ne j$ between them, which becomes a nonempty word. Otherwise they lie in $\alpha_i, \alpha_{i'}$ with $i < i'$, and the material between them contains the separator $v_i \in \Sigma^+$ of E. So every internal separator of E_j is nonempty. □

Lemma 3.2 (Composition preserves equivalence)**.** *Let $L \subseteq \Sigma^*$, let ρ be a good template of rank r, and let $\vec{x}_j, \vec{u}_j \in (\Sigma^+)^{d_j}$ satisfy $\vec{u}_j \equiv_{L,s} \vec{x}_j$ for every j. Then $\rho(\vec{u}_1, \ldots, \vec{u}_r) \equiv_{L,s} \rho(\vec{x}_1, \ldots, \vec{x}_r)$.*

Definition 3.3 (Composition witness)**.** Let K be finite and $r \geq 1$. A *composition witness of rank r* in K is a triple $(\vec{z}, \rho, (\vec{u}_1, \ldots, \vec{u}_r))$ in which ρ is a good template of rank r with output arity $e \leq f$ and child arities $d_1, \ldots, d_r \leq f$, $\vec{z} \in \mathrm{Obs}_e(K)$, $\vec{u}_j \in (\Sigma^+)^{d_j}$, and $\vec{z} = \rho(\vec{u}_1, \ldots, \vec{u}_r)$.

Equivalently, a composition witness is a segmentation of the components of an observed $\vec{z}$ into terminal gaps and nonempty intervals labelled by variables, subject to the good-template conditions. If $E[\vec{z}] \in K$, Lemma 3.1 gives $E_j[\vec{u}_j] = E[\vec{z}] \in K$ with E_j admissible, so every $\vec{u}_j \in \mathrm{Obs}(K)$. The children occupy disjoint, nonempty parts of $\vec{z}$, so

$$r \leq \textstyle\sum_j \|\vec{u}_j\|_1 \leq \|\vec{z}\|_1 \leq \max_{w \in K} |w| \leq \|K\|_+. \tag{3.1}$$

Definition 3.4 (Capped and uncapped reconstruction operators)**.** Fix f and a typing s. For $r_{\max} \geq 1$ let $\mathcal{B}_{f,s}^{\leq r_{\max}}(K) = \widehat{G}_{\leq r_{\max}}(K)$ be the grammar with start symbol $\widehat{S}$, a nonterminal $[\vec{x}]$ of arity d for each $\vec{x} \in \mathrm{Obs}_d(K)$, $1 \leq d \leq f$, and the rules

(Start) $\widehat{S} \to [(w)]$ for $w \in K \cap \Sigma^+$, and $\widehat{S} \to \varepsilon$ if $\varepsilon \in K$;

(Const) $[\vec{x}] \to \vec{x}$ for every $\vec{x} \in \mathrm{Obs}(K)$;

(Comp) $[\vec{z}] \to \rho([\vec{u}_1], \ldots, [\vec{u}_r])$ for every composition witness of rank $1 \leq r \leq r_{\max}$;

(Link) $[\vec{x}] \to [\vec{y}]$ for $\vec{x}, \vec{y} \in \mathrm{Obs}_d(K)$ with $s^{(d)}(\vec{x}) = s^{(d)}(\vec{y})$ and $E[\vec{x}], E[\vec{y}] \in K$ for some $E \in \mathcal{E}_d$.

For $K \neq \emptyset$ put $\mathcal{B}_{f,s}(K) := \widehat{G}(K) := \widehat{G}_{\leq \|K\|_+}(K)$; $\mathcal{B}_{f,s}(\emptyset)$ has no rules. Fixed encoding orders make both operators deterministic and set-driven.

By (3.1), $\mathcal{B}_{f,s}(K)$ contains a Comp rule for every composition witness in K. Both operators are sample-consistent, via $\widehat{S} \to [(w)] \to (w)$ and $\widehat{S} \to \varepsilon$. All non-start templates are λ-free, so $\widehat{S} \Rightarrow^* \varepsilon$ only by the rule $\widehat{S} \to \varepsilon$. For trivial typing and $\varepsilon \notin K$, Comp/Const, Link and Start correspond to the rule Types I–III of [14, Section 4.1], and $\mathcal{B}_{f,s}$ corresponds to the operator of the unbounded-rank learner $\mathcal{A}(p, *)$ of [14, Corollary 2].

3.2 Yield-Typed Refinement and Witnesses

Definition 3.5 (Yield-type refinement)**.** Let $G = (V, \Sigma, P, S, \mathrm{fan})$ be a finite nondeleting MCFG and s a typing. For a template ρ and $\mathbf{t}_j \in (\mathsf{S}^\sharp)^{d_j}$, let $\mathrm{out}_\rho(\mathbf{t}_1, \ldots, \mathbf{t}_r) \in (\mathsf{S}^\sharp)^{d_0}$ have ith component the image of α_i under the monoid morphism $\Gamma_\rho^* \to \mathsf{S}^\sharp$ extending $a \mapsto \widehat{s}(a)$ and $\xi_j^\kappa \mapsto (\mathbf{t}_j)_\kappa$. The *full yield-type refinement* $G^{\widehat{s}}$ has a fresh start symbol $\widetilde{S}$, nonterminals $A_{\mathbf{t}}$ for $A \in V$ and $\mathbf{t} \in (\mathsf{S}^\sharp)^{\mathrm{fan}(A)}$, start rules $\widetilde{S} \to S_{(\eta)}$ for $\eta \in \mathsf{S}^\sharp$, and for every rule $A \to \rho(B_1, \ldots, B_r)$ (including $r = 0$) and all $\mathbf{t}_j \in (\mathsf{S}^\sharp)^{\mathrm{fan}(B_j)}$ the rule $A_{\mathrm{out}_\rho(\mathbf{t}_1, \ldots, \mathbf{t}_r)} \to \rho(B_{1,\mathbf{t}_1}, \ldots, B_{r,\mathbf{t}_r})$. The *trimmed refinement* $\widetilde{G}_0$ keeps exactly the nonterminals and rules occurring in some successful derivation from $\widetilde{S}$. Unless $G^{\widehat{s}}$ is named, *typed nonterminals* and *typed rules* are the non-start nonterminals and rules of $\widetilde{G}_0$, and $L_X := \{\vec{u} : X \Rightarrow^*_{\widetilde{G}_0} \vec{u}\}$.

Typed rules have the templates of their sources, so $\widetilde{G}_0$ is good whenever G is; each L_X is nonempty, and its tuples have nonempty components if G is λ-free.

Proposition 3.6 (Yield-type invariants)**.** *(i) If $A_{\mathbf{t}} \Rightarrow^* \vec{u}$ in $G^{\widehat{s}}$, then $\widehat{s}^{(\mathrm{fan}(A))}(\vec{u}) = \mathbf{t}$, and erasing indices gives a G-derivation of $\vec{u}$.*

(ii) Every terminal derivation $A \Rightarrow^*_G \vec{u}$ lifts uniquely to $A_{\widehat{s}^{(\mathrm{fan}(A))}(\vec{u})} \Rightarrow^*_{G^{\widehat{s}}} \vec{u}$; a successful derivation of w lifts uniquely through $\widetilde{S} \to S_{(\widehat{s}(w))}$.

(iii) $L(\widetilde{G}_0) = L(G^{\widehat{s}}) = L(G)$.

Lemma 3.7 (Canonical contexts for typed nonterminals)**.** *Let G be good. Every typed nonterminal X of arity d has a context $\chi(X) \in \mathcal{E}_d$ with $\chi(X)[\vec{u}] \in L(G)$ for all $\vec{u} \in L_X$; if $\widetilde{S} \to X$ is in $\widetilde{G}_0$, one may take $\chi(X) = \square_1$.*

Lemma 3.8 (Typed nonterminals are distributionally homogeneous)**.** *Let G be good of fan-out $\le f$ with $L := L(G) \in \mathcal{C}^{\mathrm{mcf}}_{f,s}$. For every typed nonterminal X of arity d, L_X lies in a single $\equiv^d_{L,s}$-class, denoted $\mathrm{Cl}_{L,s}(X)$. This semantic class notation is distinct from the syntactic reconstruction nonterminal $[\vec{x}]$ of Definition 3.4.*

The proofs of these three statements are in Appendix D. The homogeneity of Lemma 3.8 is weaker than that of congruential MCFGs [17], whose contexts may have empty separators.

Definition 3.9 (Canonical tuples, contexts and the witness set)**.** Let G be good. For every typed nonterminal X fix a *canonical tuple* $\omega(X) \in L_X$ and a *canonical context* $\chi(X)$ as in Lemma 3.7, with $\chi(X) = \square_1$ whenever $\widetilde{S} \to X$ is in $\widetilde{G}_0$. Let $\mathsf{W}(\widetilde{G}_0)$ consist of the words

$$\chi(X)[\omega(X)] \ \ (X \text{ typed}), \qquad \chi(X)[\rho(\omega(Y_1),\dots,\omega(Y_q))] \ \ (X \to \rho(Y_1,\dots,Y_q) \text{ typed},\ q \ge 0).$$

Since $\omega(X)$ and $\rho(\omega(Y_1),\dots,\omega(Y_q))$ belong to L_X, $\mathsf{W}(\widetilde{G}_0)$ is a finite subset of $L(G)$, nonempty if $L(G) \neq \emptyset$, and $|\mathsf{W}(\widetilde{G}_0)|$ is at most the number of typed nonterminals and rules. Canonical objects are properties of the target, not inputs of the operator: qualitatively any choice works, while Section 4 chooses short ones.

3.3 Qualitative Guarantees

Proposition 3.10 (Soundness of every hypothesis)**.** *Let L be (f,s)-tuple-substitutable and $K \subseteq L$ finite. For every $r_{\max} \ge 1$, if $[\vec{x}] \Rightarrow^* \vec{u}$ in $\widehat{G}_{\le r_{\max}}(K)$, then $\vec{u} \equiv_{L,s} \vec{x}$. Consequently $L(\mathcal{B}^{\le r_{\max}}_{f,s}(K)) \subseteq L$ and $L(\mathcal{B}_{f,s}(K)) \subseteq L$.*

Proposition 3.11 (Completeness from a witness assignment)**.** *Let G be a finite good MCFG of fan-out $\le f$, let r_G be the maximum rank of a typed rule of $\widetilde{G}_0$ ($r_G := 0$ if none), and assign to each typed nonterminal $X = A_{\mathbf{t}}$ a tuple $\iota(X) \in (\Sigma^+)^{\mathrm{fan}(A)}$ with $\widehat{s}^{(\mathrm{fan}(A))}(\iota(X)) = \mathbf{t}$ and a context $\chi(X) \in \mathcal{E}_{\mathrm{fan}(A)}$, with $\chi(X) = \square_1$ if $\widetilde{S} \to X$ is in $\widetilde{G}_0$. Suppose K contains $\chi(X)[\iota(X)]$ for each X and $\chi(X)[\rho(\iota(Y_1),\dots,\iota(Y_q))]$ for each typed rule $X \to \rho(Y_1,\dots,Y_q)$, $q \ge 0$. Then*

$$L(G) \subseteq L(\mathcal{B}^{\le r_{\max}}_{f,s}(K)) \quad (r_{\max} \ge \max\{1, r_G\}), \qquad L(G) \subseteq L(\mathcal{B}_{f,s}(K)).$$

Neither substitutability nor $\iota(X) \in L_X$ is assumed; $\iota = \omega$ is the case $\mathsf{W}(\widetilde{G}_0) \subseteq K$.

Exact reconstruction needs a good presentation of the target; the following normalization controls size, fan-out and rank, but not thickness.

Proposition 3.12 (Good normalization with rank preservation)**.** *Let G_{in} be a finite standard f-MCFG of rule rank at most $r_{\max}$. There is a finite good f-MCFG G_{good} of rule rank at most $r_{\max}$ with $L(G_{\mathrm{good}}) = L(G_{\mathrm{in}}) \setminus \{\varepsilon\}$ and, for fixed $f, r_{\max}$, $\|G_{\mathrm{good}}\| = O_{f,r_{\max}}(\|G_{\mathrm{in}}\|)$. If $L(G_{\mathrm{in}}) \cap \Sigma^+ \neq \emptyset$, G_{good} is reduced.*

Theorem 3.13 (Exact reconstruction for a fixed typing)**.** *Fix $f \ge 1$ and a typing s, and let $L \in \mathcal{C}^{\mathrm{mcf}}_{f,s}$. If $L \cap \Sigma^+ = \emptyset$ put $\mathcal{F}_L := L$ and $r_G := 0$. Otherwise let G be a reduced good fan-out-$\le f$ presentation*

of $L \cap \Sigma^+$ (Proposition 3.12), let r_G be as in Proposition 3.11, and put $\mathcal{F}_L := \mathsf{W}(\widetilde{G}_0) \cup (\{\varepsilon\} \cap L)$. Then $\mathcal{F}_L \subseteq L$ is finite, and every finite K with $\mathcal{F}_L \subseteq K \subseteq L$ satisfies

$$L(\mathcal{B}_{f,s}(K)) = L, \qquad L(\mathcal{B}_{f,s}^{\leq r_{\max}}(K)) = L \quad \textit{for every } r_{\max} \geq \max\{1, r_G\}.$$

If $L \in \mathsf{MCFL}(f, r)$, G can be chosen of rank $\leq r$, so $r_{\max} = \max\{1, r\}$ suffices (also when $r = 0$).

Theorem 3.14 (Polynomial construction under a rule-rank cap)**.** *Fix $f, r_{\max} \geq 1$ and s. The encoding of $\widehat{G}_{\leq r_{\max}}(K)$ is computable in time $(1 + \|K\|_+)^{O(f r_{\max})}$; its fan-out is at most f and its rank at most $r_{\max}$.*

Definition 3.15 (Conservative sequential learner)**.** The learner $\mathcal{A}_{f,s}$ (respectively $\mathcal{A}_{f,s}^{\leq r_{\max}}$) starts with H_0, keeps its hypothesis on $\perp$ and on every datum the hypothesis generates, and otherwise replaces it by $\mathcal{B}_{f,s}(K)$ (respectively $\mathcal{B}_{f,s}^{\leq r_{\max}}(K)$), where K is the content read so far. This is the policy of [14, Algorithm 1].

Corollary 3.16 (Gold identification)**.** *Fix $f \geq 1$ and s.*

(i) *$\mathcal{A}_{f,s}$ TxtEx-identifies $\mathcal{C}_{f,s}^{\mathrm{mcf}}$.*

(ii) *For fixed $r_{\max}$, $\mathcal{A}_{f,s}^{\leq r_{\max}}$ TxtEx-identifies $\mathcal{C}_{f,s}^{\mathrm{mcf}} \cap \mathsf{MCFL}(f, r_{\max})$, and processes each datum in time polynomial in $\|K\|_+$ for the content K read so far.*

Proofs are in Appendix C (Proposition 3.12: Appendix E). None of these statements bounds the length of characteristic data or the stage of convergence.

4 Characteristic-Data Bounds

Throughout this section f, $r_{\max} \geq 1$, Σ and the typing $s : \Sigma^+ \twoheadrightarrow \mathsf{S}$ are fixed; $G = (V, \Sigma, P, S, \mathrm{fan})$ is a reduced good MCFG of fan-out $\leq f$ and rule rank $\leq r_{\max}$, and $\widetilde{G}_0$ is its trimmed $\widehat{s}$-refinement. Under a rank cap, Theorem 3.13 already supplies $O(\|G\|)$ characteristic words; the task is to bound their total length *in terms of G*. All bounds follow one scheme.

(1) *Witness length* (Lemma 4.2). If, for an auxiliary finite-monoid morphism λ refining s, every tuple of every nonterminal can be exchanged for a tuple of the same nonterminal and the same λ-vector of total length at most β, then $\mathsf{W}(\widetilde{G}_0)$ is characteristic and has total length $O(\|G\|(\bar{\theta}_G + \beta))$.

(2) *Minimal counterexamples* (Section 4.3). In a shortest derivation with a prescribed root λ-vector, a large antichain of heavy subtrees would admit a simultaneous replacement preserving that vector; so antichains are small, and Lemma 4.11 turns this into $\beta = O(\|G\|\bar{\theta}_G)$ or $O(\|G\|^2\bar{\theta}_G)$.

(3) *Regimes.* Only λ and the replacement step change: window typings keep marked boundary letters; window–group typings in addition preserve ordered group products (Proposition 4.12); local-group typings reduce to these through sliding-factor coordinates (Section 4.5).

The full yield-type refinement is not used as the primary thickness measure, since it may be exponentially thicker than G (Proposition 4.9).

4.1 Thickness Scales and Witness Length

For a rule R of G let

$$\theta_G(R) := \min\{\, |w| : w \in L(G) \text{ has a successful derivation using } R \,\},$$

the length of the derived *sentence*, not of the derivation. Put $\theta_G := \max_{R \in P} \theta_G(R)$, $\bar{\theta}_G := \max\{1, \theta_G\}$ and $t_G := \max_{A \in V} \min\{\|\vec{w}\|_1 : \vec{w} \in L_A(G)\}$ (rule and nonterminal thickness; Yoshinaka writes τ_G for θ_G [14, Section 4.3]).

If a target L contains ε, every statement below is applied to $L \cap \Sigma^+$ and the datum ε is added, as in Theorem 3.13; this adds one to the encoded length.

Lemma 4.1 (Thickness lemma)**.** *Let G be reduced and good.*

(i) *Every rule writes at most θ_G terminal symbols explicitly.*

(ii) *$t_G \le \theta_G$: every $A \in V$ has some $\vec{u} \in L_A(G)$ with $\|\vec{u}\|_1 \le \theta_G$.*

(iii) *$\theta_G \le 2\|G\| \max\{1, t_G\}$.*

(iv) *Every $A \in V$ of fan-out d has a context $E_A \in \mathcal{E}_d$ with $|E_A|_{\mathrm{fix}} \le \theta_G$ that results from a successful derivation by deleting one occurrence of an A-subtree; $E_S = \square_1$, and $E_A[\vec{u}] \in L(G)$ for all $\vec{u} \in L_A(G)$. For every typed nonterminal $X = A_{\mathbf{t}}$, the choice $\chi(X) := E_A$ meets Definition 3.9 and is* derivational *for X: it results from a successful derivation of $\widetilde{G}_0$ by deleting an occurrence of an X-subtree.*

Proof. For a rule R with left-hand side A fix a successful derivation D_R of length $\theta_G(R)$ that uses R; it exists because G is reduced. By nondeletion every terminal written by R, and every component of the tuple produced at that occurrence of A, survives exactly once in the sentence. This gives (i), and (ii) because every productive A has a rule.

(iv) For $A = S$ take $\square_1$. Otherwise delete the A-subtree of D_R and put ordered holes in place of its components; iterating Lemma 3.1 along the path to the root (siblings have nonempty yields) shows $E_A \in \mathcal{E}_d$, and its fixed letters belong to the sentence of D_R. Inserting any A-derivation gives a successful derivation, so E_A accepts $L_A(G) \supseteq L_X$ (Proposition 3.6(i)). A typed nonterminal with a start link has the form $S_{\mathbf{t}}$, and $E_S = \square_1$. For derivationality, insert the erasure of any $\widetilde{G}_0$-derivation D_X from X into the hole; the unique lift of the resulting successful derivation (Proposition 3.6(ii)) labels every node by its nonterminal and the $\widehat{s}$-vector of its yield, as D_X does, so it contains D_X at the hole, and deleting that X-subtree leaves E_A.

(iii) Follow a shortest dependency path from S to A; it repeats no nonterminal, so its rules are distinct and differ from R. Complete every sibling on the path and every child of R by a shortest terminal tuple (length $\le t_G$) and apply R. The distinct rules used write at most $\|G\|$ terminals and have at most $\|G\|$ child occurrences, so the sentence has length at most $\|G\| + \|G\| t_G \le 2\|G\| \max\{1, t_G\}$. □

Thus θ_G and t_G are polynomially equivalent on reduced good presentations; Yoshinaka's efficiency criterion measures data in target size and rule thickness [14, Definition 1]; see also [10].

For a finite-monoid morphism $\lambda : \Sigma^* \to \mathsf{N}$ we write $s \preceq \lambda$ if $\ker(\lambda|_{\Sigma^+}) \subseteq \ker s$; since $\widehat{s}$ separates ε, this is the relation of Definition 2.3 whenever λ also separates ε from Σ^+. For $\beta \ge 0$ consider

($\mathrm{R}^{\lambda}_{\beta}$) for every $B \in V$ and $\vec{u} \in L_B(G)$ there is $\vec{u}' \in L_B(G)$ with $\lambda^{(\mathrm{fan}(B))}(\vec{u}') = \lambda^{(\mathrm{fan}(B))}(\vec{u})$ and $\|\vec{u}'\|_1 \le \beta$.

Put

$$C_s(G) := |V||\mathsf{S}^\sharp|^f + |P||\mathsf{S}^\sharp|^{fr_{\max}} \;\le\; |\mathsf{S}^\sharp|^{fr_{\max}}\,\|G\|,$$

using $|V|+|P| \le \|G\|$; it bounds the numbers of typed nonterminals and typed rules together, since a typed rule is determined by its source rule and the type vectors of its at most $r_{\max}$ children.

Lemma 4.2 (Witness length)**.** *Let $\lambda : \Sigma^* \to \mathsf{N}$ be a finite-monoid morphism with $s \preceq \lambda$.*

(i) (Type-preserving exchange.) *If $X = B_{\mathbf{t}}$ is a typed nonterminal, $\vec{u} \in L_X$, $\vec{u}' \in L_B(G)$ and $\lambda^{(\mathrm{fan}(B))}(\vec{u}') = \lambda^{(\mathrm{fan}(B))}(\vec{u})$, then $\vec{u}' \in L_X$. Hence under $(\mathrm{R}^\lambda_\beta)$ there are canonical tuples with $\|\omega(X)\|_1 \le \beta$ for every X.*

(ii) (Length.) *If $\|\omega(X)\|_1 \le b$ for every X and $\chi(X) := E_B$ as in Lemma 4.1(iv), every word of $\mathsf{W}(\widetilde{G}_0)$ has length at most $2\bar{\theta}_G + r_{\max}b$, and*

$$|\mathsf{W}(\widetilde{G}_0)| \le C_s(G), \qquad \|\mathsf{W}(\widetilde{G}_0)\|_+ \le C_s(G)\,(1 + 2\bar{\theta}_G + r_{\max}b).$$

(iii) (Refined thickness.) *In (ii), each rule word $\chi(X)[\rho(\omega(Y_1),\dots,\omega(Y_q))]$ is the yield of a successful derivation of $\widetilde{G}_0$ using the typed rule $X \to \rho(Y_1,\dots,Y_q)$; hence $\theta_{\widetilde{G}_0} \le 2\bar{\theta}_G + r_{\max}b$.*

(iv) (Characteristic sample.) *If $L(G) = L \cap \Sigma^+$ for some $L \in \mathcal{C}^{\mathrm{mcf}}_{f,s}$, then $\mathsf{W}(\widetilde{G}_0) \cup (\{\varepsilon\} \cap L)$ is a characteristic sample for $\mathcal{B}^{\le r_{\max}}_{f,s}$ and for $\mathcal{B}_{f,s}$.*

Items (i)–(iii) do not use substitutability.

Proof. (i) All components are nonempty because G is λ-free, so $s \preceq \lambda$ and Proposition 3.6(i) give $\widehat{s}^{(\mathrm{fan}(B))}(\vec{u}') = \widehat{s}^{(\mathrm{fan}(B))}(\vec{u}) = \mathbf{t}$. By Proposition 3.6(ii) a G-derivation of $\vec{u}'$ lifts to a $G^{\widehat{s}}$-derivation from X. Inserted at an occurrence of X in a successful derivation of $\widetilde{G}_0$ it gives a successful derivation, because typed rules depend only on typed symbols; so all its rules are retained and $\vec{u}' \in L_X$. For the second claim apply $(\mathrm{R}^\lambda_\beta)$ to some $\vec{u} \in L_X \neq \emptyset$.

(ii) A witness word is $\chi(X)[\omega(X)]$ or a rule word. The context has at most $\bar{\theta}_G$ fixed letters, a typed rule has the template of its source and so writes at most $\bar{\theta}_G$ letters (Lemma 4.1(i)), and there are at most $r_{\max}$ child tuples. There is one word per typed nonterminal and per typed rule.

(iii) Apply the typed rule to $\widetilde{G}_0$-derivations of the $\omega(Y_j) \in L_{Y_j}$, erase the labels, and insert the result into the successful G-derivation from which E_B was obtained. As in the proof of Lemma 4.1(iv), the unique lift contains the typed derivation at the hole, and every rule of a successful derivation of $G^{\widehat{s}}$ is retained. A start link $\widetilde{S} \to S_{\mathbf{t}}$ is used by the successful derivation of $\omega(S_{\mathbf{t}})$, of length at most b.

(iv) Proposition 3.11 with $\iota = \omega$ ($r_G \le r_{\max}$, and $\chi(S_{\mathbf{t}}) = E_S = \Box_1$) and Proposition 3.10 give $L(\mathcal{B}(K)) \cap \Sigma^+ = L \cap \Sigma^+$ for every finite K with $\mathsf{W}(\widetilde{G}_0) \subseteq K \subseteq L$; the Start rule handles ε. □

4.2 Typed Thickness and Characteristic Data

For a reduced good grammar G, let $\widetilde{G}_0$ be its trimmed full yield-type refinement (Definition 3.5). Define

$$t_{\mathrm{typ}}(G,s) := \max_X \min_{\vec{u} \in L_X} \|\vec{u}\|_1, \qquad \theta_{\mathrm{typ}}(G,s) := \max\{1, \theta_{\widetilde{G}_0}\},$$

where X ranges over the non-start typed nonterminals, and let $N_{\mathrm{typ}}(G,s)$ denote their number plus the number of non-start typed rules. In particular $N_{\mathrm{typ}} \le C_s(G) \le |\mathsf{S}^\sharp|^{fr_{\max}}\|G\|$. Every rule of G has a retained typed copy, and erasing typed labels from a successful derivation recovers a G-derivation; therefore $\theta_G \le \theta_{\widetilde{G}_0} \le \theta_{\mathrm{typ}}$.

Lemma 4.3 (Comparison of the two typed thickness measures)**.** *For every reduced good G,*

$$t_{\mathrm{typ}} \le \theta_{\mathrm{typ}} \le 2\bar{\theta}_G + r_{\max} t_{\mathrm{typ}}.$$

Proof. Apply Lemma 4.1(ii) to the reduced good $\widetilde{G}_0$ for the first inequality. For the second, choose each canonical tuple $\omega(X)$ to have the shortest total length, at most t_{typ}, and use Lemma 4.2(iii). The start links are covered by that lemma as well. □

Theorem 4.4 (Characteristic data from typed thickness)**.** *Let $L \in \mathcal{C}^{\mathrm{mcf}}_{f,s}$ and G be a reduced good fan-out-$\le f$, rank-$\le r_{\max}$ grammar with $L(G) = L \cap \Sigma^+$. Take shortest canonical tuples and the derivational contexts E_B of Lemma 4.1(iv). Then $K = \mathsf{W}(\widetilde{G}_0) \cup (\{\varepsilon\} \cap L)$ is a characteristic sample for both $\mathcal{B}^{\le r_{\max}}_{f,s}$ and $\mathcal{B}_{f,s}$, with*

$$|\mathsf{W}(\widetilde{G}_0)| \le N_{\mathrm{typ}}, \qquad \|\mathsf{W}(\widetilde{G}_0)\|_+ \le N_{\mathrm{typ}}(1 + 2\bar{\theta}_G + r_{\max} t_{\mathrm{typ}}).$$

For the full sample K, the length bound has the additional term $\mathbf{1}_{\{\varepsilon \in L\}}$. The numerical length estimate holds for any typing; substitutability is used only for the characteristic-sample claim.

Proof. The chosen shortest tuples satisfy $\|\omega(X)\|_1 \le t_{\mathrm{typ}}$. Apply Lemma 4.2(ii),(iv), counting non-start typed nonterminals and rules instead of the upper bound $C_s(G)$. □

For trivial typing, every nonterminal of the λ-free G has one realized type vector, so $t_{\mathrm{typ}} = t_G$ and $N_{\mathrm{typ}} = |V| + |P| \le \|G\|$. The theorem recovers the $O(\|G\|\bar{\theta}_G)$ untyped scale, although Yoshinaka's original sample has the sharper bounds $|K_G| \le |P|$ and $\|K_G\| \le |P|\theta_G$ [14, Lemma 7 and Theorem 1]. A nontrivial typing may instead make t_{typ} exponential.

Presentation-independent lower bounds from rare types. For $L \subseteq \Sigma^*$ and $m \in \mathsf{S}$ put

$$\mu_m(L) := \min\{|w| : w \in L \cap \Sigma^+,\ s(w) = m\}, \qquad (\min \emptyset := \infty).$$

For a nonempty set of realized types $M_L := s(L \cap \Sigma^+)$, put $\mu^\star(L, s) := \max_{m \in M_L} \mu_m(L)$; set $\mu^\star(L, s) = 0$ when $M_L = \emptyset$. A typed start nonterminal $S_{(m)}$ exists for each $m \in M_L$, and $L_{S_{(m)}} = \{(w) : w \in L \cap \Sigma^+,\ s(w) = m\}$ (Proposition 3.6). Thus, whenever $L \cap \Sigma^+ \neq \emptyset$,

$$\mu^\star(L, s) \le t_{\mathrm{typ}}(G, s) \qquad \text{for every reduced good presentation } G. \tag{4.1}$$

Lemma 4.5 (Deleting an entire yield type)**.** *If $L \in \mathcal{C}^{\mathrm{mcf}}_{f,s}$ and $m \in \mathsf{S}$, then $L^{\setminus m} := L \setminus s^{-1}(m)$ also belongs to $\mathcal{C}^{\mathrm{mcf}}_{f,s}$. Moreover, $L \setminus \{\varepsilon\} \in \mathcal{C}^{\mathrm{mcf}}_{f,s}$.*

Proof. The regular language $\widehat{s}^{-1}(\mathsf{S}^\sharp \setminus \{m\})$ is (f, s)-tuple-substitutable, since equal componentwise s-types have equal total types in every admissible sentence context (item (c) after Definition 2.8). Intersections preserve tuple substitutability (Proposition 2.9(i)), and f-MCFLs are closed under regular intersection [27]. For the last assertion, intersect L with Σ^+. □

Proposition 4.6 (Rare types force positive data)**.** *Let $L \in \mathcal{C}^{\mathrm{mcf}}_{f,s}$, and let $m \in M_L$. Every characteristic sample of L for $\mathcal{B}^{\le r}_{f,s}$ ($r \ge 1$) or for $\mathcal{B}_{f,s}$ contains a word of type m. The same is true for any set-driven operator having characteristic samples for both L and $L^{\setminus m}$, in particular for an operator having characteristic samples for the entire class $\mathcal{C}^{\mathrm{mcf}}_{f,s}$. If $\varepsilon \in L$, its sample for either canonical operator must also contain ε; the same holds for a set-driven operator admitting a characteristic sample for $L \setminus \{\varepsilon\}$ as well. Consequently, for the canonical operators and for class-wide set-driven operators, every characteristic sample F satisfies*

$$\|F\|_+ \ge \sum_{m \in M_L} (\mu_m(L) + 1) + \mathbf{1}_{\{\varepsilon \in L\}}.$$

Proof. If a sample $F \subseteq L$ avoids type m, then $F \subseteq L^{\setminus m} \in \mathcal{C}^{\mathrm{mcf}}_{f,s}$ by Lemma 4.5. Soundness (Proposition 3.10) for the latter target gives $L(\mathcal{B}(F)) \subseteq L^{\setminus m} \subsetneq L$; the same applies to the bounded-rank reconstruction. For a general set-driven operator, let F' be characteristic for $L^{\setminus m}$. Then $F \cup F'$ lies inside both L and $L^{\setminus m}$, and contains the respective characteristic samples. The required outputs would be both distinct languages, a contradiction. The same argument, using $L \setminus \{\varepsilon\}$ from Lemma 4.5, forces the datum ε when it belongs to L, provided the operator has a characteristic sample for that language; the canonical operators satisfy this by Theorem 3.13. Distinct positive types demand distinct sample words, and the minimum cost of each such word is $\mu_m(L) + 1$. □

Corollary 4.7 (Characteristic-data sandwich)**.** *Suppose $L \in \mathcal{C}^{\mathrm{mcf}}_{f,s}$, $L \cap \Sigma^+ \neq \emptyset$, and G is a reduced good fan-out-$\leq f$, rank-$\leq r_{\max}$ presentation of $L \cap \Sigma^+$. Write $\mathrm{cs}^{\leq r_{\max}}_{f,s}(L)$ for the minimum total encoded length of a characteristic sample for $\mathcal{B}^{\leq r_{\max}}_{f,s}$. Then*

$$\sum_{m \in M_L} (\mu_m(L) + 1) + \mathbf{1}_{\{\varepsilon \in L\}} \;\leq\; \mathrm{cs}^{\leq r_{\max}}_{f,s}(L) \;\leq\; N_{\mathrm{typ}}(1 + 2\bar{\theta}_G + r_{\max} t_{\mathrm{typ}}(G, s)) + \mathbf{1}_{\{\varepsilon \in L\}}.$$

The lower bound is independent of the presentation and applies to every set-driven operator with characteristic samples throughout $\mathcal{C}^{\mathrm{mcf}}_{f,s}$.

Proof. Proposition 4.6 supplies the lower bound; Theorem 4.4 supplies the upper bound. □

The upper side of Corollary 4.7 grows with $\|G\|$ and the lower side does not. At fan-out one the product $\|G\|\bar{\theta}_G$ is nevertheless necessary, under every typing and for asymptotically smallest presentations.

Proposition 4.8 (The linear factor is necessary at fan-out one)**.** *Let $\mathtt{a}, \mathtt{c}, \mathtt{d} \in \Sigma$ be distinct and let s be any typing. For integers $q \geq k \geq 1$ put $M := 2^k$, $W := M + 2q + 2$ and*

$$w_i := \mathtt{a}^i \mathtt{d} \mathtt{a}^{q-i} \mathtt{c}^M \mathtt{a}^{q-i} \mathtt{d} \mathtt{a}^i \quad (0 \leq i \leq q), \qquad L_{q,k} := \{w_0, \ldots, w_q\}.$$

(i) *Every subset of $L_{q,k}$ belongs to $\mathcal{C}^{\mathrm{mcf}}_{1,s_{\mathrm{triv}}} \subseteq \mathcal{C}^{\mathrm{mcf}}_{1,s}$.*

(ii) *For $\mathcal{B}_{1,s}$, for $\mathcal{B}^{\leq r}_{1,s}$ ($r \geq 1$), and for every set-driven operator with characteristic samples throughout $\mathcal{C}^{\mathrm{mcf}}_{1,s}$, the only characteristic sample of $L_{q,k}$ is $L_{q,k}$ itself.*

(iii) *$L_{q,k}$ has a reduced good fan-out-one presentation $G_{q,k}$ of rule rank two with $\|G_{q,k}\| = 17q + 6k + 13 \leq 23(q+1)$ and $\bar{\theta}_{G_{q,k}} = t_{\mathrm{typ}}(G_{q,k}, s) = W$. Every reduced good fan-out-one presentation G of $L_{q,k}$ has $\bar{\theta}_G = W$ and $\|G\| \geq q + 1$.*

Consequently every characteristic sample F in (ii) *satisfies*

$$\|F\|_+ = (q+1)(W+1) \;\geq\; \tfrac{1}{23} \|G_{q,k}\| \, \bar{\theta}_{G_{q,k}}.$$

Proof. (i) Let $J \subseteq L_{q,k}$ and $x, y \in \Sigma^+$ with $uxv, uyv \in J$. For trivial typing and $d = f = 1$ we must show $\mathcal{D}^{(1)}_J(x) = \mathcal{D}^{(1)}_J(y)$. All words of $L_{q,k}$ have length W, so $|x| = |y|$. The two occurrences of $\mathtt{d}$ in w_i are at positions $i + 1 \in [1, q+1]$ and $W - i \in [W - q, W]$; these ranges are disjoint since $W - q = M + q + 2$, so the position of a single occurrence of $\mathtt{d}$ determines i. If u or v contains $\mathtt{d}$, that occurrence has the same position in uxv and in uyv, hence $uxv = uyv$ and $x = y$. Otherwise both occurrences of $\mathtt{d}$ lie in x; their distance $W - 2i - 1$ determines i, and the offset of the first $\mathtt{d}$ in x determines the position of x. So x has exactly one occurrence in the words of $L_{q,k}$, and

$\mathcal{D}_J^{(1)}(x) = \{(u, v)\}$; likewise $\mathcal{D}_J^{(1)}(y) = \{(u, v)\}$. Finite languages are context-free, and the inclusion is item (a) after Definition 2.8.

(ii) Let F be a characteristic sample with $w_i \notin F$. Then $F \subseteq L' := L_{q,k} \setminus \{w_i\}$, and $L' \in \mathcal{C}_{1,s}^{\mathrm{mcf}}$ by (i). Soundness (Proposition 3.10) for the target L' gives $L(\mathcal{B}(F)) \subseteq L' \subsetneq L_{q,k}$ for both canonical operators. For a class-wide set-driven operator, a characteristic sample F' of L' gives a finite $F \cup F' \subseteq L'$ on which the output must equal both $L_{q,k}$ and L', as in the proof of Proposition 4.6. Hence $F = L_{q,k}$. Conversely, $K = L_{q,k}$ is the only finite K with $L_{q,k} \subseteq K \subseteq L_{q,k}$, and the canonical operators return $L_{q,k}$ on it by sample consistency and soundness.

(iii) Take nonterminals $A_0, \dots, A_q$, $H_0, \dots, H_q$, $C_0, \dots, C_k$, start symbol A_q and the rules

$$\begin{array}{lll} A_j \to \mathtt{d}H_j\mathtt{d}\ (0 \le j \le q), & A_j \to \mathtt{a}A_{j-1}\mathtt{a}\ (1 \le j \le q), & H_0 \to C_k, \\ H_j \to \mathtt{a}H_{j-1}\mathtt{a}\ (1 \le j \le q), & C_0 \to \mathtt{c}, & C_t \to C_{t-1}C_{t-1}\ (1 \le t \le k). \end{array}$$

By induction, $L(C_t) = \{\mathtt{c}^{2^t}\}$, $L(H_j) = \{\mathtt{a}^j\mathtt{c}^M\mathtt{a}^j\}$ and $L(A_j) = \{\mathtt{a}^i\mathtt{d}\mathtt{a}^{j-i}\mathtt{c}^M\mathtt{a}^{j-i}\mathtt{d}\mathtt{a}^i : 0 \le i \le j\}$, so $L(G_{q,k}) = L_{q,k}$; the grammar is good and reduced. It has $2q + k + 3$ nonterminals and, in the order listed, $q+1, q, 1, q, 1, k$ rules of sizes $5, 5, 3, 5, 2, 5$ (Definition 2.5), which gives $\|G_{q,k}\| = 17q+6k+13$; $k \le q$ gives the bound $23(q+1)$. In a reduced presentation every rule occurs in a successful derivation, and all words of $L_{q,k}$ have length W, so $\bar{\theta}_G = W$. In $G_{q,k}$ all yields of a nonterminal, and hence of each typed nonterminal, have one length, at most W; the typed start nonterminals have yields of length W, so $t_{\mathrm{typ}}(G_{q,k}, s) = W$. Finally let G be any reduced good fan-out-one presentation of $L_{q,k}$. For trivial typing, Theorem 3.13 makes $\mathsf{W}(\widetilde{G}_0)$ characteristic for $\mathcal{B}_{1,s_{\mathrm{triv}}}$, and it has at most $|V| + |P| \le \|G\|$ words (Definition 3.9 and the remark after Theorem 4.4); by (ii) for s_{triv}, it has $q + 1$ words.

The final estimate follows from (ii), (iii) and $|w_i| = W$. □

With $q = k = n$ one has $\|G_{n,n}\| = 23n + 13$ and $\bar{\theta}_{G_{n,n}} = 2^n + 2n + 2$; with k fixed, $\bar{\theta}_{G_{q,k}} = \Theta(\|G_{q,k}\|)$. By (iii) the presentations are within a constant factor of the smallest reduced good ones, so the bound is not obtained by padding G. With trivial typing the proposition shows that Yoshinaka's bound $\|K_G\| \le |P|\theta_G$ is attained up to a constant factor. Since $t_{\mathrm{typ}} = \bar{\theta}_G$ in this family, it says nothing about the further factor $\|G\|$ in Theorems 4.15 and 4.19. The family is specific to fan-out one: in $L_{2,1}$ the pairs $(\mathtt{d}, \mathtt{ac})$ and $(\mathtt{da}, \mathtt{c})$ share the context $\square_1\mathtt{a}\square_2\mathtt{caad}$, but only the second is accepted by $\square_1\mathtt{ac}\square_2\mathtt{aad}$, so $L_{2,1}$ is not $(2, s_{\mathrm{triv}})$-tuple-substitutable.

Proposition 4.9 (Exponential blow-up of full yield-type refinement)**.** *Let $\Sigma = \{\mathtt{a}, \mathtt{c}\}$ and let $s_{\mathtt{c}}$ be the typing onto $\{1, \zeta\}$ ($\zeta^2 = \zeta$) with $s_{\mathtt{c}}(\mathtt{a}) = 1$ and $s_{\mathtt{c}}(\mathtt{c}) = \zeta$. For $n \ge 1$ the fan-out-one, rank-two grammar*

$$S \to SS \mid \mathtt{a} \mid \mathtt{c} \mid A_0 \mid \cdots \mid A_n, \qquad A_0 \to \mathtt{a}, \qquad A_i \to A_{i-1}A_{i-1} \mid \mathtt{c} \quad (1 \le i \le n)$$

is reduced and good, generates Σ^+, and has $\|G_n\| = O(n)$ and $\theta_{G_n} \le 2$, but $t_{\mathrm{typ}}(G_n, s_{\mathtt{c}}) \ge 2^n$ and $\theta_{\widetilde{G}_{n,0}} \ge 2^n$.

Proof. All templates are nonempty and use each variable once in order, so G_n is good; it is reduced and generates Σ^+ via $S \to SS \mid \mathtt{a} \mid \mathtt{c}$. Each rule has a successful use of length at most 2: complete every A_i by $\mathtt{c}$ ($i \ge 1$) or $\mathtt{a}$ ($i = 0$). A yield of A_n has type 1 iff it contains no $\mathtt{c}$, so the unique yield of $(A_n)_{(1)}$ is $\mathtt{a}^{2^n}$, and $t_{\mathrm{typ}}(G_n, s_{\mathtt{c}}) \ge 2^n$; the typed rule $S_{(1)} \to (A_n)_{(1)}$ is retained (via $S \to A_n$), and every successful derivation using it has length at least 2^n. □

This blow-up concerns the witness construction, not the data, because the tuples used for a typed nonterminal need not be its yields. For a typed nonterminal X, Lemma 3.8 places L_X in one class $\mathrm{Cl}_{L,s}(X)$; the *representative length* is

$$\tau_X := \min\{\|\vec{v}\|_1 : \vec{v} \in \mathrm{Cl}_{L,s}(X)\}, \qquad \tau(G,s) := \max_X \tau_X \quad (\max \emptyset := 0).$$

Corollary 4.10 (Characteristic data from equivalent representatives)**.** *Under the hypotheses of Theorem 4.4,*

$$\mu^\star(L,s) \le \tau(G,s) \le t_{\mathrm{typ}}(G,s),$$

and L has a characteristic sample K for both $\mathcal{B}_{f,s}^{\le r_{\max}}$ and $\mathcal{B}_{f,s}$ with

$$\|K\|_+ \le N_{\mathrm{typ}}(1 + 2\bar{\theta}_G + r_{\max}\tau(G,s)) + \mathbf{1}_{\{\varepsilon \in L\}}.$$

Proof. For each typed $X = A_{\mathbf{t}}$ choose $\iota(X) \in \mathrm{Cl}_{L,s}(X)$ with $\|\iota(X)\|_1 = \tau_X$; then $\widehat{s}^{(\mathrm{fan}(A))}(\iota(X)) = \mathbf{t}$, and $L_X \subseteq \mathrm{Cl}_{L,s}(X)$ gives $\tau \le t_{\mathrm{typ}}$. Take $\chi(X) := E_A$ as in Lemma 4.1(iv). Since $\chi(X)[\vec{u}] \in L$ for $\vec{u} \in L_X$, also $\chi(X)[\iota(X)] \in L$. For a typed rule $X \to \rho(Y_1,\dots,Y_q)$ and $\vec{u}_j \in L_{Y_j}$, Lemma 3.2 gives $\rho(\iota(Y_1),\dots,\iota(Y_q)) \equiv_{L,s} \rho(\vec{u}_1,\dots,\vec{u}_q) \in L_X$, so $\chi(X)[\rho(\iota(Y_1),\dots,\iota(Y_q))] \in L$. Let K consist of these words and $\{\varepsilon\} \cap L$. For every finite K' with $K \subseteq K' \subseteq L$, Proposition 3.11, which does not require $\iota(X) \in L_X$, gives $L \cap \Sigma^+ \subseteq L(\mathcal{B}(K'))$ for both operators (typed rules have rank at most $r_{\max}$), Start covers ε, and Proposition 3.10 gives the reverse inclusion. The length bound is counted as in Lemma 4.2(ii) with $b = \tau$, over the N_{typ} typed nonterminals and rules. Finally, for $m \in M_L$ the typed start nonterminal $X = S_{(m)}$ has $\chi(X) = \square_1$, so $\square_1 \in \mathcal{D}_L(\iota(X))$; thus $\iota(X) = (v)$ with $v \in L$ and $s(v) = m$, and $\tau_X \ge \mu_m(L)$. □

In Proposition 4.9, $L = \Sigma^+$ and every admissible context accepts every tuple, so two tuples are equivalent exactly when their types agree; single letters represent both types, and $\tau(G_n, s_{\mathsf{c}}) = 1$ although $t_{\mathrm{typ}}(G_n, s_{\mathsf{c}}) \ge 2^n$. Corollary 4.10 then gives characteristic data of length $O(\|G_n\|)$. The corollary does not bound τ itself; the proofs below bound t_{typ} and therefore work on G.

4.3 Short Representatives by Minimal Counterexamples

Let $\lambda : \Sigma^* \to \mathsf{N}$ be a finite-monoid morphism. A vector $\mathbf{t} \in \mathsf{N}^d$ is *realizable at B* (of fan-out d) if $\mathbf{t} = \lambda^{(d)}(\vec{u})$ for some $\vec{u} \in L_B(G)$. A *λ-minimal* derivation for $(B, \mathbf{t})$ is a terminal derivation from B with root λ-vector $\mathbf{t}$ that minimizes first the total yield length and then the number of nodes. For a threshold b, a node is *heavy* if its yield has total length greater than b.

Lemma 4.11 (Heavy-node count)**.** *Let $b \ge \bar{\theta}_G$ and $c \ge 0$, and let T be λ-minimal for $(B, \mathbf{t})$. If every antichain of heavy nodes of T has at most c elements, then*

$$\|\textit{root tuple of } T\|_1 \le (r_{\max} + 1)\, b\, \max\{1, c\,|V|\,|\mathsf{N}|^f\}.$$

Proof. Label each node by its nonterminal and the λ-vector of its yield; there are at most $|V|\,|\mathsf{N}|^f$ labels. If an ancestor and a proper descendant had the same label, replacing the ancestor's subtree by the descendant's would keep the root λ-vector (templates evaluate λ homomorphically), would not lengthen the root yield (by linearity the descendant's yield is part of the ancestor's), and would remove nodes, contradicting minimality. So every root-to-leaf path has at most $|V|\,|\mathsf{N}|^f$ nodes. If the root is light its yield has length at most b. Otherwise the heavy nodes form an ancestor-closed subtree whose leaves form an antichain, so there are at most $c|V|\,|\mathsf{N}|^f$ heavy nodes. Every terminal of the root yield is written by a heavy node (at most $\bar{\theta}_G \le b$ each, Lemma 4.1(i)) or lies in a light child of a heavy node (at most $r_{\max}$ children of length at most b). □

Group constants. For a finite abelian group $\mathbb{H}$, $\mathsf{d}(\mathbb{H})$ is the maximal length of a zero-sum-free sequence over $\mathbb{H}$ (one with no nonempty subsequence summing to the identity; thus $\mathsf{d}(\mathbb{H})+1$ is the Davenport constant; $\mathsf{d}(\{1\})=0$). For a finite group $\mathbb{G}$ and $f \geq 1$ put

$$N_{\mathbb{G},f} = \max\{3, |\mathbb{G}|^f\}, \quad \mathcal{P}_f = f^{f+1} f!, \quad M_{\mathbb{G},f} = \mathcal{P}_f \, N_{\mathbb{G},f}^{2^{f^2}}, \qquad c_{\mathbb{G},f} := \begin{cases} \mathsf{d}(\mathbb{G}^f), & \mathbb{G} \text{ abelian}, \\ M_{\mathbb{G},f} - 1, & \text{otherwise}. \end{cases}$$

$\mathcal{P}_f$ bounds the number of block patterns of one owner, and $N_{\mathbb{G},f} \geq 3$ is needed by the order classification in Appendix A.1. The double exponent 2^{f^2} comes from applying the Erdős–Szekeres-type extraction of Lemma A.8 once for each of up to f^2 ordered pairs of block positions; we do not know whether it is necessary. For abelian groups no order information is needed, and $c_{\mathbb{G},f} = \mathsf{d}(\mathbb{G}^f)$ is linear in f (Remark A.3).

Simultaneous block replacement. A *block-replacement family* of width f over $\mathbb{G}$ (Definition A.1) consists of $d \leq f$ rows, each an ordered product in $\mathbb{G}$ of fixed constants and *blocks*; each block has an *owner* in $[m] := \{1, \dots, m\}$, an original value and a replacement value, and each owner owns at most f blocks, which may interleave with other owners' blocks arbitrarily. $\mathrm{val}_\varrho(J)$ is the value of row ϱ when exactly the blocks of owners in $J \subseteq [m]$ take their replacement values.

Proposition 4.12 (Simultaneous block replacement)**.** *In a block-replacement family of width f over a finite group $\mathbb{G}$ with $m > c_{\mathbb{G},f}$ owners there is a nonempty $J \subseteq [m]$ with $\mathrm{val}_\varrho(J) = \mathrm{val}_\varrho(\varnothing)$ for every row ϱ. Individual block or owner values need not be preserved.*

Proof. This is Lemma A.2. For abelian $\mathbb{G}$ one selects a nonempty zero-sum subsequence of the owners' difference vectors in $\mathbb{G}^d \leq \mathbb{G}^f$. For arbitrary $\mathbb{G}$ no factors are commuted: among owners with a common block pattern (at most $\mathcal{P}_f$ patterns) one homogenizes all pairwise order comparisons (Lemma A.8), which decomposes every row into consecutive rank chunks (Lemma A.4); a pigeonhole on prefix-product boundary states then yields an interval of owners preserving all ordered row products (Lemma A.5). □

Owner contributions cannot be commuted: if the two rows are $\beta_{1,1}\beta_{2,1}$ and $\beta_{2,2}\beta_{1,2}$, with value (12) on the blocks of owner 1 and (23) on those of owner 2 in S_3, the row values are $(12)(23) \neq (23)(12)$. Example A.6 shows a replacement preserving rows of opposite owner orientation.

Lemma 4.13 (Short representatives for window–group invariants)**.** *Fix $k, \ell \geq 0$, a finite group $\mathbb{G}$ (possibly trivial) and a morphism $g : \Sigma^* \to \mathbb{G}$. Put $\lambda := (\widehat{s}_{k,\ell}, g) : \Sigma^* \to \mathsf{N} := M_{k,\ell} \times \mathbb{G}$ and $h := f(k+\ell+1)$. Then $(\mathrm{R}^\lambda_\beta)$ holds with*

$$\beta = b_{k,\ell,\mathbb{G}} := (r_{\max}+1)\,\bar{\theta}_G\,(c_{\mathbb{G},f}+h)\,|V||\mathsf{N}|^f \;\leq\; (r_{\max}+1)(c_{\mathbb{G},f}+h)|\mathsf{N}|^f\,\|G\|\,\bar{\theta}_G.$$

Proof. Let $\vec{u} \in L_B(G)$, $\mathbf{t} := \lambda^{(\mathrm{fan}(B))}(\vec{u})$, and let T be λ-minimal for $(B, \mathbf{t})$, with heavy threshold $b = \bar{\theta}_G$. In each root component of length at least $m_0 = \max\{1, k+\ell\}$ mark the first k and the last ℓ positions (one arbitrary position if $k = \ell = 0$), and in each shorter component mark all positions: at most h marks. Let Q be an antichain of heavy nodes. Incomparable subtrees have disjoint yields, so at most h nodes of Q contain a mark; let Q_0 be the others.

Suppose $|Q_0| > c_{\mathbb{G},f}$. For $v \in Q_0$ with nonterminal C_v fix a tuple of $L_{C_v}(G)$ of total length at most $\bar{\theta}_G$ (Lemma 4.1(ii)). By linearity and nondeletion each $v \in Q_0$ contributes at most f disjoint contiguous blocks to the root components, and the g-value of each root component is the ordered product of the g-values of fixed material and of these blocks. This is a block-replacement family over

$\mathbb{G}$ with owners Q_0, so Proposition 4.12 gives a nonempty $J \subseteq Q_0$ whose simultaneous replacement preserves every root g-value. The replaced blocks contain no mark: short root components are untouched, and in a long component they lie strictly between the marked prefix and suffix, which keep their letters and positions relative to the ends, while the component keeps its $k+\ell$ marked letters (at least one if $k=\ell=0$) and stays long. So the root $\widehat{s}_{k,\ell}$-vector is preserved as well, and the result is a derivation from B with root λ-vector $\mathbf{t}$ that is strictly shorter, since each replaced yield exceeded $\bar{\theta}_G$. This contradicts minimality.

Hence every heavy antichain has at most $c_{\mathbb{G},f}+h$ elements, and Lemma 4.11 with $c = c_{\mathbb{G},f}+h \geq 1$ gives the bound; the second inequality uses $|V| \leq \|G\|$. □

Corollary 4.14 (Window representatives)**.** *For $k,\ell \geq 0$, $(\mathrm{R}_\beta^{\widehat{s}_{k,\ell}})$ holds with $\beta = b_{k,\ell} := (r_{\max}+1)\bar{\theta}_G\, f(k+\ell+1)\,|V||M_{k,\ell}|^f$.*

Proof. Take $\mathbb{G}$ trivial in Lemma 4.13: then $c_{\mathbb{G},f}=0$ and $\mathsf{N} \cong M_{k,\ell}$. □

4.4 Window and Window–Group Typings

Theorem 4.15 (Polynomial characteristic data for window–finite-group typing)**.** *Let $\mathbb{G}$ be a finite group (possibly trivial), $g : \Sigma^* \to \mathbb{G}$ a morphism, and $\widehat{s} \preceq (\widehat{s}_{k,\ell}, g)$; let $L \in \mathcal{C}_{f,s}^{\mathrm{mcf}}$ and $L(G) = L \cap \Sigma^+$. Then $\mathsf{W}(\widetilde{G}_0) \cup (\{\varepsilon\} \cap L)$ is a characteristic sample for both $\mathcal{B}_{f,s}^{\leq r_{\max}}$ and $\mathcal{B}_{f,s}$ with $|\mathsf{W}(\widetilde{G}_0)| \leq C_s(G) = O(\|G\|)$ and*

$$\|\mathsf{W}(\widetilde{G}_0)\|_+ \leq K_{k,\ell,\mathbb{G}}\,\|G\|^2\bar{\theta}_G,$$
$$K_{k,\ell,\mathbb{G}} := |\mathsf{S}^\sharp|^{f r_{\max}}\big(3 + r_{\max}(r_{\max}+1)(c_{\mathbb{G},f} + f(k+\ell+1))(|M_{k,\ell}||\mathbb{G}|)^f\big).$$

The group may be noncommutative and its blocks may interleave across components. For trivial $\mathbb{G}$, i.e. $s \preceq s_{k,\ell}$, one has $c_{\mathbb{G},f}=0$.

Proof. Lemma 4.13 and Lemma 4.2(i),(ii),(iv) with $\lambda = (\widehat{s}_{k,\ell}, g)$ and $b = b_{k,\ell,\mathbb{G}}$ give the sample and $\|\mathsf{W}\|_+ \leq C_s(G)(1 + 2\bar{\theta}_G + r_{\max} b_{k,\ell,\mathbb{G}})$; insert $C_s(G) \leq |\mathsf{S}^\sharp|^{f r_{\max}}\|G\|$, $1 + 2\bar{\theta}_G \leq 3\|G\|\bar{\theta}_G$ and $|V| \leq \|G\|$. □

Remark 4.16 (Abelian groups). If $\mathbb{G}$ is abelian, Lemma A.2(a) gives $c_{\mathbb{G},f} = \mathsf{d}(\mathbb{G}^f) = O(\exp(\mathbb{G})\, f \log|\mathbb{G}|)$ (Remark A.3; $\exp(\mathbb{G})$ is the least common multiple of the element orders), so $K_{k,\ell,\mathbb{G}}$ is polynomial in f apart from the factors $|\mathsf{S}^\sharp|^{f r_{\max}}$ and $(|M_{k,\ell}||\mathbb{G}|)^f$; for nonabelian $\mathbb{G}$, $c_{\mathbb{G},f} = M_{\mathbb{G},f} - 1$ grows doubly exponentially in f.

At fan-out one, polynomial time and thick-data learning for fixed windows is due to Yoshinaka [12, Theorem 1]; the bounds above are relative to a reduced good presentation and are not claimed to improve on that result after normalization. What is new is that window and ordered group values are preserved *simultaneously* across up to f interleaved components.

4.5 An Algebraic Criterion for Polynomial Characteristic Data

LI is the pseudovariety of finite semigroups T satisfying $e\mathsf{T}e = \{e\}$ for every idempotent $e \in \mathsf{T}$, **Grp** that of finite groups, and $\mathbf{LI} \vee \mathbf{Grp}$ consists of the divisors of products $\mathsf{T} \times \mathbb{G}$ with $\mathsf{T} \in \mathbf{LI}$ and $\mathbb{G}$ a finite group [25, 26]. The following lemma makes the window–group hypothesis intrinsic to the positive image S; the semigroup facts are classical [24].

Lemma 4.17 (Locally trivial–group typings admit window–group refinements)**.** *Let $\Sigma \neq \emptyset$. If $\mathsf{S} \in \mathbf{LI} \vee \mathbf{Grp}$, there are $k \geq 1$, a finite group $\mathbb{G}$ and a morphism $g : \Sigma^* \to \mathbb{G}$ with $\widehat{s} \preceq (\widehat{s}_{k,k}, g)$; if $\mathsf{S} \in \mathbf{LI}$, $\mathbb{G}$ can be trivial.*

Proof. Choose a finite $\mathsf{S}' \in \mathbf{LI}$, a finite group $\mathbb{G}$, a subsemigroup $\mathsf{T} \leq \mathsf{S}' \times \mathbb{G}$ and a surjective morphism $\vartheta : \mathsf{T} \to \mathsf{S}$ (for $\mathsf{S} \in \mathbf{LI}$: $\mathsf{S}' = \mathsf{S}$, $\mathbb{G} = \{1\}$). Lifting each $s(a)$ to $(\tau_a, g_a) \in \mathsf{T}$ defines a morphism $\rho = (\tau, g) : \Sigma^+ \to \mathsf{T}$ with $\vartheta\rho = s$; put $g(\varepsilon) = 1_{\mathbb{G}}$. The subsemigroup $\mathsf{S}_0 := \tau(\Sigma^+) \leq \mathsf{S}'$ is locally trivial; let $k := |\mathsf{S}_0|$. Two facts:

(a) $\mathsf{S}_0^k \subseteq \mathsf{S}_0 E(\mathsf{S}_0) \mathsf{S}_0$. Let t_i be the prefix products of $v_1 \cdots v_k$. If $t_i = t_j$ with $i < j$, then $t_i = t_i u = t_i u^\omega$ for $u = v_{i+1} \cdots v_j$, so the product is $t_i u^\omega y$ with $y \in \mathsf{S}_0^1$; otherwise the t_i exhaust S_0, one of them is an idempotent e, and the product is ey. In both cases it is xey with $x, y \in \mathsf{S}_0^1$, and $xey = (xe)e(ey)$.

(b) $euf = ef$ for $e, f \in E(\mathsf{S}_0)$ and $u \in \mathsf{S}_0^1$: $euf = eu(fef) = (e(uf)e)f = ef$.

For $|w| \geq 2k$ write $w = pxz$ with $|p| = |z| = k$. By (a), $\tau(p) = aeb$ and $\tau(z) = cfd$ with $a, b, c, d \in \mathsf{S}_0$ and e, f idempotent, and by (b) $\tau(w) = ae\,(b\,\tau(x)c)\,fd = aefd$ (with $\tau(\varepsilon) := 1$), which depends only on $p = \mathrm{pref}_k(w)$ and $z = \mathrm{suf}_k(w)$. Shorter words are recorded exactly by $\widehat{s}_{k,k}$, which also separates ε. Hence equal $(\widehat{s}_{k,k}, g)$-values imply equal ρ-values and equal s-values. □

Consequently Theorem 4.15 applies to every typing with $\mathsf{S} \in \mathbf{LI} \vee \mathbf{Grp}$, with k, $\mathbb{G}$ and g depending only on s.

A finite semigroup is a *local group*, $\mathsf{T} \in \mathbf{LG}$, if $e\mathsf{T}e$ is a group for every idempotent $e \in \mathsf{T}$. This is always tested on the positive image S, never on $\mathsf{S}^\sharp$. By Proposition B.5, $\mathbf{LI} \vee \mathbf{Grp} \subsetneq \mathbf{LG}$.

Definition 4.18 (Local-group typing)**.** Fix $k \geq 1$, a finite group $\mathbb{G}$ and a map $\nu : \Sigma^{k+1} \to \mathbb{G}$ (the *sliding-factor labelling*). For $|w| \geq 2k$ let $\mathrm{fac}_k(w) = (F_1, \ldots, F_{|w|-k})$ be the factors of length $k+1$ of w in order of their starting positions, and put $\gamma_{k,\nu}(w) := \nu(F_1)\nu(F_2)\cdots\nu(F_{|w|-k})$ (ordered product in $\mathbb{G}$); put $\gamma_{k,\nu}(w) := 1_{\mathbb{G}}$ for $|w| < 2k$. The *local-group typing* is $\lambda_{k,\nu}(w) := (\widehat{s}_{k,k}(w), \gamma_{k,\nu}(w))$; by Lemma B.1 it is a morphism onto a finite monoid $M_{k,\nu}$. The typing s is *local-group-determined* if $\widehat{s} \preceq \lambda_{k,\nu}$ for some k, $\mathbb{G}$ and ν.

A local-group typing can encode a group contribution determined by a sliding factor, rather than by each letter alone. For example, $\lambda_{1,\nu}$ with $\nu(\mathtt{ab}) = 1 \in \mathbb{Z}_2$ and zero on other pairs records the parity of occurrences of $\mathtt{ab}$. Even on $\{\mathtt{a}, \mathtt{b}\}$ this is not determined by any fixed boundary window combined with a finite-group morphism (Proposition B.5(ii)). The local-group characterization below expresses this additional power without requiring a new learner or a target-specific typing.

Theorem 4.19 (Polynomial characteristic data for local-group typing)**.** *Let $\mathsf{S} \in \mathbf{LG}$. Every reduced good fan-out-$\leq f$, rank-$\leq r_{\max}$ grammar G satisfies $t_{\mathrm{typ}}(G, s) = O_{f, r_{\max}, s, \Sigma}(\|G\| \bar{\theta}_G)$. If moreover $L \in \mathcal{C}^{\mathrm{mcf}}_{f,s}$ and $L(G) = L \cap \Sigma^+$, then $\mathsf{W}(\widetilde{G}_0) \cup (\{\varepsilon\} \cap L)$ is a characteristic sample for both $\mathcal{B}^{\leq r_{\max}}_{f,s}$ and $\mathcal{B}_{f,s}$ with*

$$|\mathsf{W}(\widetilde{G}_0)| = O(\|G\|), \qquad \|\mathsf{W}(\widetilde{G}_0)\|_+ = O_{f, r_{\max}, s, \Sigma}(\|G\|^2 \bar{\theta}_G).$$

Proof. By Theorem B.4, $\widehat{s} \preceq \lambda := \lambda_{k,\nu}$ for $k = |\mathsf{S}|$ and some $\mathbb{G}, \nu$ depending only on s. Lemma B.7 gives $(\mathrm{R}^\lambda_\beta)$ with $\beta = \beta_\lambda = O(\|G\| \bar{\theta}_G)$: heavy subtrees are replaced by arbitrary short yields of their nonterminals, after the owners have been thinned so that their blocks lie at least $2k$ letters apart and each block has been enlarged by k unchanged letters on both sides (Lemma B.6); the enlarged blocks keep their window types, so the ordered sliding-factor products are controlled by Lemma B.2 and Proposition 4.12. This step uses no substitutability. Lemma 4.2(i) gives $t_{\mathrm{typ}} \leq \beta_\lambda$, and Theorem 4.4 gives $\|\mathsf{W}\|_+ \leq N_{\mathrm{typ}}(1 + 2\bar{\theta}_G + r_{\max}\beta_\lambda) = O(\|G\|^2 \bar{\theta}_G)$. □

The Rees and sliding-factor coordinates are classical; their combination with owner-wise block shortening is the quantitative step. Local-group and window–group typings thus have the same quadratic dependence on $\|G\|$; Theorem 4.15 gives explicit constants for the latter.

A bracketed doubling family. Let $\Gamma_{\mathrm{br}} = \{\mathtt{a}, \lfloor, \rfloor, \lceil, \rceil\}$, let $U_1 = \{1, \zeta\}$ with identity 1 and $\zeta^2 = \zeta$, and let $s_U : \Gamma_{\mathrm{br}}^+ \twoheadrightarrow U_1$ send $\lceil, \rceil$ to ζ and all other letters to 1; thus $s_U(w) = 1$ exactly for the nonempty ceiling-free words. For $n \geq 1$ let G_n be the fan-out-one grammar

$$S_0 \to E_n, \qquad E_0 \to \mathtt{a}, \qquad E_i \to \lfloor E_{i-1} E_{i-1} \rfloor \mid \lceil E_{i-1} \rceil \quad (1 \leq i \leq n),$$

put $L_n := L(G_n)$, $W_0 := \mathtt{a}$ and $W_i := \lfloor W_{i-1} W_{i-1} \rfloor$. This family was introduced in the companion CFG manuscript [23, Proposition 6.4]; every property used below is proved here.

Proposition 4.20 (Bracketed doubling family)**.** *For every $n \geq 1$:*

(i) *G_n is reduced and good, of fan-out one and rule rank two, with $\|G_n\| = O(n)$ and $\theta_{G_n} \leq 4n$.*

(ii) *L_n is Clark–Eyraud substitutable: for all $x, y \in \Gamma_{\mathrm{br}}^+$, $\mathcal{D}^{(1)}_{L_n}(x) \cap \mathcal{D}^{(1)}_{L_n}(y) \neq \emptyset$ implies $\mathcal{D}^{(1)}_{L_n}(x) = \mathcal{D}^{(1)}_{L_n}(y)$. Hence $L_n \in \mathcal{C}^{\mathrm{mcf}}_{1,s}$ for every typing s on Γ_{br}^+; in particular $L_n \in \mathcal{C}^{\mathrm{mcf}}_{1,s_U}$.*

(iii) *W_n is the only word of L_n of s_U-type 1, and $|W_n| = 3 \cdot 2^n - 2$.*

(iv) *Every characteristic sample F of L_n for $\mathcal{B}^{\leq r}_{1,s_U}$ ($r \geq 1$) or $\mathcal{B}_{1,s_U}$, or for any set-driven operator having characteristic samples throughout $\mathcal{C}^{\mathrm{mcf}}_{1,s_U}$, contains W_n; hence $\|F\|_+ \geq 3 \cdot 2^n - 1$.*

Proof. (i) Every template is nonempty and uses each variable once; at fan-out one, nonpermutation and nonmerging are automatic. Every nonterminal is reachable from S_0 and productive, and the $2n + 2$ rules have bounded length, so $\|G_n\| = O(n)$. Put $u_0 := \mathtt{a}$ and $u_j := \lceil u_{j-1} \rceil$, so $|u_j| = 2j + 1$. A rule with left-hand side E_i is used in the successful derivation of $\lceil^{n-i} t \rceil^{n-i}$, where t is $\mathtt{a}$, u_i, or $\lfloor u_{i-1} u_{i-1} \rfloor$; its length is at most $2n + 2i \leq 4n$. The rule $S_0 \to E_n$ is used by u_n, of length $2n + 1$.

(ii) The first claim is the second assertion of Proposition F.4 in Appendix F. It is the condition of Definition 2.8 at $f = 1$ for the trivial typing, and the finite language L_n is a 1-MCFL; item (a) after Definition 2.8 gives $L_n \in \mathcal{C}^{\mathrm{mcf}}_{1,s}$.

(iii) A derivation yields a ceiling-free word if and only if it uses no rule $E_i \to \lceil E_{i-1} \rceil$. Such a derivation from E_i is the complete binary tree of height i and yields W_i, and $|W_i| = 2|W_{i-1}| + 2$ with $|W_0| = 1$.

(iv) By (ii), $L_n \in \mathcal{C}^{\mathrm{mcf}}_{1,s_U}$, and by (iii), $1 \in M_{L_n}$ with $\mu_1(L_n) = |W_n|$. Proposition 4.6 with $m = 1$ forces a word of type 1, which can only be W_n. □

Definition 4.21 (Uniform polynomial characteristic-data length)**.** Let $f, r_{\max} \geq 1$. A pseudovariety $\mathbf{V}$ of finite semigroups has *uniform polynomial characteristic-data length at fan-out f and rank $r_{\max}$* if for every nonempty finite alphabet Σ and every surjective typing $s : \Sigma^+ \twoheadrightarrow \mathsf{S}$ with $\mathsf{S} \in \mathbf{V}$ there is a polynomial $p = p_{f, r_{\max}, s, \Sigma}$ such that for every $L \in \mathcal{C}^{\mathrm{mcf}}_{f,s}$ and every reduced good fan-out-$\leq f$, rank-$\leq r_{\max}$ presentation G of $L \cap \Sigma^+$, the operator $\mathcal{B}^{\leq r_{\max}}_{f,s}$ has a characteristic sample F for L with $\|F\|_+ \leq p(\|G\|, \bar{\theta}_G)$.

The quantifiers are ordered as written: p may depend on f, $r_{\max}$, Σ and the typing s, but not on the target L or the presentation G.

Theorem 4.22 (Fan-out one: a pseudovariety boundary for data length)**.** *Let $r_{\max} \geq 2$ and let $\mathbf{V}$ be a pseudovariety of finite semigroups.*

(i) $\mathbf{V}$ *has uniform polynomial characteristic-data length at fan-out one and rank* $r_{\max}$ *if and only if* $\mathbf{V} \subseteq \mathbf{LG}$.

(ii) *If* $\mathbf{V} \subseteq \mathbf{LG}$, *one may take* $p(\|G\|, \bar{\theta}_G) = O_{r_{\max},s,\Sigma}(\|G\|^2 \bar{\theta}_G)$ *in Definition 4.21.*

(iii) *If* $\mathbf{V} \not\subseteq \mathbf{LG}$, *the typing* s_U *of Proposition 4.20 has positive image* $U_1 \in \mathbf{V}$, *and the targets* $L_n \in \mathcal{C}^{\mathrm{mcf}}_{1,s_U}$ *have reduced good rank-two presentations* G_n *with* $\|G_n\| = O(n)$ *and* $\bar{\theta}_{G_n} = O(n)$, *while every characteristic sample of* L_n *has total encoded length at least* $3 \cdot 2^n - 1$. *This lower bound holds for* $\mathcal{B}^{\leq r_{\max}}_{1,s_U}$, *for* $\mathcal{B}_{1,s_U}$, *and for every set-driven operator having characteristic samples throughout* $\mathcal{C}^{\mathrm{mcf}}_{1,s_U}$.

Proof. (ii), and hence the *if* direction of (i), is Theorem 4.19 with $f = 1$, since every $\mathsf{S} \in \mathbf{V}$ then lies in $\mathbf{LG}$. For (iii), pick $\mathsf{S} \in \mathbf{V} \setminus \mathbf{LG}$. By Theorem B.4(ii), S has a subsemigroup isomorphic to U_1, and $U_1 \in \mathbf{V}$ because pseudovarieties are closed under subsemigroups. The typing s_U maps onto U_1; the remaining claims are Proposition 4.20(i) and (iv), using $\|G_n\| = O(n)$ and $\theta_{G_n} \leq 4n$. Since $3 \cdot 2^n - 1$ is not bounded by any polynomial in $\|G_n\|$ and $\bar{\theta}_{G_n}$, the *only if* direction of (i) follows. □

Remark 4.23 (Scope of Theorem 4.22). (a) The condition $r_{\max} \geq 2$ is needed: at rank one every typing admits polynomial data in grammar size alone (Theorem 5.1). (b) The theorem concerns pseudovarieties. For individual typings, Corollary 4.25 below gives the negative direction whenever some copy of U_1 in S is realized by two letters of its identity type and one letter of its absorbing type; this settles every typing onto U_1 except those with exactly one letter of type 1. Whether such a typing, for instance $\mathtt{a} \mapsto 1$, $\mathtt{b} \mapsto \zeta$ on $\{\mathtt{a}, \mathtt{b}\}$, admits polynomial characteristic data is open; there every finite target has at most one word of type 1, since two words $\mathtt{a}^p, \mathtt{a}^q$ with $p < q$ share the context $(\varepsilon, \varepsilon)$, which forces $\mathtt{a}^{p+k(q-p)} \in L$ for all $k \geq 0$. (c) The upper bound is for the fixed operator $\mathcal{B}^{\leq r_{\max}}_{1,s}$, whereas the lower bound holds for every set-driven operator with characteristic samples throughout the class. (d) At fan-out $f \geq 2$ the *if* direction persists (Theorem 4.19), but the *only if* direction is open at the level of pseudovarieties (Open Problem 7.1); the family L_n cannot serve there (Theorem 6.8), and for individual typings it fails: the binary typing onto U_1 admits polynomial data at fan-out two (Theorem 6.10).

Proposition 4.24 (A three-letter doubling family). *Let* $\mathtt{a}, \mathtt{b}, \mathtt{c}$ *be distinct letters, put* $K^{(0)} := \{\mathtt{a}\}$ *and* $K^{(i)} := \mathtt{b}K^{(i-1)}K^{(i-1)}\mathtt{b} \cup \mathtt{c}K^{(i-1)}\mathtt{c}$ *for* $i \geq 1$, *and let* $n \geq 1$. *Then* $K_n := K^{(n)}$ *is Clark–Eyraud substitutable, so* $K_n \in \mathcal{C}^{\mathrm{mcf}}_{1,s}$ *for every typing* s *on every alphabet containing these letters. The grammar* G'_n *with rules* $S \to E_n$, $E_0 \to \mathtt{a}$ *and* $E_i \to \mathtt{b}E_{i-1}E_{i-1}\mathtt{b} \mid \mathtt{c}E_{i-1}\mathtt{c}$ $(1 \leq i \leq n)$ *is a reduced good rank-two presentation of* K_n *with* $\|G'_n\| = 13n + 7$ *and* $\bar{\theta}_{G'_n} = 4n$. *The only word of* K_n *without* $\mathtt{c}$ *is* V_n, *where* $V_0 := \mathtt{a}$ *and* $V_i := \mathtt{b}V_{i-1}V_{i-1}\mathtt{b}$, *and* $|V_n| = 3 \cdot 2^n - 2$.

Proof. Substitutability is proved in Appendix G; letters outside $\{\mathtt{a}, \mathtt{b}, \mathtt{c}\}$ only add empty distributions, and finite languages are context-free. The grammar is good, and reduced since $E_i \Rightarrow^* \mathtt{c}^i\mathtt{a}\mathtt{c}^i$. It has $n + 2$ nonterminals and rules of sizes 3 (start), 2 (base), 7 and 5 (the two rules of each level), whence $\|G'_n\| = 13n + 7$. A shortest yield of E_i has length $2i + 1$, and the levels above E_i add at least $2(n - i)$ letters, attained by unary rules. So the start, base and unary rules have rule thickness $2n + 1$, and the binary rule of level i has $2(n - i) + 2 + 2(2i - 1) = 2n + 2i$; hence $\bar{\theta}_{G'_n} = 4n$. A word without $\mathtt{c}$ uses only binary rules, which gives V_n. □

Corollary 4.25 (A typing-level obstruction at fan-out one). *Let* $r_{\max} \geq 2$ *and let* $s : \Sigma^+ \twoheadrightarrow \mathsf{S}$ *be a typing. Suppose that distinct idempotents* $e, z \in \mathsf{S}$ *satisfy* $ez = ze = z$ *and that distinct letters* $\mathtt{a}, \mathtt{b}, \mathtt{c} \in \Sigma$ *satisfy* $s(\mathtt{a}) = s(\mathtt{b}) = e$ *and* $s(\mathtt{c}) = z$. *Then every characteristic sample of* K_n *for* $\mathcal{B}^{\leq r_{\max}}_{1,s}$, *for* $\mathcal{B}_{1,s}$, *or for any set-driven operator having characteristic samples throughout* $\mathcal{C}^{\mathrm{mcf}}_{1,s}$ *has*

total encoded length at least $3 \cdot 2^n - 1$*, while* $\|G'_n\|$ *and* $\bar{\theta}_{G'_n}$ *are* $O(n)$*. So for this single typing no polynomial bounds the characteristic-data length at fan-out one and rank* $r_{\max}$ *as in Definition 4.21. This applies to every typing onto* U_1 *with at least two letters of type* 1*.*

Proof. In $\{e, z\}$, e is an identity and z is absorbing, so a word of K_n has type e exactly when it contains no c; by Proposition 4.24 the only such word is V_n. Proposition 4.6 forces V_n into every such sample. For a typing onto U_1, surjectivity gives a letter of type ζ, since words over letters of type 1 have type 1; take $e = 1$ and $z = \zeta$. □

5 Rule Rank One

At rule rank one, grammar size alone bounds characteristic data under every typing. Notation is that of Section 4; in particular $C_s(G) = |V||\mathsf{S}^\sharp|^f + |P||\mathsf{S}^\sharp|^{fr_{\max}} \le |\mathsf{S}^\sharp|^{fr_{\max}}\|G\|$.

Theorem 5.1 (Rank-one characteristic-data bound)**.** *Fix* f *and a typing* s*.*

(i) *If* G *is a reduced good fan-out-*$\le f$*, rank-*≤ 1 *presentation of* $L \cap \Sigma^+$ *with* $L \in \mathcal{C}^{\mathrm{mcf}}_{f,s}$*, then* $\mathsf{W}(\widetilde{G}_0) \cup (\{\varepsilon\} \cap L)$ *is a characteristic sample for both* $\mathcal{B}^{\le 1}_{f,s}$ *and* $\mathcal{B}_{f,s}$*, and*

$$\|\mathsf{W}(\widetilde{G}_0)\|_+ \le |\mathsf{S}^\sharp|^f\big(3 + |\mathsf{S}^\sharp|^f\big)\|G\|^2\bar{\theta}_G \le 2|\mathsf{S}^\sharp|^f\big(3 + |\mathsf{S}^\sharp|^f\big)\|G\|^3.$$

(ii) *For any standard fan-out-*$\le f$*, rank-*≤ 1 *grammar* G_{in} *for* $L \in \mathcal{C}^{\mathrm{mcf}}_{f,s}$*, the operators* $\mathcal{B}^{\le 1}_{f,s}$ *and* $\mathcal{B}_{f,s}$ *have a common characteristic sample* K *with* $\|K\|_+ = O_{f,s}(\|G_{\mathrm{in}}\|^3)$*.*

Proof. (i) A terminal derivation in a rank-≤ 1 grammar is a chain of unary rules ending in a constant rule. In a shortest chain no nonterminal repeats (splicing out the segment between two occurrences does not lengthen the yield, by linearity), so its rules are distinct. Hence $t_G \le \|G\|$, and the proof of Lemma 4.1(iii), whose path rules together with R are distinct and whose child chain is distinct, gives $\theta_G \le 2\|G\|$, so $\bar{\theta}_G \le 2\|G\|$. In $\widetilde{G}_0$, which also has rank ≤ 1, a shortest chain from a typed nonterminal X repeats no typed nonterminal, so it has at most $|V||\mathsf{S}^\sharp|^f$ rules, each writing at most $\bar{\theta}_G$ letters (Lemma 4.1(i)). So $\|\omega(X)\|_1 \le b := |V||\mathsf{S}^\sharp|^f\bar{\theta}_G$ for suitable canonical tuples. Lemma 4.2(ii),(iv) with $r_{\max} = 1$ gives the sample and $\|\mathsf{W}\|_+ \le C_s(G)(1 + 2\bar{\theta}_G + b)$; use $C_s(G) \le |\mathsf{S}^\sharp|^f\|G\|$, $1 + 2\bar{\theta}_G \le 3\bar{\theta}_G$ and $|V| \le \|G\|$.

(ii) If $L \subseteq \{\varepsilon\}$ take $K = L$. Otherwise Proposition 3.12 gives a reduced good rank-≤ 1 presentation G of $L \cap \Sigma^+$ with $\|G\| = O_f(\|G_{\mathrm{in}}\|)$; apply (i). □

Here the special rank-one argument $\bar{\theta}_G \le 2\|G\|$ is essential for the cubic bound; applying only the general thickness lemma would lose a power of $\|G\|$. Equivalently, the shortest typed chains above give $t_{\mathrm{typ}} \le |V||\mathsf{S}^\sharp|^f\bar{\theta}_G$. No thickness parameter is needed at rank one, and the typing affects only the constant; this extends the rank-one untyped scale [14, Lemma 7]. Rank two needs thickness: the size-$O(n)$ grammar $S \to A_n$, $A_i \to A_{i-1}A_{i-1}$, $A_0 \to \mathsf{c}$ generates $\{\mathsf{c}^{2^n}\}$, which lies in $\mathcal{C}^{\mathrm{mcf}}_{f,s}$ for every f and s (over one letter, a shared sentence context forces equal total component length, hence equal distributions), and every characteristic sample for $\mathcal{B}^{\le r_{\max}}_{f,s}$ contains c^{2^n}, because it is a nonempty subset of the target ($L(\mathcal{B}(\emptyset)) = \emptyset$).

6 Comparison with Multidimensional Substitutability

Yoshinaka's multidimensional substitutability [13, 14] is the direct positive-data baseline. This section shows that (i) trivial typing recovers it and window typing recovers the CFG (k, ℓ) hierarchy

of [12], while a fixed finite typing admits non-context-free targets outside the baseline (Section 6.1); (ii) the algebraic regimes of Section 4 are separated at the level of targets (Section 6.2); and (iii) the fan-out-one lower-bound family does not lift to fan-out two (Section 6.3). Since every (f, s) condition includes arity one, $\mathcal{C}^{\mathrm{mcf}}_{f,s} \cap \mathrm{CFL} \subseteq \mathcal{C}^{\mathrm{mcf}}_{1,s}$.

6.1 Yoshinaka's Baseline and Finite Feature Licensing

Definition 6.1 (Multidimensional substitutability)**.** A language L is *fD-substitutable* if for every $1 \le d \le f$, all $E, F \in \mathcal{E}_d$ and all $\vec{x}, \vec{y} \in (\Sigma^+)^d$, $E[\vec{x}], E[\vec{y}], F[\vec{x}] \in L \Rightarrow F[\vec{y}] \in L$ (the *substitution implication*); $\mathsf{YSub}(f)$ is the class of these languages. This is Yoshinaka's definition, with nonempty components and nonempty internal separators [14, Section 3.1 and footnote 3].

The substitution implication is (f, s_{triv})-tuple substitutability, so $\mathcal{C}^{\mathrm{mcf}}_{f,s_{\mathrm{triv}}} = \mathsf{YSub}(f) \cap f\text{-MCFL} \subseteq \mathcal{C}^{\mathrm{mcf}}_{f,s}$ for every s. Yoshinaka [12, Definition 2] calls L *k,l-substitutable* if for all $x_1, y_1, z_1, x_2, y_2, z_2 \in \Sigma^*$, $v \in \Sigma^k$ and $u \in \Sigma^\ell$ with $vy_1u, vy_2u \neq \varepsilon$, $x_1vy_1uz_1, x_1vy_2uz_1, x_2vy_1uz_2 \in L$ implies $x_2vy_2uz_2 \in L$. Here y_1, y_2 may be empty; only the factors vy_iu must be nonempty. Exchanging y_1 and y_2 gives the converse implication, so the condition says that two such factors with a common context have equal distributions. Two nonempty words have equal $s_{k,\ell}$-type iff they are identical or both have length at least $\max\{1, k+\ell\}$ with the same first k and last ℓ letters, i.e. iff they are of the forms vy_1u, vy_2u above, with y_1, y_2 possibly empty, or equal. Words shorter than $k+\ell$ are therefore compared only with themselves, in line with Yoshinaka's remark that the condition says nothing about them. Hence $(1, s_{k,\ell})$-tuple substitutability is exactly Yoshinaka's (k, ℓ) condition; the componentwise version at higher fan-out is a definition of this paper.

A language L is *strictly 2-local* if there is a set Π of pairs over $\Sigma \cup \{\vdash, \dashv\}$ such that a word w lies in L iff every pair of adjacent symbols of $\vdash w \dashv$ lies in Π.

Lemma 6.2 (Strictly 2-local languages are boundary typed)**.** *Every strictly 2-local L is $(f, s_{1,1})$-tuple-substitutable for every f; in particular $L \in \mathcal{C}^{\mathrm{mcf}}_{f,s_{1,1}}$, and $L \in \mathcal{C}^{\mathrm{mcf}}_{f,s}$ whenever $s_{1,1} \preceq s$.*

Proof. Let $\vec{x}, \vec{y} \in (\Sigma^+)^d$ have equal $s_{1,1}$-types, so corresponding components are identical letters or share first and last letters, and let $E_0[\vec{x}], E_0[\vec{y}] \in L$. Suppose $E[\vec{x}] \in L$. Since the internal separators of E are nonempty, every adjacent pair of $\vdash E[\vec{y}] \dashv$ either lies inside some y_i, and then occurs in $E_0[\vec{y}] \in L$, or joins a first or last letter of some y_i to a symbol of $\vdash E \dashv$, or lies in $\vdash E \dashv$, and then occurs in $\vdash E[\vec{x}] \dashv$. So $E[\vec{y}] \in L$; symmetry and $s_{1,1} \preceq s$ finish the proof (regular languages are MCFLs). □

Proposition 6.3 (Cross-serial dependencies with finite feature licensing)**.** *Let $\Sigma_{\mathrm{feat}} = \{\mathtt{p}_0, \mathtt{p}_1, \mathtt{q}_0, \mathtt{q}_1, \mathtt{a}, \mathtt{b}, \mathtt{c}, \mathtt{d}\}$, let $g : \Sigma^*_{\mathrm{feat}} \to \mathbb{Z}_2^2$ map $\mathtt{p}_1 \mapsto (1,0)$, $\mathtt{q}_1 \mapsto (0,1)$ and all other letters to 0, and let $s := (s_{1,1}, g)|_{\Sigma^+_{\mathrm{feat}}}$ (corestricted to its image). For $R \subseteq \{0,1\}^2$ put*

$$L_R := \{\mathtt{a}^m \mathtt{p}_i \mathtt{b}^n \mathtt{c}^m \mathtt{q}_j \mathtt{d}^n : m, n \ge 1,\ (i,j) \in R\}.$$

Then for every R and every $f \ge 2$, $L_R \in \mathsf{MCFL}(2,2) \cap \mathcal{C}^{\mathrm{mcf}}_{f,s}$, and L_R is not context-free if $R \neq \emptyset$. For $\mathcal{R}_{\mathrm{feat}} = \{(0,0), (0,1), (1,0)\}$, the language $L_{\mathcal{R}_{\mathrm{feat}}}$ is neither in $\mathsf{YSub}(f)$ nor $(1, s_{1,1})$-tuple-substitutable. The typing is fixed once for the whole family.

Proof. Grammar. With fan-out-two A, B and rules $A \to (\mathtt{a}, \mathtt{c}) \mid (\mathtt{a}\xi^1_1, \mathtt{c}\xi^2_1)(A)$, $B \to (\mathtt{b}, \mathtt{d}) \mid (\mathtt{b}\xi^1_1, \mathtt{d}\xi^2_1)(B)$ and $S \to (\xi^1_1 \mathtt{p}_i \xi^1_2 \xi^2_1 \mathtt{q}_j \xi^2_2)(A, B)$ for $(i,j) \in R$, we get $L_A = \{(\mathtt{a}^m, \mathtt{c}^m)\}$, $L_B = \{(\mathtt{b}^n, \mathtt{d}^n)\}$ and $L(G) = L_R$; the grammar is good, of fan-out and rank two. For $(i,j) \in R$, intersecting with $\mathtt{a}^+ \mathtt{p}_i \mathtt{b}^+ \mathtt{c}^+ \mathtt{q}_j \mathtt{d}^+$ and erasing the markers gives the non-context-free $\{\mathtt{a}^m \mathtt{b}^n \mathtt{c}^m \mathtt{d}^n\}$.

Typed substitutability. Let $Q = \mathtt{a}^+\{\mathtt{p}_0,\mathtt{p}_1\}\mathtt{b}^+\mathtt{c}^+\{\mathtt{q}_0,\mathtt{q}_1\}\mathtt{d}^+$ (strictly 2-local), $K_R := \{w \in \Sigma^+_{\mathrm{feat}} : g(w) \in R\}$, and let φ map $\mathtt{a},\mathtt{c},\mathtt{b},\mathtt{d}$ to $(1,0),(-1,0),(0,1),(0,-1) \in \mathbb{Z}^2$ and the markers to 0. On Q, $g(w) = (i,j)$ for the markers $\mathtt{p}_i, \mathtt{q}_j$ of w, so $L_R = Q \cap K_R \cap \varphi^{-1}(0)$. The three languages are (f,s)-tuple-substitutable by Lemma 6.2, by item (c) after Definition 2.8 (K_R is a union of s-classes), and by Proposition 2.9(ii); Proposition 2.9(i) gives the intersection.

Failure without g. For $R = \mathcal{R}_{\mathrm{feat}}$, $x = \mathtt{ap_0b}$, $y = \mathtt{ap_1b}$, $E = \Box_1\mathtt{cq_0d}$ and $F = \Box_1\mathtt{cq_1d}$ we have $E[x], E[y], F[x] \in L_R$ but $F[y] \notin L_R$, since $(1,1) \notin R$. Here $s_{1,1}(x) = s_{1,1}(y)$ but $g(x) \neq g(y)$. The same four words violate the substitution implication of Definition 6.1 at $d = 1$, so $L_R \notin \mathsf{YSub}(f)$ for every $f \geq 1$. □

6.2 Separating the Typing Classes

The feature example is covered by a window and an abelian group. Two further fan-out-two targets separate the regimes of Section 4: L_{A} is admitted by a local-group typing but by no window–group-determined typing, and L_{B} by a window–S_3 typing but by no window–abelian one. Both have the form $\varsigma w\$x_\times$: a flag $\varsigma = \mathtt{r}$ leaves w unconstrained, $\varsigma = \mathtt{l}$ imposes a typing-dependent condition on w, and $x_\times \in C_\times := \{\mathtt{1}^m\mathtt{2}^n\mathtt{3}^m\mathtt{4}^n : m,n \geq 1\}$ makes the target non-context-free. Let $\varphi_\times$ map $\mathtt{1},\mathtt{3},\mathtt{2},\mathtt{4}$ to $(1,0),(-1,0),(0,1),(0,-1) \in \mathbb{Z}^2$ and all other letters to 0, and let $\#_{\mathtt{ab}}(w)$ be the number of occurrences of the factor $\mathtt{ab}$ in w.

Lemma 6.4 (Flag separation)**.** *Let s' be a typing, $u, v \in \Sigma^+$ with $s'(u) = s'(v)$, and $z \in \Sigma^*$. If L contains $\mathtt{r}uz$, $\mathtt{r}vz$ and $\mathtt{l}uz$ but not $\mathtt{l}vz$, then $L \notin \mathcal{C}^{\mathrm{mcf}}_{f,s'}$ for every $f \geq 1$.*

Proof. $\mathtt{r}\Box_1 z \in \mathcal{E}_1$ accepts u and v, while $\mathtt{l}\Box_1 z$ accepts u but not v; so $\mathcal{D}^{(1)}_L(u) \neq \mathcal{D}^{(1)}_L(v)$, and every (f,s') condition includes arity one. □

The targets. Let $\Sigma_{\mathrm{A}} = \{\mathtt{l},\mathtt{r},\mathtt{a},\mathtt{b},\mathtt{c},\$,\mathtt{1},\mathtt{2},\mathtt{3},\mathtt{4}\}$,

$$L_{\mathrm{A}} := \{\varsigma w\$x_\times : \varsigma \in \{\mathtt{l},\mathtt{r}\},\ w \in \{\mathtt{a},\mathtt{b},\mathtt{c}\}^+,\ x_\times \in C_\times,\ \varsigma = \mathtt{r} \text{ or } \#_{\mathtt{ab}}(w) \text{ even}\},$$

and $s_{\mathrm{A}} := \lambda_{1,\nu}|_{\Sigma^+_{\mathrm{A}}}$ with $\nu : \Sigma^2_{\mathrm{A}} \to \mathbb{Z}_2$, $\nu(\mathtt{ab}) = 1$ and $\nu = 0$ otherwise (Definition 4.18); it records the first and last letters and, for $|w| \geq 2$, the parity of $\#_{\mathtt{ab}}(w)$. Let $\Sigma_{\mathrm{B}} = \Sigma_{\mathrm{A}} \setminus \{\mathtt{c}\}$, $\alpha = (12)$, $\beta = (23) \in S_3$, $g_{\mathrm{B}} : \Sigma^*_{\mathrm{B}} \to S_3$ with $\mathtt{a} \mapsto \alpha$, $\mathtt{b} \mapsto \beta$ and all other letters $\mapsto 1$,

$$L_{\mathrm{B}} := \{\varsigma w\$x_\times : \varsigma \in \{\mathtt{l},\mathtt{r}\},\ w \in \{\mathtt{a},\mathtt{b}\}^+,\ x_\times \in C_\times,\ \varsigma = \mathtt{r} \text{ or } g_{\mathrm{B}}(w) = 1\},$$

and $s_{\mathrm{B}} := (s_{1,1}, g_{\mathrm{B}})|_{\Sigma^+_{\mathrm{B}}}$, corestricted. **Ab** is the pseudovariety of finite abelian groups, and $\mathbf{LI} \vee \mathbf{Ab}$ consists of the divisors of $\mathsf{T} \times \mathbb{G}$ with $\mathsf{T} \in \mathbf{LI}$ and $\mathbb{G}$ finite abelian.

Proposition 6.5 (A target separating local-group and window–group typings)**.** (i) $L_{\mathrm{A}} \in \mathsf{MCFL}(2,2) \setminus \mathrm{CFL}$.

(ii) $L_{\mathrm{A}} \in \mathcal{C}^{\mathrm{mcf}}_{f,s_{\mathrm{A}}}$ *for every $f \geq 2$, and* $s_{\mathrm{A}}(\Sigma^+_{\mathrm{A}}) \in \mathbf{LG}$.

(iii) *If $s' \preceq (s_{k,\ell}, g'|_{\Sigma^+_{\mathrm{A}}})$ for some k,ℓ and a morphism g' into a finite group (in particular if $s'(\Sigma^+_{\mathrm{A}}) \in \mathbf{LI} \vee \mathbf{Grp}$), then $L_{\mathrm{A}} \notin \mathcal{C}^{\mathrm{mcf}}_{f,s'}$ for all $f \geq 1$.*

Hence $s_{\mathrm{A}}(\Sigma^+_{\mathrm{A}}) \in \mathbf{LG} \setminus (\mathbf{LI} \vee \mathbf{Grp})$, and L_{A} is covered by Theorem 4.19 but, for no typing, by Theorem 4.15, even with Lemma 4.17.

Proposition 6.6 (A target separating nonabelian and window–abelian typings)**.** (i) $L_{\mathrm{B}} \in \mathsf{MCFL}(2,2) \setminus \mathrm{CFL}$.

(ii) $L_{\mathrm{B}} \in \mathcal{C}^{\mathrm{mcf}}_{f,s_{\mathrm{B}}}$ *for every* $f \geq 2$*, and* s_{B} *is window–group-determined with group* S_3*.*

(iii) *If* $s' \preceq (s_{k,\ell}, g'|_{\Sigma_{\mathrm{B}}^+})$ *for some* k, ℓ *and a morphism* g' *into a finite* abelian *group (in particular if* $s'(\Sigma_{\mathrm{B}}^+) \in \mathbf{LI} \vee \mathbf{Ab}$*), then* $L_{\mathrm{B}} \notin \mathcal{C}^{\mathrm{mcf}}_{f,s'}$ *for all* $f \geq 1$*.*

Hence L_{B} *is covered by Theorem 4.15 with the nonabelian constant* $c_{S_3,f} = M_{S_3,f} - 1$*, but for no typing by its trivial-group or abelian-group cases (Remark 4.16).*

Proof of Propositions 6.5 and 6.6. Write $\diamond \in \{\mathrm{A}, \mathrm{B}\}$, let $\Sigma_\diamond^\circ$ be $\{\mathtt{a}, \mathtt{b}, \mathtt{c}\}$ or $\{\mathtt{a}, \mathtt{b}\}$, and let $P_\diamond(w)$ be "$\#_{\mathtt{ab}}(w)$ even" or "$g_{\mathrm{B}}(w) = 1$".

(i) Let W be a right-linear (fan-out-one, rank-one, good) grammar for the regular language $\{\varsigma w : \varsigma \in \{\mathtt{l}, \mathtt{r}\}, w \in (\Sigma_\diamond^\circ)^+, \varsigma = \mathtt{r} \text{ or } P_\diamond(w)\}$; it tracks the last letter and the parity, respectively the value in S_3. Add

$$C \to (\xi_1^1\$\mathtt{1}, \mathtt{3})(W) \mid (\xi_1^1\mathtt{1}, \xi_1^2\mathtt{3})(C), \quad D \to (\mathtt{2}, \mathtt{4}) \mid (\xi_1^1\mathtt{2}, \xi_1^2\mathtt{4})(D), \quad S \to (\xi_1^1\xi_2^1\xi_1^2\xi_2^2)(C, D).$$

Then $L_C = \{(\varsigma w\$\mathtt{1}^m, \mathtt{3}^m)\}$, $L_D = \{(\mathtt{2}^n, \mathtt{4}^n)\}$, and S generates $L_\diamond$; the grammar is good (adjacent variables in the start template belong to different children). Intersecting with $\mathtt{ra}\$\mathtt{1}^+\mathtt{2}^+\mathtt{3}^+\mathtt{4}^+$ and erasing $\mathtt{ra}\$$ gives the non-context-free $C_\times$.

(ii) $L_\diamond = Q_\diamond \cap \Pi_\diamond \cap \varphi_\times^{-1}(0)$ with $Q_\diamond = \{\mathtt{l}, \mathtt{r}\}(\Sigma_\diamond^\circ)^+\$\mathtt{1}^+\mathtt{2}^+\mathtt{3}^+\mathtt{4}^+$ and $\Pi_\diamond$ the set of nonempty words that begin with $\mathtt{r}$, or begin with $\mathtt{l}$ and satisfy $P_\diamond$ for the whole word. On $Q_\diamond$ the condition $P_\diamond$ of the whole word is that of w: no factor $\mathtt{ab}$ meets ς, \$ or the digits, and g_{B} is trivial on all other letters. $Q_\diamond$ is strictly 2-local (its pairs are $\vdash \varsigma$, ςx, xy, $x\$$, $\$\mathtt{1}$, $\mathtt{11}, \mathtt{12}, \mathtt{22}, \mathtt{23}, \mathtt{33}, \mathtt{34}, \mathtt{44}$, $\mathtt{4} \dashv$ for $\varsigma \in \{\mathtt{l}, \mathtt{r}\}$, $x, y \in \Sigma_\diamond^\circ$) and $s_{1,1} \preceq s_\diamond$, so Lemma 6.2 applies. $\Pi_\diamond$ is a union of $s_\diamond$-classes: $s_\diamond$ records the first letter, and for words of length at least two the parity of $\#_{\mathtt{ab}}$, respectively the value of g_{B} (the one-letter words $\mathtt{l}, \mathtt{r}$ lie in $\Pi_\diamond$); so item (c) after Definition 2.8 applies. Proposition 2.9 handles $\varphi_\times^{-1}(0)$ and the intersection; with (i), $L_\diamond \in \mathcal{C}^{\mathrm{mcf}}_{f,s_\diamond}$ for $f \geq 2$. s_{A} is local-group-determined, so its positive image lies in $\mathbf{LG}$ by Theorem B.4.

(iii) for A. Let $\hat{k} = \max\{k, \ell, 1\}$, $q \geq 2$ an even multiple of the exponent of the group of g', $u = \mathtt{a}^{\hat{k}}(\mathtt{abc})^q\mathtt{a}^{\hat{k}}$ and $v = \mathtt{a}^{\hat{k}}(\mathtt{bac})^q\mathtt{a}^{\hat{k}}$. Both are long for $s_{k,\ell}$ with prefix $\mathtt{a}^k$ and suffix $\mathtt{a}^\ell$, and $g'(u) = g'(\mathtt{a})^{2\hat{k}} = g'(v)$; so $s'(u) = s'(v)$. But $\#_{\mathtt{ab}}(u) = q$ is even and $\#_{\mathtt{ab}}(v) = 1$ (only at $\mathtt{a}^{\hat{k}}\mathtt{b}$). With $z = \$\mathtt{1234}$, Lemma 6.4 applies. A typing with positive image in $\mathbf{LI} \vee \mathbf{Grp}$ satisfies the hypothesis by Lemma 4.17.

(iii) for B. Let $\hat{k} \geq \max\{k, \ell, 1\}$ be even, $u = \mathtt{a}^{\hat{k}}(\mathtt{ab})^3\mathtt{a}^{\hat{k}}$ and $v = \mathtt{a}^{\hat{k}}(\mathtt{ab})^2\mathtt{ba}\,\mathtt{a}^{\hat{k}}$. They have the same prefix $\mathtt{a}^k$, suffix $\mathtt{a}^\ell$ and Parikh vector, so $s'(u) = s'(v)$ for abelian g'. Since $\alpha^2 = \beta^2 = 1$, $\alpha\beta$ has order 3 and $\beta\alpha = (\alpha\beta)^{-1}$, we get $g_{\mathrm{B}}(u) = (\alpha\beta)^3 = 1$ and $g_{\mathrm{B}}(v) = (\alpha\beta)^2(\alpha\beta)^{-1} = \alpha\beta \neq 1$. With $z = \$\mathtt{1234}$, Lemma 6.4 applies. For $\mathbf{LI} \vee \mathbf{Ab}$, the proof of Lemma 4.17 with an abelian group factor gives an abelian g'.

Consequences. If $s_{\mathrm{A}}(\Sigma_{\mathrm{A}}^+)$ were in $\mathbf{LI} \vee \mathbf{Grp}$, (iii) with $s' = s_{\mathrm{A}}$ would contradict (ii). Theorem 4.15 and its trivial-group and abelian cases assume a typing of the excluded kind. □

6.3 Why the fan-out-one lower bound does not lift

Lemma 6.7 (Power transfer across two tuple components)**.** *Let* $f \geq 2$*,* $L \in \mathcal{C}^{\mathrm{mcf}}_{f,s}$*,* $u \in \Sigma^+$*, and let* $i_0, q \geq 1$ *be an eventual index and period of* $s(u)$*, i.e.* $s(u)^{i+q} = s(u)^i$ *for* $i \geq i_0$*. Suppose* $a, b \geq i_0$*,* $d \in q\mathbb{Z}_{>0}$*, and some word of* L *contains the factor* $u^{a+d+b+2}$*. Then for all* $w_{\mathrm{L}}, w_{\mathrm{R}} \in \Sigma^*$ *and* $w_{\mathrm{M}} \in \Sigma^+$*,*

$$w_{\mathrm{L}}u^a w_{\mathrm{M}} u^{d+b} w_{\mathrm{R}} \in L \quad \Longleftrightarrow \quad w_{\mathrm{L}}u^{a+d} w_{\mathrm{M}} u^b w_{\mathrm{R}} \in L.$$

Proof. Write the given word as $p'u^{a+d+b+2}q'$. The tuples $\vec{x} = (u^a, u^{d+b})$ and $\vec{y} = (u^{a+d}, u^b)$ have the same componentwise s-types by eventual periodicity. They also have the common admissible accepting context $E = p'u\Box_1 u\Box_2 q'$, since $E[\vec{x}] = E[\vec{y}] = p'u^{a+d+b+2}q' \in L$. The (f, s)-condition thus equates their two-component distributions. Test them in the admissible context $w_{\mathrm{L}}\Box_1 w_{\mathrm{M}}\Box_2 w_{\mathrm{R}}$; the condition $w_{\mathrm{M}} \neq \varepsilon$ is essential. □

Theorem 6.8 (The bracketed doubling family does not lift)**.** *Let $s : \Gamma_{\mathrm{br}}^{+} \twoheadrightarrow \mathsf{S}$ be any fixed finite-semigroup typing and let L_n be the family of Proposition 4.20. If $n \geq 2|\mathsf{S}| + 3$, then $L_n \notin \mathcal{C}_{f,s}^{\mathrm{mcf}}$ for every $f \geq 2$.*

Proof. Choose an eventual index $i_0 \geq 1$ and period $q \geq 1$ for $s(\lfloor)$, with $i_0 + q \leq |\mathsf{S}| + 1$. Thus $n \geq 2i_0 + q + 2$. Put $a = i_0$, $d = q$, $b = n - a - d - 2 \geq i_0$. The unique ceiling-free word $W_n = \lfloor W_{n-1}W_{n-1}\rfloor \in L_n$ starts with exactly n occurrences of $\lfloor$, so contains the factor $\lfloor^{a+d+b+2}$. In this same W_n, let $\lfloor^a$ be the a letters immediately after the first $\lfloor$, and let $\lfloor^{d+b}$ be the last $d + b$ letters of the initial run of the second copy of W_{n-1}. This is possible since $d + b = n - a - 2 \leq n - 1$. We can write

$$W_n = w_{\mathrm{L}}\lfloor^a w_{\mathrm{M}}\lfloor^{d+b} w_{\mathrm{R}},$$

where $w_{\mathrm{L}} = \lfloor$ and $w_{\mathrm{M}} \neq \varepsilon$ contains the first $\mathtt{a}$. By Lemma 6.7, membership in L_n would force $w_{\mathrm{L}}\lfloor^{a+d} w_{\mathrm{M}}\lfloor^b w_{\mathrm{R}} \in L_n$. The resulting word is ceiling-free, but it begins with $\lfloor^{n+d}\mathtt{a}$ rather than $\lfloor^n\mathtt{a}$. This contradicts uniqueness of the ceiling-free member W_n (Proposition 4.20(iii)). Higher f include the arity-two substitutability condition. □

Remark 6.9. Power transfer preserves the total length and Parikh vector of the word. It does not prove that all rare-type representatives are short; rather it shows why the particular exponential lower-bound family of Proposition 4.20 is excluded by every fixed finite typing at fan-out two.

6.4 A typing outside local groups with polynomial data at fan-out two

At fan-out one, typings onto U_1 with two letters of type 1 need exponential characteristic data (Corollary 4.25). At fan-out two the arity-two condition is strong enough to remove this obstruction for the binary typing. Let $\Sigma = \{\mathtt{a}, \mathtt{b}\}$ and let $s_{U,2} : \Sigma^+ \twoheadrightarrow U_1$ send $\mathtt{a}$ to 1 and $\mathtt{b}$ to ζ, so that a word has type ζ exactly when it contains $\mathtt{b}$.

Theorem 6.10 (Polynomial data for the binary typing onto U_1)**.** *Let $r_{\max} \geq 2$ and put $C_{r_{\max}} := 42r_{\max}(r_{\max} + 1) + 96(r_{\max} + 1)$. For every $L \in \mathcal{C}_{2,s_{U,2}}^{\mathrm{mcf}}$ and every reduced good fan-out-≤ 2, rank-$\leq r_{\max}$ presentation G of $L \cap \Sigma^+$, the operators $\mathcal{B}_{2,s_{U,2}}^{\leq r_{\max}}$ and $\mathcal{B}_{2,s_{U,2}}$ have a common characteristic sample K with*

$$\|K\|_+ \leq 3^{2r_{\max}}(3 + r_{\max}C_{r_{\max}})\,\|G\|^3\bar{\theta}_G^5 + \mathbf{1}_{\{\varepsilon\in L\}}.$$

In particular, the first question of Open Problem 7.1 has a positive answer at $f = 2$ for every $r_{\max} \geq 2$, although $U_1 \notin \mathbf{LG}$.

Outline. Say that L has a *deep insertion* if some word of L has a factor $\mathtt{a}^6 z \mathtt{a}^6$ with z containing $\mathtt{b}$. Without one, marking the first and last six letters of every pure root component keeps the type of every shortened tuple and bounds the representative length τ by $O(\|G\|\bar{\theta}_G)$ (Lemma H.3). With one, two-hole contexts around the run $\mathtt{a}^6$ identify $(\mathtt{a}^i, \mathtt{a}^j)$ and $(\mathtt{a}^{i'}, \mathtt{a}^{j'})$ with $i + j = i' + j'$, which forces a factor $\mathtt{abaa}$ and then $\mathcal{D}(\mathtt{ab}) = \mathcal{D}(\mathtt{ba})$, so L is Parikh-saturated (Lemma H.5). For Parikh-saturated targets an analysis of node-minimal derivations bounds τ by $O(\|G\|^2\bar{\theta}_G^5)$ (Lemma H.6), and Corollary 4.10 converts the bound on τ into the sample bound. The details are in Appendix H. □

Remark 6.11 (Scope of Theorem 6.10). (a) The theorem concerns one typing and fan-out two. It does not decide the pseudovariety question at fan-out two: whether typings onto U_1 with two letters of type 1, such as the typing of Corollary 4.25, admit polynomial data at fan-out two is open, and so is whether the binary typing needs superpolynomial data at fan-out one (Remark 4.23(b)). (b) The proof uses the arity-two condition only through runs of the single neutral letter; with two neutral letters a long pure factor need not contain a long run. Lemma H.3 holds for every typing onto U_1, so any superpolynomial family for such a typing must contain deep insertions whose pure flanks grow superpolynomially. (c) The proof treats tuples of arity at most two; fan-out $f \geq 3$ is open.

7 Conclusion

For a reduced good presentation G and fixed alphabet, fan-out, rank and typing, Theorem 4.4 reduces the total encoded length of characteristic samples to the shortest typed yield $t_{\mathrm{typ}}(G, s)$ and to $N_{\mathrm{typ}} = O(\|G\|)$ witness words. Owner-wise simultaneous replacement (Proposition 4.12) then gives $O(\|G\|^2 \bar{\theta}_G)$ for window–group typings, for noncommutative groups and interleaved tuple components alike (Theorem 4.15), and the same order $O(\|G\|^2 \bar{\theta}_G)$ for local-group typings, after thinning and padding the owners (Theorem 4.19); rank one gives $O(\|G\|^3)$ under every typing (Theorem 5.1). For characteristic-data length itself, the rare-type lower bound (Proposition 4.6) makes **LG** an exact boundary at fan-out one, at the level of pseudovarieties (Theorem 4.22) and for every typing that realizes U_1 by letters as in Corollary 4.25, and at fan-out one the product $\|G\|\bar{\theta}_G$ is necessary under every typing (Proposition 4.8). At fan-out two the exponential family behind that boundary is not tuple-substitutable under any fixed finite typing (Theorem 6.8); moreover the binary typing onto U_1 admits polynomial characteristic data there (Theorem 6.10), so local groups are not necessary at fan-out two for individual typings.

Open Problem 7.1 (Characteristic-data length at higher fan-out). Fix $f \geq 2$ and $r_{\max} \geq 2$. For which typings s with $\mathsf{S} \notin \mathbf{LG}$ is there a polynomial p such that, for every $L \in \mathcal{C}^{\mathrm{mcf}}_{f,s}$ and every reduced good fan-out-$\leq f$, rank-$\leq r_{\max}$ presentation G of $L \cap \Sigma^+$, the operator $\mathcal{B}^{\leq r_{\max}}_{f,s}$ has a characteristic sample of total length at most $p(\|G\|, \bar{\theta}_G)$? At $f = 2$ the binary typing onto U_1 is such a typing (Theorem 6.10); typings onto U_1 with two letters of type 1, which need exponential data at fan-out one (Corollary 4.25), and all $f \geq 3$ are open. Conversely, for a typing with positive image U_1, are there targets $L_n \in \mathcal{C}^{\mathrm{mcf}}_{f,s}$ with $\|G_n\|\bar{\theta}_{G_n} = n^{O(1)}$ whose characteristic samples need length $n^{\omega(1)}$? By Proposition 4.6, a family with $\mu^\star(L_n, s) = n^{\omega(1)}$ would suffice, for every set-driven operator with characteristic samples throughout the class; Theorem 6.8 excludes the bracketed doubling family. By Corollary 4.10, a positive answer to the first question would follow from a polynomial bound on the representative length $\tau(G, s)$, which lies between $\mu^\star(L, s)$ and $t_{\mathrm{typ}}(G, s)$.

Tightness of the exponents. In $O(\|G\|^2 \bar{\theta}_G)$, for window–group and local-group typings alike, one factor $\|G\|$ counts the witness words and the other bounds root-to-leaf paths in a λ-minimal derivation by the number of node labels (Lemma 4.11). The path factor is genuine for the witness construction: the grammars $S \to SS \mid \mathtt{a} \mid \mathtt{b} \mid A_0 \mid \cdots \mid A_n$, $A_0 \to \mathtt{b}$, $A_i \to \mathtt{a} \mid A_{i-1}\mathtt{b}$ $(1 \leq i \leq n)$ generate $\{\mathtt{a}, \mathtt{b}\}^+$ with $\|G_n\| = 10n + 16$ and $\bar{\theta}_{G_n} = 2$, but $\mathtt{b}^{n+1}$ is the only yield of A_n of $s_{1,1}$-type $\langle \mathtt{b}, \mathtt{b} \rangle$, so $t_{\mathrm{typ}}(G_n, s_{1,1}) \geq n + 1$. For this target, however, $\tau(G_n, s_{1,1}) \leq 2$ (Corollary 4.10), so the example bounds the witness method, not the data. At fan-out one the product $\|G\|\bar{\theta}_G$ is necessary under every fixed typing, even for asymptotically smallest presentations and for every set-driven operator with characteristic samples throughout $\mathcal{C}^{\mathrm{mcf}}_{1,s}$ (Proposition 4.8); there the remaining gap is a single factor $\|G\|$ for every typing with $\mathsf{S} \in \mathbf{LG}$. At fan-out two or more that family is not

tuple-substitutable, and the known examples force only $\Omega(\bar{\theta}_G)$: the size-$O(n)$ grammar for $\{\mathsf{c}^{2^n}\}$ of Section 5 has $\bar{\theta}_G = 2^n$, which leaves a factor $\|G\|^2$. These are gaps between the known examples and the bounds, not proved ratios. Neither lower-bound argument closes them: for a fixed typing the rare-type argument yields at most $|\mathsf{S}|(t_{\mathrm{typ}}+1)$, and the family of Proposition 4.8 consists of $O(\|G\|)$ words of length $\bar{\theta}_G = t_{\mathrm{typ}}$. For noncommutative groups the constant $M_{\mathbb{G},f}$ is doubly exponential in f because of the Ramsey-type homogenization (Lemma A.8); whether $\mathrm{poly}(|\mathbb{G}|^f)$ owners suffice, as for abelian groups, is open.

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Data availability

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A Simultaneous Block Replacement in Finite Groups

This appendix proves Proposition 4.12, used in Lemmas 4.13 and B.7. The abelian case is a zero-sum argument. The general case never commutes factors; it has three parts: an order classification for homogeneous point configurations (Lemma A.4), an interval selection by boundary states (Lemma A.5), and a Ramsey-type extraction of homogeneous indices (Lemmas A.7 and A.8).

Definition A.1 (Block-replacement family)**.** Let $\mathbb{G}$ be a finite group and $f \geq 1$. A *block-replacement family* of width f over $\mathbb{G}$ with index set $[m]$ consists of $d \leq f$ *lines*; on line ϱ, a left-to-right sequence of blocks $\beta_{\varrho,1}, \dots, \beta_{\varrho,q_\varrho}$ and constants $a_{\varrho,0}, \dots, a_{\varrho,q_\varrho} \in \mathbb{G}$; for every block β an *owner* $o(\beta) \in [m]$ and two values $x_\beta, x'_\beta \in \mathbb{G}$; every index owns at most f blocks. For $J \subseteq [m]$ put

$$\mathrm{val}_\varrho(J) := a_{\varrho,0}\, y_{\varrho,1}\, a_{\varrho,1} \cdots y_{\varrho,q_\varrho}\, a_{\varrho,q_\varrho}, \qquad y_{\varrho,j} := \begin{cases} x'_{\beta_{\varrho,j}}, & o(\beta_{\varrho,j}) \in J, \\ x_{\beta_{\varrho,j}}, & o(\beta_{\varrho,j}) \notin J. \end{cases}$$

Lemma A.2 (Block replacement)**.** *In a block-replacement family of width f over $\mathbb{G}$ there is a nonempty $J \subseteq [m]$ with $\mathrm{val}_\varrho(J) = \mathrm{val}_\varrho(\emptyset)$ for every line ϱ, provided that either* (a) *$\mathbb{G}$ is abelian and $m > \mathsf{d}(\mathbb{G}^f)$, or* (b) *$\mathbb{G}$ is arbitrary and $m \geq M_{\mathbb{G},f}$, with d and $M_{\mathbb{G},f}$ as in Section 4.3.*

Proof of (a). Write $\mathbb{G}$ additively. By commutativity $\mathrm{val}_\varrho(J) - \mathrm{val}_\varrho(\emptyset) = \sum_{i\in J} \delta_{i,\varrho}$, where $\delta_{i,\varrho}$ is the sum of $x'_\beta - x_\beta$ over the blocks β of line ϱ owned by i. The sequence $(\delta_i)_{i\in[m]}$ with $\delta_i := (\delta_{i,\varrho})_\varrho \in \mathbb{G}^d$ has length $m > \mathsf{d}(\mathbb{G}^f) \geq \mathsf{d}(\mathbb{G}^d)$ ($\mathbb{G}^d$ embeds in $\mathbb{G}^f$), so it has a nonempty zero-sum subsequence $(\delta_i)_{i\in J}$. □

Remark A.3 (Size of the zero-sum constant). The prefix-sum pigeonhole gives $\mathsf{d}(\mathbb{G}^f) \leq |\mathbb{G}|^f - 1$, but much more is true. Write $\mathbb{G} \cong C_{q_1} \oplus \cdots \oplus C_{q_\ell}$ and $q = \exp(\mathbb{G}) \geq 2$. Then

$$f\sum_{i=1}^{\ell}(q_i - 1) \;\leq\; \mathsf{d}(\mathbb{G}^f) \;\leq\; (q-1) + q\log\frac{|\mathbb{G}|^f}{q},$$

where the lower bound is attained by a basis sequence and the upper bound is due to van Emde Boas and Kruyswijk [31, Proposition 5.1.8 and Theorem 5.5.5]; for a p-group the lower bound is an equality [31, Theorem 5.5.9]. Hence $\mathsf{d}(\mathbb{G}^f) = O(\exp(\mathbb{G})\, f \log|\mathbb{G}|)$; for $\mathbb{G} = C_2$ the antichain bound drops from $2^f - 1$ to f. Since zero-sum-free sequences of length $\mathsf{d}(\mathbb{G}^d)$ exist, the argument cannot do better; we do not claim that a grammar realizes them.

A.1 Arbitrary Finite Groups: Order Homogenization

Lemma A.4 (A homogeneous block-order lemma)**.** *Let $e \geq 1$ and $n \geq 3$, and let (t,i), $t \in [e]$, $i \in [n]$, be distinct points on a line. Suppose that* (i) *the order of (t,i) and (u,i) does not depend on i, and* (ii) *for each ordered pair (t,u), including $t = u$, the order of (t,i) and (u,j) is the same for all $i < j$. Then $[e]$ is partitioned into classes such that: the points of each class occupy a consecutive region containing no other points; each class has a direction (increasing or decreasing in i); and its region consists of n successive* chunks*, one per index in that direction, each containing exactly one point of every type of the class, in an order independent of the index.*

Proof. By (ii) with $t = u$, the points of each type are monotone in i. For distinct t, u let X mean $(t,i) < (u,j)$ and Y mean $(u,i) < (t,j)$ for $i < j$; both are constant by (ii).

Both increasing. $X \wedge Y$ puts all points of index i of both types before those of index $j > i$ (*interleaved by rank*). Under $X \wedge \neg Y$ all comparisons of different indices put t left of u, and so do equal indices, since $(u,i) < (t,i)$ would give $(t,1) < (t,2) < (u,1) < (t,1)$; so t and u are *separated*. $\neg X \wedge Y$ is symmetric, and $\neg X \wedge \neg Y$ gives $(t,1) < (t,2) < (u,1) < (u,2) < (t,1)$.

t increasing, u decreasing. $X \wedge Y$ gives $(t,2) < (u,3) < (u,1) < (t,2)$ and $\neg X \wedge \neg Y$ gives $(t,3) < (u,2) < (t,1) < (t,3)$. Under $X \wedge \neg Y$ the comparisons of different indices put t left of u, and $(u,i) < (t,i)$ would give $(t,1) < (u,2) < (u,1) < (t,1)$; so they are separated, and $\neg X \wedge Y$ is symmetric. Reversing indices covers two decreasing types.

Thus two types are separated, or have the same direction and are interleaved. Interleaving is transitive: if t, u and u, v are interleaved (all increasing, say) but t, v are separated, then $(v,1) < (u,2) < (t,3) < (v,1)$ if t is left of v, and $(t,1) < (u,2) < (v,3) < (t,1)$ otherwise. So interleaving is an equivalence relation. A type outside a class is separated from every type of the class and lies on the same side of all of them, since otherwise two interleaved types would be separated; hence class regions contain no other points. Inside a class, interleaving gives the chunks, and (i) gives the within-chunk order. □

Lemma A.5 (Simultaneous finite-group interval shortening)**.** *Let $\mathbb{G}$ be a finite group and consider a block-replacement family of width f over $\mathbb{G}$ with $n \geq \max\{3, |\mathbb{G}|^f\}$ indices, in which all indices have the same number of blocks, on the same lines, in the same within-index order, and in which, for every two block positions t, u on a common line, the truth value of "block t of index i lies left*

of block u of index j" is the same for all $i < j$. Then a nonempty interval J of indices satisfies $\mathrm{val}_\varrho(J) = \mathrm{val}_\varrho(\emptyset)$ *for every line ϱ.*

Proof. Apply Lemma A.4 on each line; there are $c \le f$ class regions in total. Fix a region $\mathcal{Q}$ and number its chunks $1, \dots, n$ from left to right. Assign constants between chunks to the chunk on their left, and absorb fixed constants at the ends of the region into its first or last chunk. Each chunk has original and replacement products with the *same* fixed constants in the same order. Let P_p be the original value of chunks $1, \dots, p$ and Q_p their value when all of them are replaced ($P_0 = Q_0 = 1_{\mathbb{G}}$), and put $\kappa_p^{\mathcal{Q}} := P_p Q_p^{-1}$. Replacing exactly chunks $p+1, \dots, p'$ gives the ordered value $P_p\,(Q_p^{-1}Q_{p'})\,(P_{p'}^{-1}P_n)$, which equals P_n iff $\kappa_p^{\mathcal{Q}} = \kappa_{p'}^{\mathcal{Q}}$; no factors are commuted. If $\mathcal{Q}$ is increasing, the owners in $(a, b]$ occupy exactly chunks $(a, b]$; if it is decreasing, chunks $(n-b, n-a]$. For $j = 0, \dots, n$ let the state be the vector in $\mathbb{G}^c$ whose $\mathcal{Q}$-entry is $\kappa_j^{\mathcal{Q}}$ for increasing and $\kappa_{n-j}^{\mathcal{Q}}$ for decreasing regions. Since $n + 1 > |\mathbb{G}|^c$, two states coincide, at $a < b$. Replacing the owners $a+1, \dots, b$ then preserves every region value, and nothing outside the regions changes. □

Example A.6 (Opposite orientations and noncommuting values). In S_3 let $\alpha = (12)$, $\beta = (23)$. Three owners have blocks U_i, V_i on row 1 and W_i on row 2:

$$\text{row 1: } (U_1\beta V_1)\,\alpha\,(U_2\beta V_2)\,\alpha\,(U_3\beta V_3), \qquad \text{row 2: } W_3\,\beta\,W_2\,\beta\,W_1,$$

so row 1 is one increasing class and row 2 one decreasing class. Originally $U_i = \beta$, $V_i = W_i = 1$, and both rows have value 1. Let owner 1 have the trivial replacement $U_1' = U_1$, $W_1' = W_1$, and replace exactly owners $J = \{2, 3\}$ by $U_2' = 1$, $W_2' = \beta\alpha\beta$, $U_3' = \alpha\beta\alpha\beta$, $W_3' = \alpha$, $V_i' = 1$. Then row 1 becomes $\alpha \cdot \beta\alpha \cdot \alpha\beta\alpha = 1$ and row 2 becomes $\alpha\beta \cdot \beta\alpha\beta \cdot \beta = 1$, computed in order. In the notation of the proof, $(a, b] = (1, 3]$ and $\kappa_1 = \kappa_3 = 1$ on row 1, while $\kappa_{3-b} = \kappa_0 = 1 = \kappa_2 = \kappa_{3-a}$ on row 2: one owner interval matches two different boundary-index orientations, although neither replaced owner preserves its own contribution.

Lemma A.7 (Two monotone sequences)**.** *Let $m \ge 3$, $m' \ge (m-1)^2 + 1$ and $\mathrm{lt} : \{(i, j) : 1 \le i < j \le m'\} \to \{0, 1\}$ be monotone in each argument with fixed directions: for each fixed i it is monotone in j, for each fixed j monotone in i, and each of these two directions is the same for all values of the fixed argument. Then* lt *is constant on the pairs $i < j$ of some $I \subseteq [m']$ with $|I| = m$.*

Proof. Replacing lt by $1 - \mathrm{lt}$ if necessary, rows are nondecreasing in j. It suffices to find an index set of size at least m, since constancy persists in subsets.

If columns are nonincreasing in i, let $J(i) := \min\{j > i : \mathrm{lt}(i, j) = 1\}$ (∞ if none); J is nondecreasing. The chain $i_1 = 1$, $i_{r+1} = J(i_r)$ satisfies $\mathrm{lt}(i_a, i_b) = 1$ for $a < b$; if it has m terms we are done. Otherwise its at most $m - 1$ terms cut $[m']$ into at most $m - 1$ intervals $[i_r, i_{r+1})$, $[i_L, m']$, one of which has at least $m'/(m-1) > m - 1$, hence at least m, indices, and inside it $j < J(i_r) \le J(i)$ gives $\mathrm{lt}(i, j) = 0$.

If columns are nondecreasing, let κ be the least index with $\mathrm{lt}(\kappa, \kappa + 1) = 1$ ($\kappa = m'$ if none). Then $\mathrm{lt}(i, j) = 1$ for $\kappa \le i < j$ and $\mathrm{lt}(i, j) \le \mathrm{lt}(j - 1, j) = 0$ for $i < j \le \kappa$, and one of $[1, \kappa]$, $[\kappa, m']$ has at least $(m' + 1)/2 \ge m$ indices because $(m-1)^2 + 1 \ge 2m - 1$ for $m \ge 3$. □

Lemma A.8 (Polynomial order homogenization)**.** *Let $N \ge 3$, $m_0 := N$ and $m_{\ell+1} := (m_\ell - 1)^2 + 1$, so $m_k \le N^{2^k}$. Let $\mathcal{T}$ be a finite set of types, each assigned to a line, and for $i \in [n]$ and $t \in \mathcal{T}$ let $p_t(i)$ be a point on the line of t, the points on each line being distinct. Let $\mathcal{O}$ be the set of ordered pairs (t, u), $t = u$ allowed, of types on a common line, and $k := |\mathcal{O}|$. If $n \ge m_k$, there is $I \subseteq [n]$ with $|I| = N$ such that for every $(t, u) \in \mathcal{O}$ the truth value of $p_t(i) < p_u(j)$ is the same for all $i < j$ in I.*

Proof. Treat the pairs one at a time, the pairs (t,t) first; each step passes from an index set of size at least $(m-1)^2+1$ to a subset of size at least m, and earlier homogeneity persists in subsets. For (t,t), Erdős–Szekeres gives a monotone subsequence of length m. For $t \neq u$ (both now monotone), relabel the current indices $1,\dots,m'$ and put $\mathrm{lt}(i,j) := [p_t(i) < p_u(j)]$ for $i<j$. For fixed i, lt is monotone in j in the direction of p_u; for fixed j, monotone in i in the direction opposite to p_t. Since $m \geq m_0 = N \geq 3$, Lemma A.7 applies. □

Proof of Lemma A.2(b). An index owning no block can be chosen alone, so assume every index owns a block. The *pattern* of an index records the number of its blocks, the line of each, and their order on shared lines; there are at most $f \cdot f^f \cdot f! = \mathcal{P}_f$ patterns, so at least $N_{\mathbb{G},f}^{2^{f^2}}$ indices share one. Absorb the blocks of all other indices into the constants: these indices are never replaced, so their blocks keep their original values in every row. The blocks of the common pattern give at most f types and at most f^2 ordered pairs on common lines, so Lemma A.8 with $N = N_{\mathbb{G},f}$ yields $N_{\mathbb{G},f}$ indices with homogeneous order comparisons. The blocks of the indices discarded in this extraction are absorbed into the constants in the same way. The remaining $N_{\mathbb{G},f}$ indices then satisfy the hypotheses of Lemma A.5, whose interval is the required J. □

For fixed f, $M_{\mathbb{G},f} = \mathcal{P}_f \max\{3,|\mathbb{G}|^f\}^{2^{f^2}}$ is polynomial in $|\mathbb{G}|$; its degree $f \cdot 2^{f^2}$ is exponential in f^2, so $M_{\mathbb{G},f}$ itself is doubly exponential in f.

B Local-Group Coordinates and MCFG Block Shortening

LG is always tested on the positive image $\mathsf{S} = \widehat{s}(\Sigma^+)$. The classical identity $\mathbf{LG} = \mathbf{Grp} * \mathbf{D}$ places it within the sliding-window and delay theory [33, 34], and the exclusion of $U_1 = \{1,\zeta\}$ ($\zeta^2 = \zeta$) is [32, Proposition 2.3]. The coordinates below are a self-contained implementation of this algebra, not a new characterization; what is specific to MCFG learning is the factorization of group values into fixed separators and replaceable blocks (Lemma B.2). Throughout, $k \geq 1$, $\mathbb{G}$ is a finite group, $\lambda_{k,\nu}$ is as in Definition 4.18, a word is *short* if $|w| < 2k$ (recorded exactly by $\widehat{s}_{k,k}$) and *long* otherwise, and for $x,y \in \Sigma^k$ we put $J_\nu(x,y) := \prod_{i=1}^{k} \nu\big((xy)[i..i+k]\big)$, the ordered product over the k factors of length $k+1$ of xy that start in x.

Lemma B.1 (Local-group typings are morphisms)**.** *$\lambda_{k,\nu}(uv)$ is a function of $\lambda_{k,\nu}(u)$ and $\lambda_{k,\nu}(v)$; hence $\lambda_{k,\nu}$ is a morphism onto a finite monoid $M_{k,\nu}$ with at most $|\Sigma|^{<2k} + |\Sigma|^{2k}|\mathbb{G}|$ elements. For long u,v,*

$$\lambda_{k,\nu}(uv) = \big(\langle \mathrm{pref}_k\, u, \mathrm{suf}_k\, v\rangle,\ \gamma_{k,\nu}(u)\, J_\nu(\mathrm{suf}_k\, u, \mathrm{pref}_k\, v)\, \gamma_{k,\nu}(v)\big).$$

Proof. The window component is a morphism. For long u,v the factors of length $k+1$ of uv are, in order, those of u, the k factors starting in the last k positions of u (inside $\mathrm{suf}_k(u)\,\mathrm{pref}_k(v)$), and those of v. If u is short and v long, the factors starting in u lie inside $u\,\mathrm{pref}_k(v)$ and the others are those of v; the case of long u and short v is symmetric; and if both are short, uv is determined exactly. □

Lemma B.2 (Block decomposition)**.** *Let $w = x_0\beta_1 x_1 \cdots \beta_q x_q$ and $w' = x_0\beta'_1 x_1 \cdots \beta'_q x_q$ with $\widehat{s}_{k,k}(\beta'_j) = \widehat{s}_{k,k}(\beta_j)$ for all j. Then $\widehat{s}_{k,k}(w') = \widehat{s}_{k,k}(w)$, and if w is long there are $z_0,\dots,z_q \in \mathbb{G}$, depending only on the x_j and the window types of the β_j, with*

$$\gamma_{k,\nu}(w) = z_0\,\gamma_{k,\nu}(\beta_1)\,z_1 \cdots \gamma_{k,\nu}(\beta_q)\,z_q, \qquad \gamma_{k,\nu}(w') = z_0\,\gamma_{k,\nu}(\beta'_1)\,z_1 \cdots \gamma_{k,\nu}(\beta'_q)\,z_q.$$

Proof. The window claim holds because $\widehat{s}_{k,k}$ is a morphism. A short block equals its replacement and is absorbed into the fixed words, so let all blocks be long. A factor of length $k+1$ either lies inside some block or meets every block it touches in a prefix or suffix of length at most k; factors of the second kind are spelled by the fixed words and the first and last k letters of the blocks, so they are the same in w and w' and occur in the same order. Ordered by starting position, the factors inside β_j form one run with product $\gamma_{k,\nu}(\beta_j)$; let z_j be the ordered product of the factors between consecutive runs. □

Lemma B.3 (Sliding-factor coordinates)**.** *Let* S *be a finite semigroup whose idempotents all lie in its minimal ideal* $\mathsf{K} = \mathcal{M}(\mathbb{H}; I, \Lambda; \mathsf{P})$ *(a Rees matrix semigroup with* P *over* $\mathbb{H}$*), let* $k := |\mathsf{S}|$*, and let* $h : \Sigma^+ \to \mathsf{S}$ *be a morphism. There is* $\nu : \Sigma^{k+1} \to \mathbb{H}$ *such that every* w *with* $|w| \geq 2k$ *satisfies*

$$h(w) = \big(\mathrm{row}(h(\mathrm{pref}_k\, w)),\ \mathrm{grp}(h(\mathrm{pref}_k\, w))\,\gamma_{k,\nu}(w),\ \mathrm{col}(h(\mathrm{suf}_k\, w))\big), \tag{$\star$}$$

where row, grp, col *are the coordinates of* $(i, a, \lambda) \in \mathsf{K}$*. In particular* $h(w)$ *is determined by its first and last* k *letters and an ordered product in* $\mathbb{H}$*.*

Proof. (1) Products of k *letters lie in* K*.* The Rees quotient S/K has no idempotent but 0, so each element has a power equal to 0. If a product of $m = |\mathsf{S}/\mathsf{K}| \leq k$ elements were nonzero, two of its m nonzero prefix products would coincide, $t_i = t_j$ with $i < j$; then $t_i = t_i x^q$ for $x = v_{i+1} \cdots v_j$ and all q, impossible for $x^q = 0$.

(2) Coordinates. With $(i, a, \lambda)(j, b, \mu) = (i, ap_{\lambda j}b, \mu)$, fix $i_0 \in I$ and $e_\lambda := (i_0, p^{-1}_{\lambda i_0}, \lambda)$, a right identity for elements with Λ-coordinate λ. For $y \in \mathsf{S}$ write $e_\lambda y = (j, c, \mu)$ and $\mathrm{act}(\lambda, y) := p_{\lambda j}c$; then $(i, a, \lambda)y = (i, a\,\mathrm{act}(\lambda, y), \mu)$, so right multiplication fixes the I-coordinate. Dually, left multiplication fixes the Λ-coordinate of an element of K: for any $\mu \in \Lambda$ the idempotent $f := (i, p^{-1}_{\mu i}, \mu) \in \mathsf{K}$ satisfies $f(i, a, \lambda) = (i, a, \lambda)$, so $y(i, a, \lambda) = (yf)(i, a, \lambda)$ with $yf \in \mathsf{K}$, and the Rees product of two elements of K has the Λ-coordinate of its right factor.

(3) Group coordinate. Let $w = a_1 \cdots a_m$, $m \geq 2k$, and $h(a_1 \cdots a_k) = (i, a, \lambda_k)$. For $j \geq k$ the Λ-coordinate λ_j of $h(a_1 \cdots a_j)$ equals that of $h(a_{j-k+1} \cdots a_j)$ by (1) and (2). Put $\nu(F) := \mathrm{act}(\mathrm{col}(h(F[1..k])), h(F[k+1]))$. Applying the right-action identity for $j = k, \ldots, m-1$ in order gives $h(w) = (i,\ a \prod_{j=k}^{m-1} \nu(a_{j-k+1} \cdots a_{j+1}),\ \lambda_m)$, and the product is $\gamma_{k,\nu}(w)$. □

Theorem B.4 (Local-group criteria in sliding-factor coordinates)**.** *For a typing* s *with positive image* S *the following are equivalent:* (i) $\mathsf{S} \in \mathbf{LG}$*;* (ii) S *has no subsemigroup isomorphic to* U_1*;* (iii) *every idempotent of* S *lies in the minimal ideal of* S*;* (iv) s *is local-group-determined, and one may take* $k = |\mathsf{S}|$*.*

Proof. By [32, Proposition 2.3, items (1), (3) and (4)], (i) and (ii) are each equivalent to $\mathsf{S}^m = \mathsf{K}$ for some m. The implication (ii)⇒(iii) is also direct: if an idempotent e lies outside K, pick $y \in \mathsf{K}$ and let f be the idempotent power of eye; then $f \in \mathsf{K}$, so $f \neq e$, and $ef = fe = f$, so $\{e, f\} \cong U_1$. If $\mathsf{S}^m = \mathsf{K}$ then $e = e^m \in \mathsf{K}$ for each idempotent e; conversely, if $E(\mathsf{S}) \subseteq \mathsf{K}$ then $\mathsf{S}^k = \mathsf{S}E(\mathsf{S})\mathsf{S} \subseteq \mathsf{K}$ for $k = |\mathsf{S}|$ by the pumping lemma [32, Lemma 2.2], and $\mathsf{S}^k \supseteq \mathsf{K}$ since S^k is an ideal. So (i)–(iii) are equivalent.

(iii)⇒(iv): Lemma B.3 with $h = s$ and $k = |\mathsf{S}|$ determines $s(w)$ for $|w| \geq 2k$ from $\lambda_{k,\nu}(w)$, and $\widehat{s}_{k,k}$ records shorter words and ε exactly; so $\widehat{s} \preceq \lambda_{k,\nu}$.

(iv)⇒(i): S is a homomorphic image of $\mathsf{T} := \lambda_{k,\nu}(\Sigma^+)$ and $\mathbf{LG}$ is closed under homomorphic images, so it suffices that $\mathsf{T} \in \mathbf{LG}$. No short element is idempotent (squaring changes the exactly recorded word or makes it long). Let $e = (\langle p, q\rangle, \gamma_e)$ be idempotent. By Lemma B.1, every element of $e\mathsf{T}e$ has window $\langle p, q\rangle$, and long elements with this window multiply by $(\langle p, q\rangle, \alpha)(\langle p, q\rangle, \beta) = (\langle p, q\rangle, \alpha J \beta)$, $J := J_\nu(q, p)$. This operation makes $\{\langle p, q\rangle\} \times \mathbb{G}$ a group isomorphic to $\mathbb{G}$ via $\alpha \mapsto \alpha J$, and $e\mathsf{T}e$ is a finite nonempty subsemigroup of it, hence a subgroup. □

Proposition B.5 (Window–group typings are local-group typings, strictly)**.** (i) *If $\widehat{s} \preceq (\widehat{s}_{k,\ell}, g)$ for a morphism g into a finite group, then $\widehat{s} \preceq \lambda_{\hat{k},\nu}$ with $\hat{k} = \max\{k, \ell, 1\}$ and $\nu(F) := g(\text{last letter of } F)$. Hence* $\mathbf{LI} \vee \mathbf{Grp} \subseteq \mathbf{LG}$.

(ii) *Already for $\Sigma = \{\mathtt{a}, \mathtt{b}\}$ and $h_{\mathtt{ab}} := \lambda_{1,\nu}$ with $\nu(\mathtt{ab}) = 1 \in \mathbb{Z}_2$ and $\nu = 0$ otherwise, $h_{\mathtt{ab}}(\Sigma^+) \in \mathbf{LG} \setminus (\mathbf{LI} \vee \mathbf{Grp})$.*

Proof. (i) For $|w| \geq 2\hat{k}$, $g(w) = g(\mathrm{pref}_{\hat{k}}\, w)\, \gamma_{\hat{k},\nu}(w)$, and $\mathrm{pref}_{\hat{k}}, \mathrm{suf}_{\hat{k}}$ determine $\widehat{s}_{k,\ell}(w)$; shorter words are recorded exactly. For the inclusion, a semigroup $\mathsf{S} \in \mathbf{LI} \vee \mathbf{Grp}$ is the positive image of the evaluation typing over the alphabet S, which is window–group-determined by Lemma 4.17, hence local-group-determined, so $\mathsf{S} \in \mathbf{LG}$ by Theorem B.4. (ii) Membership in $\mathbf{LG}$ follows from Theorem B.4. Let $k, \ell \geq 0$ and let $g' : \Sigma^* \to \mathbb{H}$ map into an arbitrary finite group. Choose $m \geq k + \ell + 1$ and $N_0 \geq 1$ divisible by the order of $g'(\mathtt{a})$, and put $u = \mathtt{b}^m \mathtt{a}^{N_0} \mathtt{b}^m$ and $v = \mathtt{b}^{2m}$. They have equal length-k prefix and length-ℓ suffix, are long for $s_{k,\ell}$, and satisfy $g'(u) = g'(\mathtt{b})^{2m} = g'(v)$, but $\#_{\mathtt{ab}}(u) = 1$ and $\#_{\mathtt{ab}}(v) = 0$. Thus $h_{\mathtt{ab}}$ is not window–group-determined, and Lemma 4.17 excludes $\mathbf{LI} \vee \mathbf{Grp}$. □

Lemma B.6 (Thinned and padded owners)**.** *Fix $k \geq 1$. Consider pairwise disjoint nonempty intervals in at most f words (*rows*), each interval belonging to one of m owners and each owner holding at most f intervals, and suppose that no interval meets the first or last k letters of its row. Then there is a set I of at least $m/(4kf + 1)$ owners with the following property. For arbitrary nonempty replacement words of the intervals of the owners in I, there are pairwise disjoint* enlarged blocks*, at most f per owner, such that* (a) *each enlarged block contains intervals of exactly one owner in I, and every interval of an owner in I lies in exactly one enlarged block;* (b) *replacing, inside an enlarged block, the intervals of its owner yields a word of length at least $2k + 1$ with the same first and last k letters as the block, which itself has length at least $2k + 1$; and* (c) *for every $J \subseteq I$, replacing the intervals of the owners in J equals replacing their enlarged blocks as in* (b).

Proof. Join two distinct owners if some interval of one and some interval of the other lie in the same row with fewer than $2k$ letters between them. A neighbouring interval to the right of an interval $[l, r)$ (in cut positions) starts at one of the $2k$ cuts $r, \ldots, r + 2k - 1$, and disjointness and nonemptiness allow at most one interval per cut; the same holds on the left. So each owner has at most $4kf$ neighbours, and greedily choosing an owner and deleting its closed neighbourhood gives an independent set I of at least $m/(4kf + 1)$ owners.

Enlarge every interval of an owner in I by k letters on each side; the boundary hypothesis keeps the enlargement inside its row. Intervals of distinct owners in I are at least $2k$ letters apart, so their enlargements have disjoint interiors; merge overlapping enlargements of the same owner. This gives (a) and does not increase the number of blocks per owner. Let $[l, r)$ be the union of the merged enlargements and l_0 the smallest left end of an interval in it, so $l = l_0 - k$. No interval of an owner in I meets $[l, l_0)$: one of the same owner would have been merged with a smaller left end, and one of another owner in I would end at least $2k$ letters before l_0. Symmetrically for the last k letters. Replacing only the owner's intervals therefore keeps the first and last k letters, and the replacement contains at least one nonempty word besides these $2k$ letters, which gives (b). Letters of owners outside I inside a block are never replaced, so (c) holds for every $J \subseteq I$. □

Lemma B.7 (Short tuples with prescribed local-group type)**.** *Let G be reduced and good of fan-out $\leq f$ and rank $\leq r_{\max}$, and $\lambda = \lambda_{k,\nu}$ with group $\mathbb{G}$. Put $C^{\mathrm{buf}}_{k,\mathbb{G},f} := 2kf + (4kf + 1)c_{\mathbb{G},f}$ and $n_\lambda := |V||M_{k,\nu}|^f$. For every $B \in V$ and every realizable $\mathbf{t} \in M^{\mathrm{fan}(B)}_{k,\nu}$ there is $\vec{u} \in L_B(G)$ with $\lambda^{(\mathrm{fan}(B))}(\vec{u}) = \mathbf{t}$ and*

$$\|\vec{u}\|_1 \leq \beta_\lambda := (r_{\max} + 1)\, \bar{\theta}_G\, C^{\mathrm{buf}}_{k,\mathbb{G},f}\, n_\lambda.$$

Substitutability is not used.

Proof. Let T be λ-minimal for $(B, \mathbf{t})$ with heavy threshold $b = \bar{\theta}_G$. In each root component of length at least $2k$ mark the first and the last k letters, and in each shorter component mark every letter: at most $2kf$ marks. Let Q be an antichain of heavy nodes. By linearity and nondeletion their components occupy disjoint nonempty intervals of the root components, at most f per node, and at most $2kf$ nodes contain a mark; let Q_0 be the others, so their intervals avoid the first and last k letters of every root component and lie in components of length at least $2k$.

Suppose $|Q| > C^{\mathrm{buf}}_{k,\mathbb{G},f}$. Then $|Q_0| > (4kf+1)c_{\mathbb{G},f}$, and Lemma B.6 gives a set $I \subseteq Q_0$ of more than $c_{\mathbb{G},f}$ owners. For $v \in I$ with nonterminal C_v fix a tuple of $L_{C_v}(G)$ of total length at most $\bar{\theta}_G$ (Lemma 4.1(ii)); its components are nonempty, and no window or group condition is imposed on it. By Lemma B.6(b) every enlarged block and its replacement are long and have the same first and last k letters, hence the same $\widehat{s}_{k,k}$-type. So Lemma B.2, applied with the enlarged blocks, writes $\gamma_{k,\nu}$ of every long root component as an ordered product of the block values and constants that do not depend on which owners are replaced. This is a block-replacement family of width f over $\mathbb{G}$ with more than $c_{\mathbb{G},f}$ owners, and Proposition 4.12 gives a nonempty $J \subseteq I$ preserving every root $\gamma_{k,\nu}$-value.

Replace the subtrees at the nodes of J by derivations of the chosen tuples. This is a derivation from B because J is an antichain, and by Lemma B.6(c) its root components are those of the enlarged-block replacement. Short root components contain no replaced letter and are unchanged; long ones keep their marked first and last k letters and stay long, so their $\widehat{s}_{k,k}$-types are unchanged, and their $\gamma_{k,\nu}$-values are preserved by the choice of J. Hence the root λ-vector is $\mathbf{t}$. Every replaced node had total yield length greater than $\bar{\theta}_G$ and its replacement has at most $\bar{\theta}_G$, so the new derivation is strictly shorter, contradicting minimality. Therefore every heavy antichain has at most $C^{\mathrm{buf}}_{k,\mathbb{G},f} \geq 1$ elements, and Lemma 4.11 with $b = \bar{\theta}_G$ and $\mathsf{N} = M_{k,\nu}$ gives the bound. □

C Auxiliary Verification of Qualitative Reconstruction

Proof of Lemma 3.2. Put $\vec{z}_j := \rho(\vec{u}_1, \ldots, \vec{u}_j, \vec{x}_{j+1}, \ldots, \vec{x}_r)$. For $E \in \mathcal{E}_e$, Lemma 3.1 gives an admissible E_j with $E[\vec{z}_{j-1}] = E_j[\vec{x}_j]$ and $E[\vec{z}_j] = E_j[\vec{u}_j]$; since $\vec{u}_j \equiv_{L,s} \vec{x}_j$, E accepts $\vec{z}_{j-1}$ iff it accepts $\vec{z}_j$. So $\mathcal{D}_L(\vec{z}_0) = \mathcal{D}_L(\vec{z}_r)$. The types agree because $\widehat{s}^{(e)}(\rho(\cdots)) = \mathrm{out}_\rho$ of the children's types. □

Proof of Proposition 3.10. Induction on the height of a derivation from $[\vec{x}]$. Const derives $\vec{x}$. Link $[\vec{x}] \to [\vec{y}]$ has same-type $\vec{x}, \vec{y}$ with a common context in $K \subseteq L$, so $\vec{x} \equiv_{L,s} \vec{y}$ by substitutability; combine with the hypothesis for $[\vec{y}]$. For Comp $[\vec{z}] \to \rho([\vec{u}_1], \ldots, [\vec{u}_r])$ with $\vec{z} = \rho(\vec{u}_1, \ldots, \vec{u}_r)$, the children derive tuples equivalent to the $\vec{u}_j$, and Lemma 3.2 transfers this to $\vec{z}$. For $\widehat{S} \to [(w)]$ with $w \in K \subseteq L$, a derived (v) satisfies $(v) \equiv_{L,s} (w)$, and $\square_1$ accepts (w), so $v \in L$; $\widehat{S} \to \varepsilon$ occurs only if $\varepsilon \in K \subseteq L$. The uncapped grammar is the case of cap $\|K\|_+$. □

Proof of Proposition 3.11. This is the argument of [15, Lemma 3] for the concrete grammar. Each $\iota(X)$ and each $\vec{z}_R := \rho(\iota(Y_1), \ldots, \iota(Y_q))$ for a typed rule $R : X \to \rho(Y_1, \ldots, Y_q)$ is observed, via $\chi(X)$. They have the same type (out_ρ of the children's types is the type of X) and the common context $\chi(X)$, so Link gives $[\iota(X)] \to [\vec{z}_R]$. If $q = 0$, Const gives $[\vec{z}_R] \to \vec{z}_R$; otherwise $(\vec{z}_R, \rho, (\iota(Y_j))_j)$ is a composition witness of rank $q \leq r_G$ (and $q \leq \|K\|_+$ by (3.1)), so Comp gives $[\vec{z}_R] \to \rho([\iota(Y_1)], \ldots, [\iota(Y_q)])$. By induction, $X \Rightarrow^*_{\widetilde{G}_0} \vec{u}$ implies $[\iota(X)] \Rightarrow^* \vec{u}$. For $w \in L(G) = L(\widetilde{G}_0)$ (Proposition 3.6(iii)) a successful derivation starts with a start link $\widetilde{S} \to X$; then $\chi(X) = \square_1$, so $\iota(X) = (w')$ with $w' \in K$, and Start gives $\widehat{S} \to [\iota(X)] \Rightarrow^* (w)$. □

Proof of Theorem 3.13. If $L \cap \Sigma^+ = \emptyset$, then $K = L \subseteq \{\varepsilon\}$ and Definition 3.4 gives the claim. Otherwise $\mathcal{F}_L \subseteq L$ is finite, Proposition 3.11 with $\iota = \omega$ (canonical tuples have the right type by Proposition 3.6(i)) gives $L \cap \Sigma^+ \subseteq L(\mathcal{B}(K))$, Proposition 3.10 gives $L(\mathcal{B}(K)) \subseteq L$, and $\varepsilon \in L(\mathcal{B}(K))$ iff $\varepsilon \in K$ iff $\varepsilon \in L$. Rank-preserving normalization gives the last sentence. □

Proof of Theorem 3.14. Put $n_K = \max\{1, \|K\|_+\}$. An occurrence is a sample word with at most $2f$ cut positions, so there are $n_K^{O(f)}$ of them, each typed in linear time; grouping occurrences by context and type lists the Link rules. A composition witness of rank $\leq r_{\max}$ is a parent occurrence with at most $2fr_{\max}$ further cut positions and a labelling whose number depends only on $f, r_{\max}$; goodness is checked directly. This gives $n_K^{O(fr_{\max})}$ Comp rules, and all other rules are fewer. □

Proof of Corollary 3.16. Let $L \in \mathcal{C}^{\mathrm{mcf}}_{f,s}$ (for (ii), also $L \in \mathsf{MCFL}(f, r_{\max})$), let $\mathcal{F}_L$ be the characteristic sample of Theorem 3.13 for the operator used, and consider a text for L; membership in MCFLs is decidable [27], so the learner is computable. Let n_0 be a stage by which all of $\mathcal{F}_L$ has appeared. If the learner changes its hypothesis at some stage $n \geq n_0$, the new hypothesis $\mathcal{B}(K_n)$ generates exactly L, so it is never changed again. Otherwise the hypothesis H held at n_0 is final. If $H = H_0$, every datum was generated by $\emptyset$, so $L = \emptyset$. Else $H = \mathcal{B}(K_m)$ for some $m \leq n_0$, so $L(H) \subseteq L$ by Proposition 3.10; every word of L appears in the text, before m (then it lies in $K_m \subseteq L(H)$ by sample consistency) or after it (then it is generated by H, which is never changed). So $L(H) = L$. For (ii), each update is a construction (Theorem 3.14) plus a membership test, both polynomial in $\|K\|_+$ for fixed f and $r_{\max}$ [14, Proposition 4]. □

These bounds concern the work per datum. They do not bound the stage of convergence, since a text may postpone any example arbitrarily long; short characteristic data alone gives no presentation-independent convergence stage.

D Auxiliary Proofs for Typed Witness Construction

Proof of Proposition 3.6. At every rule application the morphism property gives

$$\widehat{s}^{(\mathrm{fan}(A))}(\rho(\vec{u}_1, \ldots, \vec{u}_r)) = \mathrm{out}_\rho(\widehat{s}^{(d_1)}(\vec{u}_1), \ldots, \widehat{s}^{(d_r)}(\vec{u}_r)).$$

Induction on height gives (i) and the existence and uniqueness of the lift in (ii), since each typed label is forced by its yield. Erasing labels gives $L(G^{\widehat{s}}) \subseteq L(G)$, (ii) gives the converse, and trimming keeps every rule of every successful derivation. □

Definitions 2.8, 3.4 and 3.5 apply verbatim with any finite-monoid morphism $h : \Sigma^* \to \mathsf{N}$ in place of $\widehat{s}$ (tuple types being $h^{(d)}$); write $\mathcal{C}^{\mathrm{mcf}}_{f,h}$ for the class. The class and the operator use types of nonempty components only, so they depend only on $h|_{\Sigma^+}$.

Lemma D.1 (Monoid implementations)**.** *Let $h_0 : \Sigma^* \to M_0$ be a finite-monoid morphism, $\mathsf{S}_0 := h_0(\Sigma^+)$ and $s_0 := h_0|_{\Sigma^+}$.*

(i) *On $(\Sigma^+)^d$, equality of $h_0^{(d)}$-, $s_0^{(d)}$- and $\widehat{s_0}^{(d)}$-values coincide. Hence $\mathcal{C}^{\mathrm{mcf}}_{f,h_0} = \mathcal{C}^{\mathrm{mcf}}_{f,s_0}$, and the operators built with h_0 and with $\widehat{s_0}$ return the same grammar on every sample.*

(ii) *For every good MCFG G, the trimmed refinements of G under h_0 and under $\widehat{s_0}$ coincide (the untrimmed ones need not); hence the same canonical tuples, contexts and witness words may be chosen.*

Proof. (i) The three maps agree on Σ^+. (ii) By λ-freeness and Proposition 3.6(i), every typed nonterminal of a successful derivation under either morphism is $A_{\mathbf{t}}$ with $\mathbf{t} \in \mathsf{S}_0^{\mathrm{fan}(A)}$, and every start link used is $\widetilde{S} \to S_{(\eta)}$ with $\eta \in \mathsf{S}_0$. For children indexed in S_0, each nonempty template component is evaluated in S_0 identically by both morphisms, so both refinements have the same rules among S_0-indexed symbols, which carry all successful derivations. □

Proof of Lemma 3.7. Take a successful typed derivation containing X and follow the path from the root to an occurrence of X, starting with $\square_1$; at each rule apply Lemma 3.1 with the (nonempty) sibling yields fixed. This gives $\chi(X) \in \mathcal{E}_d$, and replacing the X-subtree by any X-derivation gives a successful derivation of $\chi(X)[\vec{u}]$. If $\widetilde{S} \to X$ is a rule, take the path of length zero. □

Proof of Lemma 3.8. For $\vec{u}, \vec{v} \in L_X$, Proposition 3.6(i) gives equal types, $\chi(X)$ accepts both, and the arity is at most f; substitutability gives $\mathcal{D}_L(\vec{u}) = \mathcal{D}_L(\vec{v})$. □

E Details of the Good Normalization

The four conversions below give Proposition 3.12; each satisfies a tuple-language identity from which the size bound is read off. Correctness of the nondeleting and nonmerging conversions is standard [14,27,29]; the size bounds are not stated there and are derived here. For $\vec{w} = (w_1, \ldots, w_d)$ and nonempty $I \subseteq [d]$, $\pi_I(\vec{w})$ keeps the components indexed by I in order, $\sigma\vec{w}$ permutes components and $\mu(\vec{w})$ concatenates consecutive blocks of components; so $\|\pi_I(\vec{w})\|_1 \le \|\vec{w}\|_1 = \|\sigma\vec{w}\|_1 = \|\mu(\vec{w})\|_1$.

Lemma E.1 (Stagewise tuple languages)**.** *Let G_0 be a reduced f-MCFG of rank at most $r_{\max}$. Each conversion below preserves linearity, does not increase fan-out, rank or any template, and multiplies the numbers of nonterminals and rules by a factor depending only on f and $r_{\max}$.*

(a) Nondeletion *(construction (f3) in the proof of [27, Lemma 2.2]). Introduce A^I for nonempty $I \subseteq [\mathrm{fan}(A)]$. For $R : A \to \rho(B_1, \ldots, B_q)$ let J_j be the components of B_j used by the I-indexed outputs of ρ; keep these outputs, use $B_j^{J_j}$ if $J_j \neq \emptyset$ and omit B_j otherwise. The rules are nondeleting and $L_{A^I} = \pi_I(L_A(G_0))$.*

(b) Nonpermutation. *Decorate a fan-out-d nonterminal by $\sigma \in \mathfrak{S}_d$ (the symmetric group on $\{1, \ldots, d\}$), the child's permutation being determined by the order in which the parent template uses its variables. The decorated nonterminal generates σL_A.*

(c) Empty components. *Let $\mathsf{Pat}(A)$ be the set of support patterns of L_A, computed by saturation. For realizable child patterns, replace variables of empty components by ε, delete empty parent components, omit all-empty children, and introduce A^I for each nonempty $I \in \mathsf{Pat}(A)$. Then $L_{A^I} = \pi_I\{\vec{u} \in L_A : \mathrm{supp}(\vec{u}) = I\}$, the result is λ-free, and $S^{\{1\}}$ generates the nonempty part of the language.*

(d) Nonmerging *(construction in the proof of [14, Section 2.3, Lemma 1]). Decorate a nonterminal by a composition μ of its fan-out, the child's composition being given by the maximal runs of its consecutive variables in the parent template. The decorated nonterminal generates the image $\mu(L_A)$.*

Stages (b)–(d) preserve the properties established by earlier stages.

Proof. (a) For any child tuples, $\pi_I(\rho(\vec{v}_1, \ldots, \vec{v}_q)) = \rho^I((\pi_{J_j}\vec{v}_j)_{J_j \neq \emptyset})$; induction gives $\subseteq$, and for $\supseteq$ the omitted children are completed by terminal derivations, which exist because G_0 is reduced.

There are at most 2^f copies of each nonterminal and rule. (b) Linearity and nondeletion make the child permutations well defined; induction gives σL_A. There are at most $f!$ copies. (c) Retained variables occur once and in order; induction gives the identity, and omitted all-empty children are completed by derivations of their empty pattern. A retained child B^J has only the components indexed by J, and each of them is used by the parent template exactly once and in the original order, so nondeletion and nonpermutation persist. There are at most 2^f patterns per nonterminal and $2^{f r_{\max}}$ choices per rule. (d) Nonpermutation makes the runs compatible with each child's order and nondeletion makes them cover its components; then $\rho^{\mu}(\mu_1(\vec{v}_1), \ldots, \mu_q(\vec{v}_q)) = \mu(\rho(\vec{v}_1, \ldots, \vec{v}_q))$, and induction gives the image identity (merging need not be injective, so no inverse-image claim is made). Concatenating nonempty components yields nonempty components, and each merged block of a child is used once and in order, so λ-freeness, nondeletion and nonpermutation persist. The size is linear for fixed $f, r_{\max}$ [14, Section 2.3, after Example 2]. □

Proof of Proposition 3.12. If $L(G_{\text{in}}) \cap \Sigma^+ = \emptyset$, take G_{good} to consist of the start symbol alone, with no rules; it is good and generates $\emptyset = L(G_{\text{in}}) \setminus \{\varepsilon\}$. Otherwise apply (a)–(d) to the trimmed input and trim again. Each stage multiplies the size by a factor depending only on $f, r_{\max}$, so $\|G_{\text{good}}\| = O_{f,r_{\max}}(\|G_{\text{in}}\|)$, and $L(G_{\text{good}}) = L(G_{\text{in}}) \setminus \{\varepsilon\}$ by (c). If this language is nonempty, the final trimming leaves a reduced grammar, and trimming preserves goodness. □

F Substitutability of the Bracketed Doubling Family

This appendix proves Proposition 4.20(ii), adapting the deterministic-parser argument of [23]. Fix $n \geq 1$, put $L^{(0)} := \{\mathtt{a}\}$ and $L^{(i)} := \lceil L^{(i-1)} \rceil \cup \lfloor L^{(i-1)} L^{(i-1)} \rfloor$ for $i \geq 1$, so that $L_{E_i}(G_n) = L^{(i)}$ and $L_n = L^{(n)}$. For $x, y \in \Gamma_{\text{br}}^*$ write $x \equiv y$ if $\{(u, v) : uxv \in L_n\} = \{(u, v) : uyv \in L_n\}$; this is a monoid congruence.

The parser. A *stack* is a word over $E_0, \ldots, E_n, \rceil, \rfloor$, with its top on the left; ε is the empty stack. For every stack $\mathcal{Z}$ and $1 \leq i \leq n$ the moves are

$$E_0\mathcal{Z} \xrightarrow{\mathtt{a}} \mathcal{Z}, \quad E_i\mathcal{Z} \xrightarrow{\lceil} E_{i-1}\rceil\mathcal{Z}, \quad E_i\mathcal{Z} \xrightarrow{\lfloor} E_{i-1}E_{i-1}\rfloor\mathcal{Z},$$
$$\rceil\mathcal{Z} \xrightarrow{\rceil} \mathcal{Z}, \quad \rfloor\mathcal{Z} \xrightarrow{\rfloor} \mathcal{Z},$$

and there are no others; in particular no move leaves ε. Write $\mathcal{X} \xrightarrow{w} \mathcal{Y}$ if reading w from $\mathcal{X}$ ends in $\mathcal{Y}$. Three properties are used.

(P1) *Determinism and injectivity.* For each letter c, the c-moves form a partial injective map on stacks: an opening bracket reveals the expanded E_i and the untouched suffix, and $\mathtt{a}$, $\rceil$, $\rfloor$ identify the popped top symbol. Hence every word acts as a partial injection on stacks.

(P2) *Weight.* Give E_i weight 4^i and each closing marker weight 1, and let $\mathrm{wt}(\mathcal{X})$ be the total weight. Every move lowers wt, since $4^i > 2 \cdot 4^{i-1} + 1$; hence $\mathcal{X} \xrightarrow{w} \mathcal{X}$ only for $w = \varepsilon$.

(P3) *Locality.* Moves read and rewrite only the top symbol, and closing markers are only popped. Hence $\mathcal{X} \xrightarrow{w} \mathcal{Y}$ implies $\mathcal{X}\mathcal{Z} \xrightarrow{w} \mathcal{Y}\mathcal{Z}$. Conversely, if $\mathcal{X}_1 \neq \varepsilon$ and $\mathcal{X}_1\mathcal{Z} \xrightarrow{w} \mathcal{Y}$, then either (a) the run reaches $\mathcal{Z}$: $w = w_1 w_2$ with $\mathcal{X}_1 \xrightarrow{w_1} \varepsilon$ and $\mathcal{Z} \xrightarrow{w_2} \mathcal{Y}$, so $\mathrm{wt}(\mathcal{Y}) \leq \mathrm{wt}(\mathcal{Z})$; or (b) it does not: $\mathcal{Y} = \mathcal{Y}_1\mathcal{Z}$ with $\mathcal{X}_1 \xrightarrow{w} \mathcal{Y}_1 \neq \varepsilon$, so $\mathrm{wt}(\mathcal{Y}) > \mathrm{wt}(\mathcal{Z})$, and $\mathcal{Y}_1$ ends with the last symbol of $\mathcal{X}_1$ whenever that symbol is a closing marker. For fixed $\mathcal{Z}$, comparing $\mathrm{wt}(\mathcal{Y})$ with $\mathrm{wt}(\mathcal{Z})$ decides the case.

Lemma F.1. *For $0 \le i \le n$ and every stack $\mathcal{Z}$, $E_i\mathcal{Z} \xrightarrow{w} \mathcal{Z}$ if and only if $w \in L^{(i)}$. Moreover, if $\mathcal{X} \xrightarrow{w} \mathcal{Y}$ and $w \in L^{(i)}$, then $\mathcal{X} = E_i\mathcal{Y}$.*

Proof. A run $E_i\mathcal{Z} \xrightarrow{w} \mathcal{Z}$ is in case (a) of (P3), since case (b) ends in $\mathcal{Y}_1\mathcal{Z} \neq \mathcal{Z}$; so $w = w_1w_2$ with $E_i \xrightarrow{w_1} \varepsilon$ and $\mathcal{Z} \xrightarrow{w_2} \mathcal{Z}$, and $w_2 = \varepsilon$ by (P2). With the lifting in (P3), it suffices to show $E_i \xrightarrow{w} \varepsilon$ iff $w \in L^{(i)}$, by induction on i. From E_0 only the $\mathtt{a}$-move applies, and ε admits none. For $i \ge 1$, the first letter is $\lceil$ or $\lfloor$. After $\lceil$, the stack $E_{i-1}\rceil$ must be emptied; case (a) of (P3) with $\mathcal{X}_1 = E_{i-1}$, $\mathcal{Z} = \rceil$ splits the rest as zz' with $E_{i-1} \xrightarrow{z} \varepsilon$ and $z' = \rceil$. After $\lfloor$, two such splittings of $E_{i-1}E_{i-1}\rfloor$ give $z_1z_2\rfloor$ with $E_{i-1} \xrightarrow{z_j} \varepsilon$. Induction gives $E_i \xrightarrow{w} \varepsilon$ iff $w \in L^{(i)}$. For the last claim, $E_i\mathcal{Y} \xrightarrow{w} \mathcal{Y}$ by the first part, and (P1) gives $\mathcal{X} = E_i\mathcal{Y}$. □

Lemma F.2. *If $0 \le i \le n$ and $w, w' \in L^{(i)}$, then $w \equiv w'$.*

Proof. If $pwq \in L_n$, Lemma F.1 gives a run $E_n \xrightarrow{p} \mathcal{X} \xrightarrow{w} \mathcal{Y} \xrightarrow{q} \varepsilon$ with $\mathcal{X} = E_i\mathcal{Y}$, and also $E_i\mathcal{Y} \xrightarrow{w'} \mathcal{Y}$; so $pw'q \in L_n$. Exchange w and w' for the converse. □

Lemma F.3 (Branch synchronization)**.** *Let $\mathcal{X} \xrightarrow{x} \mathcal{Y}$ and $\mathcal{X} \xrightarrow{y} \mathcal{Y}$, where x, y are nonempty and begin with different letters. Then $\mathcal{X} = E_i\mathcal{Z}$ for some $1 \le i \le n$ and stack $\mathcal{Z}$, and $x = wx'$, $y = w'y'$ with $w, w' \in L^{(i)}$, $\mathcal{Z} \xrightarrow{x'} \mathcal{Y}$ and $\mathcal{Z} \xrightarrow{y'} \mathcal{Y}$.*

Proof. Only a top E_i with $i \ge 1$ admits two different letters, namely $\lceil$ and $\lfloor$. Say $x = \lceil x_0$ and $y = \lfloor y_0$, so $E_{i-1}\rceil\mathcal{Z} \xrightarrow{x_0} \mathcal{Y}$ and $E_{i-1}E_{i-1}\rfloor\mathcal{Z} \xrightarrow{y_0} \mathcal{Y}$. By (P3) both runs are in the same case. Case (b) is impossible: it would give $\mathcal{Y} = \mathcal{Y}_1\mathcal{Z} = \mathcal{Y}_2\mathcal{Z}$ with $\mathcal{Y}_1$ ending in $\rceil$ and $\mathcal{Y}_2$ in $\rfloor$. In case (a), $x_0 = x_1x'$ with $E_{i-1}\rceil \xrightarrow{x_1} \varepsilon$ and $\mathcal{Z} \xrightarrow{x'} \mathcal{Y}$, so $w := \lceil x_1$ satisfies $E_i \xrightarrow{w} \varepsilon$ and $w \in L^{(i)}$ by Lemma F.1. The same argument gives $y = w'y'$. □

Proposition F.4. *If $\mathcal{X} \xrightarrow{x} \mathcal{Y}$ and $\mathcal{X} \xrightarrow{y} \mathcal{Y}$, then $x \equiv y$. Consequently L_n is Clark–Eyraud substitutable.*

Proof. Induct on $|x|+|y|$. If x or y is empty, then $\mathcal{X} = \mathcal{Y}$, and the other is empty by (P2). If $x = cx'$ and $y = cy'$ for a letter c, determinism gives $\mathcal{X} \xrightarrow{c} \mathcal{X}'$ with $\mathcal{X}' \xrightarrow{x'} \mathcal{Y}$ and $\mathcal{X}' \xrightarrow{y'} \mathcal{Y}$, and induction applies. Otherwise Lemma F.3 gives $x = wx'$, $y = w'y'$; induction gives $x' \equiv y'$, Lemma F.2 gives $w \equiv w'$, and $\equiv$ is a congruence.

For the consequence, let $x, y \in \Gamma_{\mathrm{br}}^{+}$ and $uxv, uyv \in L_n$. Lemma F.1 gives runs $E_n \xrightarrow{u} \mathcal{X} \xrightarrow{x} \mathcal{Y} \xrightarrow{v} \varepsilon$ and $E_n \xrightarrow{u} \mathcal{X}' \xrightarrow{y} \mathcal{Y}' \xrightarrow{v} \varepsilon$. By (P1), $\mathcal{X} = \mathcal{X}'$ and $\mathcal{Y} = \mathcal{Y}'$, so $x \equiv y$. □

G Substitutability of the Three-Letter Doubling Family

This appendix proves that K_n in Proposition 4.24 is Clark–Eyraud substitutable. Fix $n \ge 1$ and write $x \equiv y$ if $\{(u, v) : uxv \in K_n\} = \{(u, v) : uyv \in K_n\}$, a monoid congruence. Stacks are words over $E_0, \dots, E_n$ and the closing markers $\hat{\mathtt{b}}_i, \hat{\mathtt{c}}_i$ $(1 \le i \le n)$, top on the left, and for every stack $\mathcal{Z}$ and $1 \le i \le n$ the moves are

$$E_0\mathcal{Z} \xrightarrow{\mathtt{a}} \mathcal{Z}, \quad E_i\mathcal{Z} \xrightarrow{\mathtt{b}} E_{i-1}E_{i-1}\hat{\mathtt{b}}_i\mathcal{Z}, \quad E_i\mathcal{Z} \xrightarrow{\mathtt{c}} E_{i-1}\hat{\mathtt{c}}_i\mathcal{Z},$$
$$\hat{\mathtt{b}}_i\mathcal{Z} \xrightarrow{\mathtt{b}} \mathcal{Z}, \quad \hat{\mathtt{c}}_i\mathcal{Z} \xrightarrow{\mathtt{c}} \mathcal{Z}$$

and no others. The moves are deterministic, and properties (P2) and (P3) of Appendix F hold with the same proofs. So does the first part of Lemma F.1: $E_i\mathcal{Z} \xrightarrow{w} \mathcal{Z}$ iff $w \in K^{(i)}$. Property (P1) fails, because $\mathtt{b}$ and $\mathtt{c}$ both open and close: for instance $\mathtt{b}$ maps $E_i\mathcal{Z}$ and $\hat{\mathtt{b}}_jE_{i-1}E_{i-1}\hat{\mathtt{b}}_i\mathcal{Z}$ to the same stack. We replace it by injectivity on reachable stacks, those $\mathcal{X}$ with $E_n \xrightarrow{u} \mathcal{X}$ for some u.

Lemma G.1. *Put $\mathcal{F}_i := \{E_{i-1}\hat{\mathtt{b}}_i, \hat{\mathtt{b}}_i, \hat{\mathtt{c}}_i\}$. Every reachable stack is ε or $\mathcal{Y}F_{j+1}\cdots F_n$ with $F_i \in \mathcal{F}_i$ and $\mathcal{Y} \in \{E_j, \hat{\mathtt{b}}_j, \hat{\mathtt{c}}_j\}$ ($\mathcal{Y} = E_j$ with $0 \le j \le n$, or a marker with $1 \le j \le n$). Each letter, and hence each word, acts as a partial injection on reachable stacks.*

Proof. The set of such stacks contains E_n and is closed under the moves: from $E_jF_{j+1}\cdots F_n$, $\mathtt{b}$ gives $E_{j-1}(E_{j-1}\hat{\mathtt{b}}_j)F_{j+1}\cdots F_n$ and $\mathtt{c}$ gives $E_{j-1}\hat{\mathtt{c}}_jF_{j+1}\cdots F_n$; popping the top leaves $F_{j+1}\cdots F_n$, which is ε or again of the stated form, a first frame $E_j\hat{\mathtt{b}}_{j+1}$ being read as $\mathcal{Y} = E_j$ followed by the frame $\hat{\mathtt{b}}_{j+1}$. For injectivity, $\mathtt{a}$ only pops E_0. A $\mathtt{b}$-opening produces a stack beginning with two symbols E, from which i and $\mathcal{Z}$ are recovered; a $\mathtt{b}$-closing produces a product of frames, which begins with a marker or with $E_j\hat{\mathtt{b}}_{j+1}$, never with two symbols E, and whose first marker (or emptiness, for $i = n$) determines the popped level. The same holds for $\mathtt{c}$, whose openings begin with $E_{i-1}\hat{\mathtt{c}}_i$. □

Lemma G.2. *If $\mathcal{X}$ is reachable, $w \in K^{(i)}$ and $\mathcal{X} \xrightarrow{w} \mathcal{Y}$, then $\mathcal{X} = E_i\mathcal{Y}$. Consequently $w \equiv w'$ for all $w, w' \in K^{(i)}$.*

Proof. Induct on i; for $i = 0$, $\mathtt{a}$ only pops E_0. Let $i \ge 1$; all stacks met are reachable. If $w = \mathtt{c}v\mathtt{c}$ with $v \in K^{(i-1)}$ and the first $\mathtt{c}$ opens E_j, induction applied to v at $E_{j-1}\hat{\mathtt{c}}_j\mathcal{Z}$ gives $j = i$, and the last $\mathtt{c}$ pops $\hat{\mathtt{c}}_i$. If the first $\mathtt{c}$ pops $\hat{\mathtt{c}}_j$, the rest is a product of frames; to read v it must begin with E_{i-1}, so its first frame is $E_{i-1}\hat{\mathtt{b}}_i$, and after v the top is $\hat{\mathtt{b}}_i$, on which $\mathtt{c}$ has no move. If $w = \mathtt{b}v_1v_2\mathtt{b}$ and the first $\mathtt{b}$ opens E_j, induction gives $j = i$ and the two child words pop the two copies of E_{i-1} before the last $\mathtt{b}$ pops $\hat{\mathtt{b}}_i$. If the first $\mathtt{b}$ pops $\hat{\mathtt{b}}_j$, then as before v_1 leaves the top $\hat{\mathtt{b}}_i$, from which v_2 cannot be read. This proves the first claim. For the second, if $uwv \in K_n$, the run $E_n \xrightarrow{u} \mathcal{X} \xrightarrow{w} \mathcal{Y} \xrightarrow{v} \varepsilon$ has $\mathcal{X} = E_i\mathcal{Y}$, and $E_i\mathcal{Y} \xrightarrow{w'} \mathcal{Y}$; exchange w, w' for the converse. □

Proposition G.3. *K_n is Clark–Eyraud substitutable.*

Proof. Lemma F.3 holds with the same proof, with $\mathtt{c}, \mathtt{b}$ in place of $\lceil, \lfloor$ and $K^{(i)}$ in place of $L^{(i)}$, since it uses only determinism, (P3) and the recognition of complete words from E_i. The proof of the first claim of Proposition F.4 then gives: if $\mathcal{X} \xrightarrow{x} \mathcal{Y}$ and $\mathcal{X} \xrightarrow{y} \mathcal{Y}$, then $x \equiv y$, with Lemma G.2 in place of Lemma F.2. Now let x, y be nonempty with $uxv, uyv \in K_n$. Determinism gives runs $E_n \xrightarrow{u} \mathcal{X} \xrightarrow{x} \mathcal{Y} \xrightarrow{v} \varepsilon$ and $E_n \xrightarrow{u} \mathcal{X} \xrightarrow{y} \mathcal{Y}' \xrightarrow{v} \varepsilon$; $\mathcal{Y}$ and $\mathcal{Y}'$ are reachable, so Lemma G.1 gives $\mathcal{Y} = \mathcal{Y}'$, and $x \equiv y$. □

H Polynomial Data for the Binary Typing onto U_1

This appendix proves Theorem 6.10. Throughout, $G = (V, \Sigma, P, S, \mathrm{fan})$ is reduced and good of fan-out $\le f$ and rank $\le r$, $r := r_{\max}$, $v := |V|$, $\theta := \bar{\theta}_G$, and for each $A \in V$ the context E_A is that of Lemma 4.1(iv), so $|E_A|_{\mathrm{fix}} \le \theta$ and $E_A[L_A(G)] \subseteq L(G)$. For each $B \in V$ fix $\omega_B \in L_B(G)$ with $\|\omega_B\|_1 \le \theta$. In a derivation tree, the *yield* of a node is the tuple it derives, and a node *writes* the terminals that occur in its template.

Lemma H.1. *Let $L \in \mathcal{C}^{\mathrm{mcf}}_{f,s}$ with $L(G) = L \cap \Sigma^+$, and let $X = A_{\mathbf{t}}$ be a typed nonterminal of arity d. Then $\mathrm{Cl}_{L,s}(X) = \{\vec{z} \in (\Sigma^+)^d : s^{(d)}(\vec{z}) = \mathbf{t},\ E_A[\vec{z}] \in L\}$; hence τ_X is the least total length of such a tuple.*

Proof. Fix $\vec{u} \in L_X$. By Lemma 3.8, $\mathrm{Cl}_{L,s}(X)$ is the set of tuples with the type and the distribution of $\vec{u}$; they are accepted by $E_A \in \mathcal{D}_L(\vec{u})$. A tuple $\vec{z}$ of type $\mathbf{t}$ accepted by E_A shares this context with $\vec{u}$, so $\mathcal{D}_L(\vec{z}) = \mathcal{D}_L(\vec{u})$ by substitutability. □

Lemma H.2. *In a derivation tree of G, let u be a proper ancestor of u' with the same nonterminal, and suppose their yields have the same total Parikh vector. Then every rule on the path from u to u' has rank one, writes no terminal and is a componentwise identity, so replacing the subtree at u by the subtree at u' changes no derived tuple and removes at least one node.*

Proof. By linearity and nondeletion every letter of the yield of u' occurs in the yield of u, so equal Parikh vectors exclude terminals written on the path and siblings off the path, since siblings have nonempty yields (λ-freeness). So every rule on the path has one child of some fan-out d', each of its d' variables occurs once (nondeletion) and in order (nonpermutation), and each parent component is nonempty (λ-freeness). Two variables in one component would be consecutive, which nonmerging forbids; hence every component is a single variable, the fan-outs agree, and the rule is the identity. □

For a typing s onto U_1 put $\Sigma_1 := \{x \in \Sigma : s(x) = 1\}$; words of Σ_1^+ are *pure*, and a word has type ζ iff it is not pure. For $c \geq 1$, L has a *c-deep insertion* if some word of L has a factor xzy with $x, y \in \Sigma_1^c$ and z not pure.

Lemma H.3. *Let s be a typing onto U_1, $c \geq 1$, and suppose $L(G) = L \cap \Sigma^+$ has no c-deep insertion. Then $\tau(G, s) \leq 2^{f+1} cf(r+1)v\theta$.*

Proof. Let $X = A_{\mathbf{t}}$ be a typed nonterminal of arity d and T a derivation of $\widetilde{G}_0$ from X with the fewest nodes. No typed label repeats on a path of T, since cutting between two equal typed labels gives a smaller derivation from X; as all components are nonempty, there are at most $2^f v$ typed labels, which bounds the length of every path. In each pure root component mark the first and the last $\min\{c, |u_i|\}$ letters, and in each other component one letter outside Σ_1; there are at most $2cf$ marks. Let H be the set of nodes on the paths from the root to the nodes writing marked letters, so $|H| \leq 2cf \cdot 2^f v$. Erase the type indices and replace, one at a time, each subtree hanging off H by a derivation of ω_B for its root nonterminal B.

Every intermediate root tuple lies in $L_A(G)$. We claim that the pure root components stay pure. A replaced subtree contributes contiguous blocks to root components, and these blocks contain no mark. A pure component of length at most $2c$ is entirely marked and receives no block; in a longer one every block lies between the first and the last c letters, which are unchanged and pure. If a replacement made such a component impure, inserting the root tuple into E_A would give a word of L with a c-deep insertion. Marked letters outside Σ_1 also stay, so the final tuple has type $\mathbf{t}$ and lies in L_X by Lemma 4.2(i). Each node of H writes at most θ letters and has at most r replaced children of length at most θ, so the tuple has length at most $(r+1)\theta|H|$. □

From now on $\Sigma = \{\mathtt{a}, \mathtt{b}\}$, $s = s_{U,2}$, $\Psi(w) := (|w|_{\mathtt{a}}, |w|_{\mathtt{b}})$, and L is *Parikh-saturated* if membership of a word of Σ^+ in L depends only on its Parikh vector.

Lemma H.4. *Let $L \in \mathcal{C}_{2,s}^{\mathrm{mcf}}$ and let some word of L contain $\mathtt{a}^m$.* (i) *If $i, j, i', j' \geq 1$ and $i + j = i' + j' \leq m - 1$, then $U\mathtt{a}^i Z\mathtt{a}^j V \in L \iff U\mathtt{a}^{i'} Z\mathtt{a}^{j'} V \in L$ for all $U, V \in \Sigma^*$ and $Z \in \Sigma^+$.* (ii) *If $m \geq 4$ and some word of L contains $\mathtt{aaba}$ or $\mathtt{abaa}$, then L is Parikh-saturated.*

Proof. (i) Let $p\mathtt{a}^m q \in L$ and $k := m - i - j \geq 1$. The tuples $(\mathtt{a}^i, \mathtt{a}^j)$ and $(\mathtt{a}^{i'}, \mathtt{a}^{j'})$ of type $(1, 1)$ share the admissible context $p\Box_1\mathtt{a}^k\Box_2 q$, so they have the same distribution; test it on the admissible context $U\Box_1 Z\Box_2 V$. (ii) Say $U\mathtt{aaba}V \in L$ (the other case is symmetric). By (i) with $i + j = 3$, $U\mathtt{abaa}V \in L$. The context $(U\mathtt{a}, \mathtt{a}V)$ accepts $\mathtt{ab}$ and $\mathtt{ba}$, both of type ζ, so $\mathcal{D}(\mathtt{ab}) = \mathcal{D}(\mathtt{ba})$: in every context $\mathtt{ab}$ may be replaced by $\mathtt{ba}$ and conversely. Adjacent transpositions connect all words with one Parikh vector. □

Lemma H.5. *Let $L \in \mathcal{C}_{2,s}^{\mathrm{mcf}}$. If some word of L has a factor $\mathtt{a}^6 z \mathtt{a}^6$ with z containing $\mathtt{b}$, then L is Parikh-saturated.*

Proof. The first maximal run $\mathtt{b}^e$ of z is preceded by at least six letters $\mathtt{a}$ and followed by at least one, so the word is $P\mathtt{a}^i\mathtt{b}^e\mathtt{a}^j R$ with $i \geq 6$, $j \geq 1$. Choose $i_0 + j_0 = 5$ with $1 \leq i_0 \leq i$, $1 \leq j_0 \leq j$, and put $P_0 := P\mathtt{a}^{i-i_0}$, $R_0 := \mathtt{a}^{j-j_0}R$. Lemma H.4(i) with $m = 6$ gives $w_k := P_0\mathtt{a}^k\mathtt{b}^e\mathtt{a}^{5-k}R_0 \in L$ for $1 \leq k \leq 4$. If $e = 1$, w_2 contains $\mathtt{aaba}$. If $e \geq 2$, the tuples $\vec{x} = (\mathtt{a}, \mathtt{b}^{e-1}\mathtt{a})$ and $\vec{y} = (\mathtt{aa}, \mathtt{b}^{e-1})$ of type $(1, \zeta)$ are accepted by $P_0\Box_1\mathtt{b}\Box_2\mathtt{a}^3R_0$ (giving w_1 and w_2), so they have the same distribution; the context $P_0\Box_1\mathtt{a}\Box_2\mathtt{ba}^2R_0$ accepts $\vec{y}$ (giving w_3) and hence $\vec{x}$, which gives a word containing $\mathtt{abaa}$. Apply Lemma H.4(ii). □

Lemma H.6. *Let $f \leq 2$, let L be Parikh-saturated and $(1, s)$-tuple-substitutable, and $L(G) = L \cap \Sigma^+$. Then $\tau(G, s) \leq (42r(r+1) + 12(r+1))v^2\theta^5$.*

Proof. Fix $X = A_{\mathbf{t}}$, put $(p, q) := \Psi(E_A)$, so $p + q \leq \theta$, and $\mathcal{P} := \Psi(L \cap \Sigma^+)$. By Lemma H.1 and Parikh saturation, τ_X is the least $\|\vec{z}\|_1$ over tuples $\vec{z}$ of type $\mathbf{t}$ with $(p, q) + \sum_i \Psi(z_i) \in \mathcal{P}$. Write $(x_B, y_B) := \Psi(\omega_B)$. For a node u of a derivation tree let (a_u, b_u) be the Parikh vector of its yield and B_u its nonterminal. If u lies in a successful derivation of a word w, replacing its subtree by a derivation of ω_{B_u} gives a word of L with Parikh vector $\Psi(w) - (a_u, b_u) + (x_{B_u}, y_{B_u})$; for a proper descendant u' with $B_{u'} = B_u$, replacing the subtree at u by that at u' gives one with $\Psi(w) - (a_u - a_{u'}, b_u - b_{u'})$.

Type ζ only. Mark one letter $\mathtt{b}$ in each component and argue as in Lemma H.3: $\tau_X \leq 2 \cdot 4v(r+1)\theta$.

Mixed type, say $\mathbf{t} = (1, \zeta)$ (the other order is symmetric). The fillers may be taken as $(\mathtt{a}^c, \mathtt{b}^y)$, so τ_X is the least $c + y$ with $c, y \geq 1$ and $(p + c, q + y) \in \mathcal{P}$. Fix such c, y and a successful derivation T of $W := \mathtt{b}^{q+y}\mathtt{a}^{p+c}$ with the fewest nodes. Call u heavy if $a_u + b_u > \theta$. If $a_u < c$ and $b_u < y$ for a heavy u, its replacement gives $(p + c', q + y') \in \mathcal{P}$ with $c', y' \geq 1$ and $c' + y' < c + y$; so every heavy node has $a_u \geq c$ or $b_u \geq y$. If u' is a proper descendant of u with the same nonterminal, then $a_u - a_{u'} \geq c$ or $b_u - b_{u'} \geq y$: a zero difference contradicts node-minimality (Lemma H.2), and another small difference contradicts minimality of $c + y$. Along a path, $(\lfloor a_u/c \rfloor, \lfloor b_u/y \rfloor)$ is componentwise nonincreasing and takes at most $p + q + 3 \leq \theta + 3$ values, and no nonterminal repeats while it is constant; so paths have at most $v(\theta + 3)$ nodes. Heavy nodes at one depth have disjoint yields, so there are at most $(p + c)/c + (q + y)/y \leq \theta + 2$ of them. Heavy nodes are closed under ancestors, and every letter is written by a heavy node or lies in one of its at most r non-heavy children; hence $c + y \leq |W| \leq \max\{\theta, (r+1)\theta(\theta+2)(\theta+3)v\} \leq 12(r+1)v\theta^3$.

Type 1 only, with $h := d \in \{1, 2\}$. Let $K := \min\{k \geq h : (p + k, q) \in \mathcal{P}\}$; then $\tau_X = K$, and we may assume $K > \theta + 2 \geq p + 2$. Fix a successful derivation T of $W := \mathtt{b}^q\mathtt{a}^{p+K}$ with the fewest nodes.

(P1) *Row rigidity*: there are no $(c_0, y), (c_0 + \delta, y) \in \mathcal{P}$ with $c_0 \geq 1$ and $1 \leq \delta \leq K - h$. Otherwise the context $(\mathtt{b}^y\mathtt{a}^{c_0-1}, \varepsilon)$ accepts $\mathtt{a}$ and $\mathtt{a}^{1+\delta}$, so $\mathcal{D}(\mathtt{a}) = \mathcal{D}(\mathtt{a}^{1+\delta})$ by $(1, s)$-substitutability; the context $(\mathtt{b}^q\mathtt{a}^{p+K-\delta-1}, \varepsilon)$ accepts $\mathtt{a}^{1+\delta}$, hence $\mathtt{a}$, and $(p + K - \delta, q) \in \mathcal{P}$ with $h \leq K - \delta < K$, contradicting the choice of K.

(P2) *Height.* If u' is a proper descendant of u with the same nonterminal and $b_u = b_{u'}$, then $a_u - a_{u'} \geq K - h + 1$: the value 0 contradicts node-minimality (Lemma H.2), and a value in $[1, K - h]$ contradicts the choice of K. Along a path b_u takes at most $q + 1$ values, and three nodes with one nonterminal and one value of b_u would give $p + K \geq a_u \geq 2(K - h + 1)$, false as $K > p + 2$. So paths have at most $2v(q + 1)$ nodes. A node is *mixed* if its yield contains $\mathtt{b}$; each mixed node lies on the path to one of the q nodes writing a letter $\mathtt{b}$, so there are $M \leq 2vq(q + 1) \leq 4v\theta^2$ of them.

(P3) *Heavy pure nodes.* Call a pure node u heavy if $\ell_u := a_u > \theta + 1$, and put $g_u := \ell_u - x_{B_u} \geq 2$. Two comparable heavy pure nodes have different nonterminals, since by (P2) the upper one would

have $\ell_u \geq K - h + 1 + \theta + 2 > p + K = |W|_{\mathtt{a}}$. In particular heavy pure nodes with one nonterminal have disjoint yields. For an antichain J of heavy pure nodes, replacing all of them gives a word with Parikh vector $(p + K - \sigma(J), q + Y(J))$, where $\sigma(J) := \sum_{u \in J} g_u$ and $Y(J) := \sum_{u \in J} y_{B_u}$; by (P1), two antichains with $\sigma(J), \sigma(J') \leq K - h$ and $Y(J) = Y(J')$ have $\sigma(J) = \sigma(J')$, because both $\mathtt{a}$-counts are at least $p + h \geq 1$.

(P4) *Repeated nonterminals.* If B labels $m_B \geq 2$ heavy pure nodes, their yields are disjoint, so each has $\ell_u \leq p + K - (\theta + 2) \leq K - 2$ and $g_u \leq K - 2 \leq K - h$; comparing two singletons by (P3) gives a common value g_B, and then a common ℓ_B. If moreover $y_B = 0$, one replacement gives $(p + K - g_B, q) \in \mathcal{P}$ with $h \leq K - g_B < K$, which is impossible; so $y_B \geq 1$. Call B *frequent* if $m_B > \theta$. For frequent B, C, choose y_C nodes labelled B and, separately, y_B nodes labelled C; both sets are antichains with $Y = y_B y_C$, and $y_C g_B \leq (m_B - 1)\ell_B \leq p + K - \ell_B \leq K - 2$, and likewise for C. By (P3), $y_C g_B = y_B g_C$, so all frequent B share a ratio $\alpha := g_B / y_B$, and $\ell_B = x_B + \alpha y_B \leq (1 + \alpha)\theta$.

(P5) *The ratio.* Put $\alpha_* := (r + 1)(\theta + 1)(2\theta + 1)v \leq 6(r + 1)v\theta^2$; we show $\alpha \leq \alpha_*$ when frequent nonterminals exist. This is clear if $\alpha \leq \theta$. Otherwise fix a node u labelled by a frequent B, and let H_u be the number of heavy pure nodes in its subtree. These nodes are closed under ancestors within the subtree, and each writes at most θ letters and has at most r other children, of length at most $\theta + 1$; so $\ell_B \leq (r + 1)(\theta + 1)H_u$. Group them by their number of heavy proper ancestors in the subtree: by (P3) there are at most v groups, each an antichain. In one group the nodes with frequent labels have length at least α and disjoint yields inside that of u, so there are at most $(1+\alpha)\theta/\alpha < \theta+1$ of them; nodes with other labels number at most $v\theta$ in all of T. So $H_u \leq v(2\theta+1)$, and $\alpha \leq g_B \leq \ell_B \leq \alpha_*$.

(P6) *Counting.* Call a pure node *large* if $\ell_u > T_* := (\alpha_* + 1)\theta \leq 7(r + 1)v\theta^3$. By (P4)–(P5) no frequent nonterminal labels a large node, so there are at most $v\theta$ large nodes, and they are closed under ancestors within pure subtrees. Every letter $\mathtt{a}$ of W is written by a mixed node, is written by a large node, or lies in a maximal non-large pure subtree, whose root is the root of T or a child of a mixed or a large node. Hence

$$p + K \leq \theta M + (rM + rv\theta + 1)T_* + \theta \cdot v\theta \leq 5v\theta^3 + 6rv\theta^2 \cdot 7(r + 1)v\theta^3 \leq (42r(r + 1) + 5)v^2\theta^5,$$

using $M \leq 4v\theta^2$ and $4rv\theta^2 + rv\theta + 1 \leq 6rv\theta^2$. In all three cases $\tau_X \leq (42r(r+1)+12(r+1))v^2\theta^5$. □

Proof of Theorem 6.10. If L has no 6-deep insertion, Lemma H.3 with $f = 2$, $c = 6$ gives $\tau \leq 96(r + 1)v\theta$. Otherwise L is Parikh-saturated by Lemma H.5, and Lemma H.6 applies. In both cases $\tau \leq C_r v^2\theta^5$. By Corollary 4.10 there is a common characteristic sample with $\|K\|_+ \leq N_{\mathrm{typ}}(1 + 2\theta + r\tau) + \mathbf{1}_{\{\varepsilon \in L\}}$, and $N_{\mathrm{typ}} \leq |V||\mathsf{S}^\sharp|^2 + |P||\mathsf{S}^\sharp|^{2r} \leq 3^{2r}\|G\|$ since $|\mathsf{S}^\sharp| = 3$. With $v \leq \|G\|$ and $1 + 2\theta \leq 3\|G\|^2\theta^5$ this gives the stated bound. □

Declaration of generative AI and AI-assisted technologies in the manuscript preparation process

During preparation of this work, the author used ChatGPT (OpenAI) and Claude (Anthropic) to assist with organization of the manuscript, expression and translation of text, cross-checks of references and notation, and editing of LaTeX source. After using these tools, the author reviewed and edited the material as needed and takes full responsibility for the content of the publication.

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